The Functions in the Package gamlss.prepdata

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Introduction

This booklet introduces the `gamlss.prepdata` package and its functionality. It aims to describe the available functions and how they can be used.

The latest versions of the packages **gamlss**, **gamlss2** and **gamlss.prepdata** are shown below:

```
rm(list=ls())
library(gamlss)
library(gamlss2)
library(ggplot2)
library(gamlss.ggplots)
library(gamlss.prepdata)
library("dplyr")
packageVersion("gamlss")
```

```
[1] '5.5.1'
```

```
packageVersion("gamlss2")
```

```
[1] '0.1.0'
```

```
packageVersion("gamlss.prepdata")
```

```
[1] '0.1.17'
```

The package: `gamlss.prepdata`

The `gamlss.prepdata` package originated from the `gamlss.ggplots` package. As `gamlss.ggplots` became too large, for easy maintenance, it was split into two separate packages, and `gamlss.prepdata` was created.

Since `gamlss.prepdata` is still at an experimental stage, some of functions are remained hidden to allow time for thorough checking and validation. These hidden functions can still be accessed using the triple colon notation, for example: `gamlss.prepdata:::`.

The functions available in `gamlss.prepdata` are intended for pre-fitting — that is, to be used before applying the `gamlss()` or `gamlss2()` fitting functions. The available functions can be grouped into the following categories:

Information functions

These functions provide information about:

- the size of the dataset;
- the extent of missing values;
- the structure of the dataset;
- whether the class of variables is appropriate for analysis.

Plotting functions

These functions allow plotting;

- individual variables and
- pairwise relationships of variables

Features functions

These functions can assist in:

- detecting outliers;
- applying transformations to variables and
- scaling variables.

Data Partition functions

Functions that facilitate partitioning the data to improve inference and avoid overfitting during model selection.

Distributional Regression

The aim of this vignette is to demonstrate how to manipulate and prepare data before applying a distributional regression analysis.

The general form a distributional regression model can be written as;

$$
\begin{aligned}
\mathbf{y} &\stackrel{ind}{\sim} \mathcal{D}(\theta_1, \dots, \theta_k) \\
g_1(\theta_1) &= \mathcal{ML}_1(\mathbf{x}_{11}, \mathbf{x}_{21}, \dots, \mathbf{x}_{p1}) \\
\dots &= \dots \\
g_k(\theta_k) &= \mathcal{ML}_k(\mathbf{x}_{1k}, \mathbf{x}_{2k}, \dots, \mathbf{x}_{pk}).
\end{aligned}
$$

where we assume that the response variable y_i for $i = 1, \dots, n$, is independently distributed having a distribution $\mathcal{D}(\theta_1, \dots, \theta_k)$ with k parameters and where all parameters could be effected by the explanatory variables $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p$. The $\mathcal{ML}$ represents **any** regression type machine learning algorithm i.e. LASSO, Neural networks etc.

When only **additive smoothing** terms are used in the fitting the model can be written as;

$$
\begin{aligned}
\mathbf{y} &\stackrel{ind}{\sim} \mathcal{D}(\theta_1, \dots, \theta_k) \\
g_1(\theta_1) &= b_{01} + s_1(\mathbf{x}_{11}) + \cdots, +s_p(\mathbf{x}_{p1}) \\
\dots &= \dots \\
g_k(\theta_k) &= b_{0k} + s_1(\mathbf{x}_{1k}) + \cdots, +s_p(\mathbf{x}_{pk}).
\end{aligned}
$$

which is the GAMLSS model introduced by Rigby and Stasinopoulos (2005).

There are three books on GAMLSS, D. M. Stasinopoulos et al. (2017), Rigby et al. (2019) and M. D. Stasinopoulos et al. (2024) and several 'GAMLSS lecture materials, available from GitHUb, https://mstasinopoulos.github.io/Porto_short_course/ and https://mstasinopoulos.github.io/ShortCourse/. The latest **R** packages related to GAMLSS can be found in https://gamlss-dev.r-universe.dev/builds.

The aim of this package is to prepare data and extract useful information that can be utilized during the modeling stage.

Information functions

The functions for obtaining information from a dataset are described in this section and summarized in Table 1. These functions can provide:

- General information about the dataset such as the dimensions (number of rows and columns) and the percentage of omitted (missing) observations
- Information about the variables in the dataset
- Information about the observations in the dataset
- Changing the variables into suitable **R** classes for futher analysis.

The functions which use a `data.frame` as their first argument are staring from `data_` while ones variable whixg take only a variable start fro, `y_`.

Here is a table of the information functions;

Table 1: A summary table of the functions to obtain information

Functions	Usage
`data_dim()`	Returns the number of rows and columns, and show the percentage of omitted (missing) observations
`data_names()`	Lists the names of the variables in the dataset
`data_rename()`	Allows renaming of one or more variables in the dataset
`data_shorter_names()`	Allows shortening the names of one or more variables in the dataset
`data_distinct()`	Displays the number of distinct values for each variable in the dataset
`data_na_vars()`	Displays the count of NA (missing) values for each variable in the dataset
`data_na_obs()`	Displays which observation are missing for each variable
`data_omit()`	Removes all rows (observations) that contain NA's (missing values)
`data_str()`	Displays the class for each variable (e.g., numeric, factor, etc.) along with additional detail
`data_cha2fac()`	Converts all character variables in the dataset to factors

Functions	Usage
`data_few2fac()`	Converts variables with a small number of distinct values (e.g., binary or categorical) to factors
`data_int2num()`	Converts integer variables with several distinct values to numeric
`data_fac2num()`	Converts factors or characters to to numeric
`data_rm()`	Removes one or more variables (columns) from the dataset
`data_rmNAvars()`	Removes all variables with NA's as their only value.
`data_rm1val()`	Removes all variables/factors with a single value/level
`data_select()`	Allows selection of one or more variables (columns)
`data_exclude_class()`	Removes all variables of a specified class (e.g., factor, numeric) from the dataset
`data_only_continuous()`	Retains only the numeric variables (excluding factors or other variable types)
`data_factor()`	For all factors in the data the first level becomes the level with `lower` of `higher` number of observations

Next we demonstrate simple use of the functions.

`data_dim()`

This function provides detailed information about the dimensions of a `data.frame`. It is similar to the R function `dim()`, but with additional details. The output is the original data frame, allowing it to be used in a series of piping commands.

```r
rent99 |> data_dim()
```

```
*************************************************************
*************************************************************
the R class of the data is: data.frame
the dimensions of the data are: 3082 by 9
number of observations with missing values: 0
% of NA's in the data: 0 %
*************************************************************
*************************************************************
```

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data_names()

This function provides the names of the variables in the `data.frame`, similar to the `R` function `names()`.

```
rent99 |> data_names()
```

```
*************************************************************
*************************************************************
the names of variables
[1] "rent"     "rentsqm"  "area"     "yearc"    "location" "bath"     "kitchen"
[8] "cheating" "district"
*************************************************************
*************************************************************
```

The output of the function is the original data frame.

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data_rename()

Renames one or more variables (columns) in the `data.frame` using the function `data_rename()`;

```
da<- rent99 |> data_rename(oldname="rent", newname="R")
head(da)
```

```
         R   rentsqm area yearc location bath kitchen cheating district
1 109.9487  4.228797   26  1918        2    0       0        0      916
2 243.2820  8.688646   28  1918        2    0       0        1      813
3 261.6410  8.721369   30  1918        1    0       0        1      611
4 106.4103  3.547009   30  1918        2    0       0        0     2025
5 133.3846  4.446154   30  1918        2    0       0        1      561
6 339.0256 11.300851   30  1918        2    0       0        1      541
```

The output of the function is the original data frame with new names.

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data_shorter_names()

If the variables in the dataset have very long names, they can be difficult to handle in formulae during modelling. The function `data_shorter_names()` abbreviates the names of the explanatory variables, making them easier to use in formulas.

```
rent99 |> data_shorter_names()
```

```
**************************************************************
**************************************************************
the names of variables
[1] "rent"  "rents" "area"  "yearc" "locat" "bath"  "kitch" "cheat" "distr"
**************************************************************
**************************************************************
```

If no long variable names exist in the dataset, the function data_shorter_names() does nothing. However, when applicable, the function abbreviates long names and returns the original data frame with the new shortened names.

> **Warning**
>
> Note that there is a risk when using a small value for the `max` option, as it may result in identical names for different variables. This could lead to confusion or errors in the modelling process. It is important to carefully choose an appropriate value for `max` to avoid this issue.

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data_distinct()

The distinct values for each variable are shown below.

```
rent99 |> data_distinct()
```

```
    rent  rentsqm     area    yearc location     bath  kitchen cheating
    2723     3053      132       68        3        2        2        2
district
     336
```

The output of the function is the original data frame.

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`data_na_vars()`

This function provides information about which variables have missing observations and how many missing values there are for each variable;

```
rent99 |> data_na_vars()
```

```
    rent  rentsqm     area    yearc location     bath  kitchen cheating
       0        0        0        0        0        0        0        0
district
       0
```

The output of the function is the original data frame nothing is changing.

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`data_na_obs()`

This function provides information about which variables have missing observations and how many missing values there are for each variable;

```
rent99 |> data_na_obs()
```

```
$rent
integer(0)

$rentsqm
integer(0)

$area
integer(0)

$yearc
integer(0)

$location
integer(0)

$bath
integer(0)

$kitchen
```

```
integer(0)

$cheating
integer(0)

$district
integer(0)
```

The output of the function is the original data frame nothing is changing.

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data_omit()

The function `data_omit()` omits all observations with missing values.

```r
rent99 |> data_omit()
```

```
**************************************************************
**************************************************************
the R class of the data is: data.frame
the dimensions of the data before omition are: 3082 x 9
the dimensions of the data saved after omition are: 3082 x 9
the number of observations omited: 0
**************************************************************
**************************************************************
```

The output of the function is a new data frame withll NA's omitted.

> **Warning**
>
> It is important to select the relevant variables for the analysis before using the `data_omit()` function, as some unwanted variables may contain many missing values that that could lead to unnecessary omissions.

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data_str()

The function `data_str()` (similar to the **R** function `str()`) provides information about the types of variable exist in the data.

```
rent99 |> data_str()
```

```
************************************************************
************************************************************
the structure of the data
'data.frame':   3082 obs. of  9 variables:
 $ rent    : num  110 243 262 106 133 ...
 $ rentsqm : num  4.23 8.69 8.72 3.55 4.45 ...
 $ area    : int  26 28 30 30 30 30 31 31 32 33 ...
 $ yearc   : num  1918 1918 1918 1918 1918 ...
 $ location: Factor w/ 3 levels "1","2","3": 2 2 1 2 2 2 1 1 1 2 ...
 $ bath    : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ kitchen : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 2 1 1 ...
 $ cheating: Factor w/ 2 levels "0","1": 1 2 2 1 2 2 1 2 1 1 ...
 $ district: int  916 813 611 2025 561 541 822 1713 1812 152 ...
************************************************************
************************************************************
table of the class of variabes

 factor integer numeric
      4       2       3
************************************************************
************************************************************
distinct values in variables
    rent  rentsqm     area    yearc location     bath  kitchen cheating
    2723     3053      132       68        3        2        2        2
district
     336
consider to make those characters vectors into factors:
location bath kitchen cheating
************************************************************
************************************************************
```

The output of the function is the original data frame.

The next set of functions manipulate variables withing `data.frame`s.

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`data_cha2fac()`

Often, variables in datasets are read as character vectors, but for analysis, they may need to be treated as factors. This function transforms any character vector (with a relatively small number of distinct values) into a factor.

```
rent99 |> data_cha2fac()  -> da
```

```
**************************************************************
not character vector was found
```

Since no character were found nothing have changed. The output of the function is a new data frame.

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`data_few2fac()`

There are occasions when some variables have very few distinct observations, and it may be better to treat them as factors. The function `data_few2fac()` converts (numeric) vectors with a small number of distinct values into factors.

```
rent99 |>  data_few2fac() -> da
```

```
**************************************************************
    rent  rentsqm     area    yearc location     bath  kitchen cheating
    2723     3053      132       68        3        2        2        2
district
     336
**************************************************************
4 vectors with fewer number of values than 5 were transformed to factors
**************************************************************
**************************************************************
```

```
str(da)
```

```
'data.frame':   3082 obs. of  9 variables:
 $ rent    : num  110 243 262 106 133 ...
 $ rentsqm : num  4.23 8.69 8.72 3.55 4.45 ...
 $ area    : int  26 28 30 30 30 30 31 31 32 33 ...
 $ yearc   : num  1918 1918 1918 1918 1918 ...
```

```
 $ location: Factor w/ 3 levels "1","2","3": 2 2 1 2 2 2 1 1 1 2 ...
 $ bath    : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
 $ kitchen : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 2 1 1 ...
 $ cheating: Factor w/ 2 levels "0","1": 1 2 2 1 2 2 1 2 1 1 ...
 $ district: int  916 813 611 2025 561 541 822 1713 1812 152 ...
```

The output of the function is a new data frame. The results can be seen by using the **R** function `str()`;

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`data_int2num()`

Occasionally, we need to convert integer variables with a very large range of values into numeric vectors, especially for plots. The function `data_int2num()` performs this conversion.

```
rent99 |> data_int2num() -> da
```

```
**************************************************************
    rent  rentsqm     area    yearc location     bath  kitchen cheating
    2723     3053      132       68        3        2        2        2
district
     336
2 integer vectors with more number of values than 50 were transformed to numeric
**************************************************************
```

The output of the function is a new data frame.

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`data_fac2num()`

Occasionally, we need to convert integer variables with a very large range of values into numeric vectors, especially for plots. The function `data_int2num()` performs this conversion.

```
da <- data_fac2num(rent, vars=c("B"))
```

```
**************************************************************
**************************************************************
the R class of the data is: data.frame
the dimensions of the data before omition are: 1969 x 9
```

```
the dimensions of the data saved after omition are: 1969 x 9
the number of observations omited: 0
**********************************************************
**********************************************************
```

```r
class(da$B)
```

```
[1] "numeric"
```

The output of the function is a new data frame.

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`data_rm()`

For analysisng datasets, it's easier to keep only the relevant variables. The function `data_rm()` allows you to remove unnecessary variables, making the dataset more manageable for analysis.

```r
data_rm(rent99, c(2,9)) -> da
dim(rent99)
```

```
[1] 3082    9
```

```r
dim(da)
```

```
[1] 3082    7
```

The output of the function is a new data frame. Note that this could be also done using the function `select()` of the package **dplyr**, or our own `data_select()` function.

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`data_rmNAvars()`

Occutionally when we read data from files in `R`some extra variables are accidentaly produced with no values but NA's. The function `data_rmNAvars()` removes those variables from the data;

```
data_rmNAvars(da)
```

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`data_rm1val()`

This function searches for variables with only a single distinct value (often factors left over from a previous `subset()` operation) and removes them from the dataset.

```
da |> data_rm1val()
```

The output of the function is a new data frame.

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`data_select()`

This function selects specified variables from a dataset.

```
da1<-rent |> data_select( vars=c("R", "Fl", "A"))
head(da1)
```

```
       R Fl    A
1  693.3 50 1972
2  422.0 54 1972
3  736.6 70 1972
4  732.2 50 1972
5 1295.1 55 1893
6 1195.9 59 1893
```

The output of the function is a new data frame.

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`data_exclude_class()`

This function searches for variables (columns) of a specified `R` class and removes them from the dataset. By default, the class to remove is factor.

```r
da |> data_exclude_class()  -> da1
head(da1)
```

```
      rent area yearc
1 109.9487   26  1918
2 243.2820   28  1918
3 261.6410   30  1918
4 106.4103   30  1918
5 133.3846   30  1918
6 339.0256   30  1918
```

The output of the function is a new data frame.

`data_only_continuous()`

This function keeps only the continuous variables in the dataset, removing all non-continuous variables.

```r
da |> data_only_continuous()  -> da1
head(da1)
```

```
      rent area yearc
1 109.9487   26  1918
2 243.2820   28  1918
3 261.6410   30  1918
4 106.4103   30  1918
5 133.3846   30  1918
6 339.0256   30  1918
```

The output of the function is a new data frame.

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`data_factor()`

This function changes the reference level of factors to be the the `lower` or `higher` of the number of obserevations.

```r
da |> gamlss.prepdata:::data_factor()  -> da1
levels(da1$loc)
```

```
[1] "3" "1" "2"
```

It moved the level 3 into the first place. The output of the function is a new data frame.

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Graphical functions

Graphical methods are used in pre-analysis to examine individual variables or pair-wise relationships between variables.

Table 2: A summary table of the graphical functions

Functions	Usage
data_plot()	Plots each variable in the dataset to visualize its distribution or other important characteristics (see also Section
data_bucket()	Generates bucket plots for all numerical variables in the dataset to visualize their skewness and kurtosis
data_response()	Plots the response variable alongside its z-score, providing a standardized version of the response for comparison
data_zscores()	Plots the z-scores for all continuous variables in the dataset, allowing for easy visualization of the standardized values (see also Section)
data_xyplot()	Generates pairwise plots of the response variable against all other variables in the dataset to visualize their relationships
data_cor()	Calculates and plots the pairwise correlations for all continuous variables in the dataset to assess linear relationships between them
data_void()	Searches for pairwise empty spaces across all continuous variables in the dataset to identify problems with interpretation or prediction
data_pcor()	Calculates and plots the pairwise patrial-correlations for all continuous variables in the dataset
data_inter()	Searches for potential pairwise interactions between variables in the dataset to identify relationships or dependencies that may be useful for modelling

Functions	Usage
`data_leverage()`	Detects outliers in the continuous explanatory variables (x-variables) as a group to highlight unusual observations
`data_Ptrans_plot()`	Plots the response variable against various power transformations of the continuous x-variables to explore potential relationships and model suitability

Next we give examples of the graphical functions.

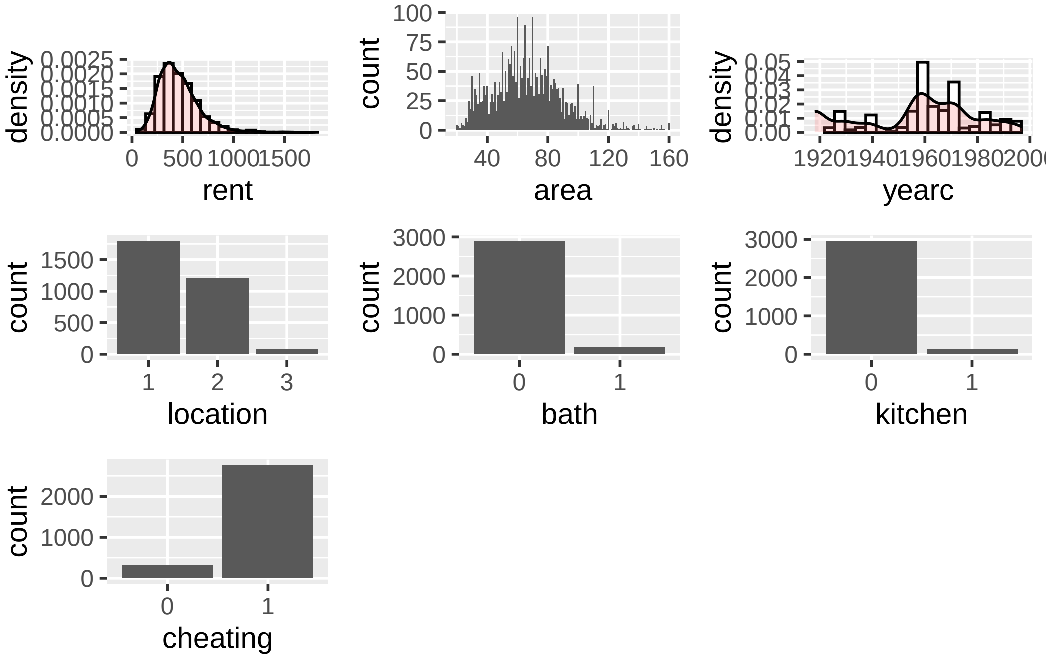
`data_plot()`

The function `data_plot` plots all the variables of the data individually; It plots the continuous variable as *histograms* with a density plots superimposed, see the plot for `rent` and `yearc`. As an alternative a `dot plots` can be requested see for an example in **?@sec-data_response**. For integers the function plots *needle* plots, see `area` below and for categorical the function plots *bar* plots, see `location`, `bath kitchen` and `cheating` below.

```
da |> data_plot()
```

```
 100 % of data are saved,
that is, 3082 observations.
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_bar()`).
Removed 2 rows containing missing values or values outside the scale range
(`geom_bar()`).
```



The function could saves the `ggplot2` figures.

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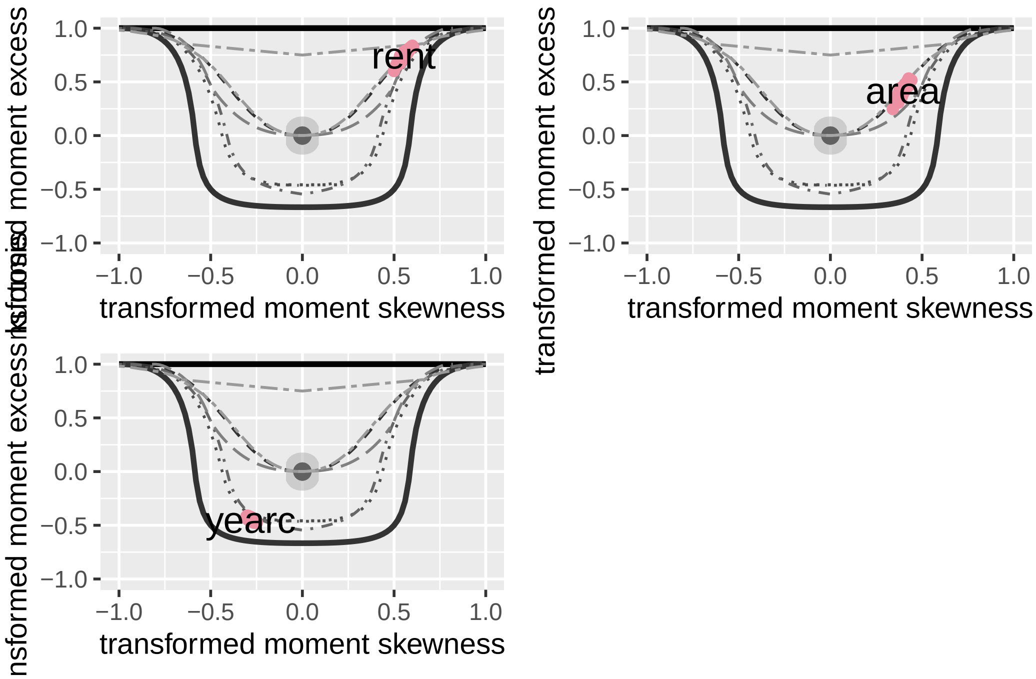
`data_bucket()`

The function `data_bucket` plots bucket plots (which identify skewness and kurosid) for all variables in the data set.

```
da |> data_bucket()
```

```
 100 % of data are saved,
that is, 3082 observations.
    rent      area     yearc location      bath   kitchen cheating
    2723       132        68        3         2         2        2
```



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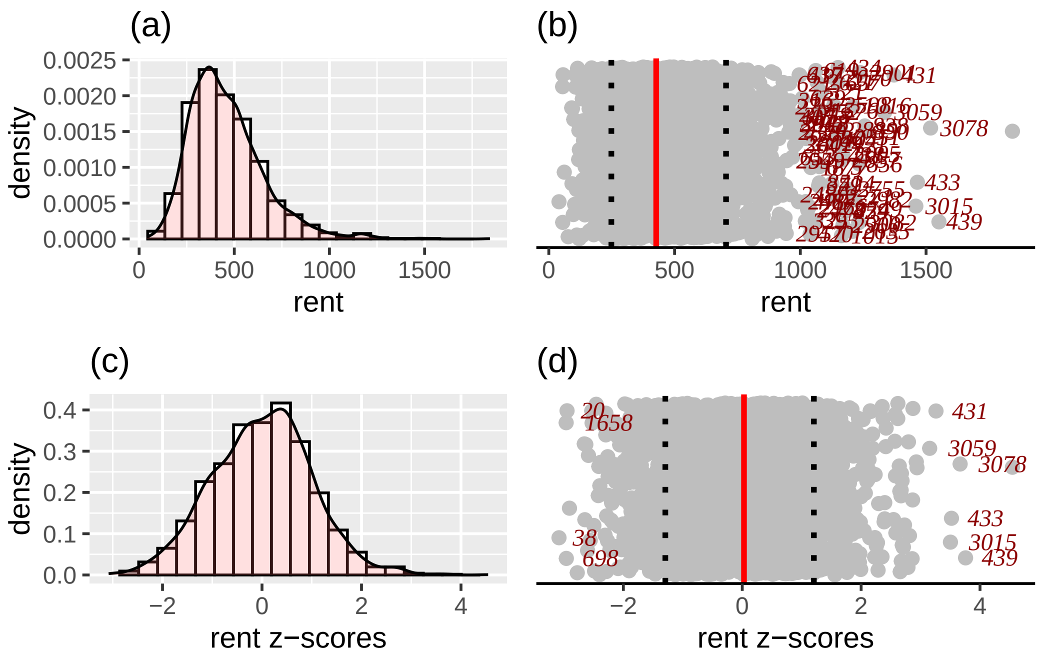
`data_response()`

The function `data_response()` plots four different plots related to a continuous response variable. It plots; i) a histogram (and the density function); ii) a dot plot of the response in the original scale iii) a histogram (and the density function) of the response at the z-scores scale and vi) s dot-plot in the z-score scale.

The z-score scale is defined by fitting a distribution to the variable (normal or SHASH) and then taking the residuals, see also next section. The dot plots are good in identify highly skew variables and unusual observations. They display the median and inter quantile range of the data. The y-axis of a dot plot is a randomised uniform variable (therefore the plot could look slightly different each time.)

```
da |> data_response(, response=rent)
```

```
 100 % of data are saved,
that is, 3082 observations.
the class of the response is numeric is this correct?
a continuous distribution on (0,inf) could be used
```



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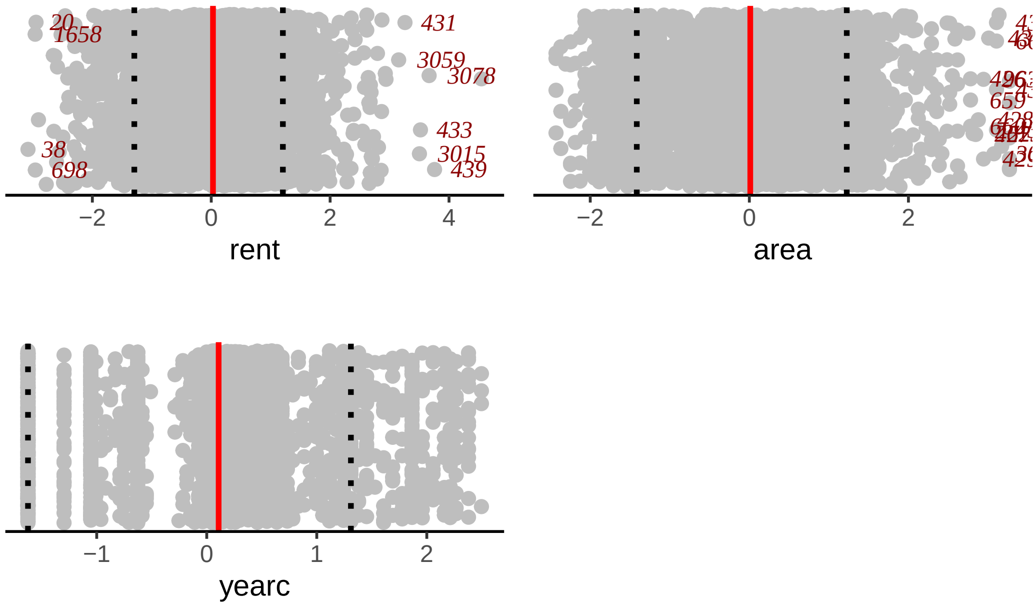
`data_zscores()`

One could fit any four parameter (GAMLSS) distribution, defined on $-\infty$ to ∞, to any continuous variable, where skewness and kurtosis is suspected and take the quantile residuals (z-scores) as the transformed values of the x-values. The function `y_zscores()` performs just this. It takes a continuous variable and fits a continuous (four parameter `SHASHo`) distribution and gives the z-scores. The fitting distribution is specified by the user, but the default is SHASHo. The function `data_zscores()` takes all continuous variables and plots their z-scores. The methodology is use also to identify outliers in `data_outliers()`.

```
da |> data_zscores()
```

```
 100 % of data are saved,
that is, 3082 observations.
```



In order to see how the z-scores are calculated consider the function `y_zscores()` which is taking an individual variables z-scores;

```
z <- y_zscores(rent99$rent, plot=FALSE)
```

The function is equivalent of fitting a constant model to all the parameters of a given distribution and than taking the quantile residuals (or z=scores) as he variable of interest. The default distribution is the four parameter `SHASHo` distribution.

```
library(gamlss2)
m1 <- gamlssML(rent99$rent, family=SHASHo) # fitting a 4 parameter distribution
cbind(z,resid(m1))[1:5,]#  and taking the residuals
```

```
          z
1 -2.496097 -2.496097
2 -1.340464 -1.340464
3 -1.182014 -1.182014
4 -2.526326 -2.526326
5 -2.295069 -2.295069
```

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data_xyplot()

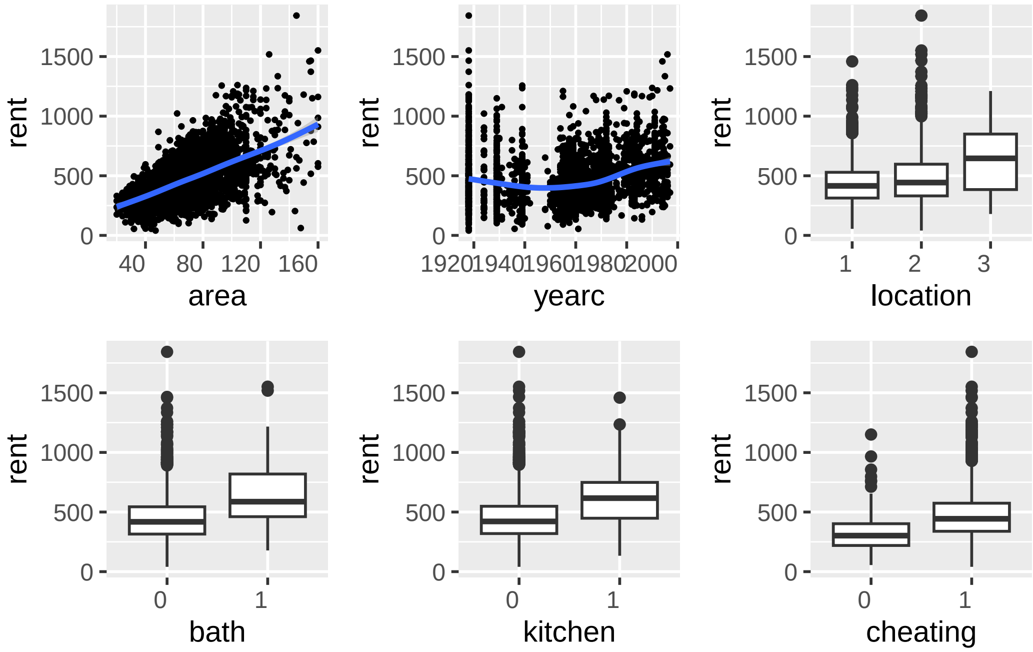
The functions `data_xyplot()` plots the response variable against each of the independent explanatory variables. It plots the continuous against continuous as *scatter* plots and continuous variables against categorical as *box* plot.

> **Note**
>
> At the moment there is no provision for categorical response variables.

```
da |> data_xyplot(response=rent )
```

```
 100 % of data are saved,
that is, 3082 observations.
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



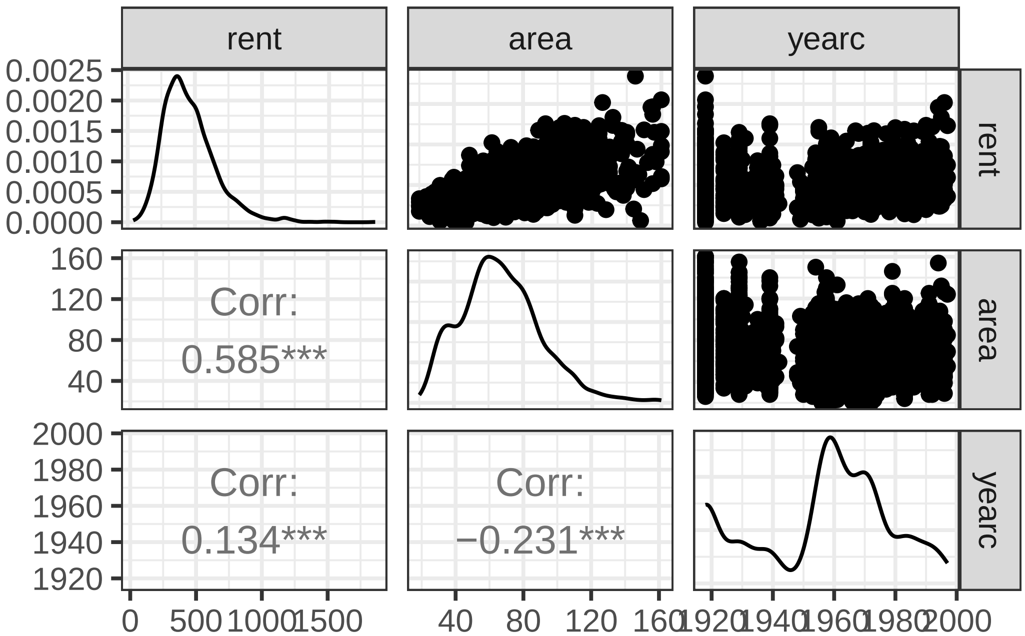
The output of the function saves the `ggplot2` figures.

Note hat the package `gamlss.prepdata` do not provides pairwise plot of the explanatory variables themself but the package `GGally` does. here is an example ;

```r
library(GGally)
```

```
Registered S3 method overwritten by 'GGally':
  method from
  +.gg   ggplot2
```

```r
dac <- gamlss.prepdata::data_only_continuous(da)
ggpairs(dac, lower=list(continuous = "cor",
      combo = "box_no_facet", discrete = "count",
      na = "na"),upper=list(continuous = "points",
      combo = "box_no_facet", discrete = "count", na = "na")) +
theme_bw(base_size =15)
```



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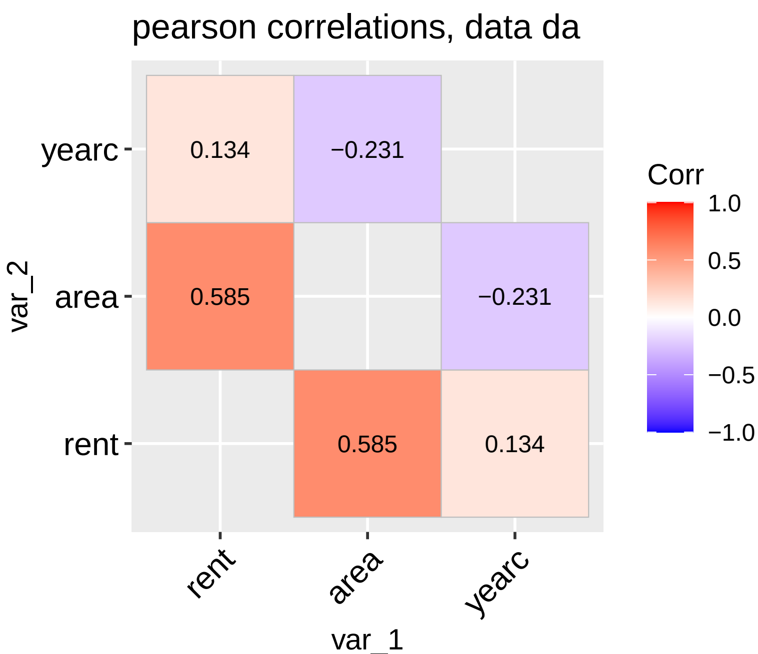
data_cor()

The function `data_corr()` is taking a `data.frame` object and plot the correlation coefficients of all its continuous variables.

```r
data_cor(da, lab=TRUE)
```

```
 100 % of data are saved,
that is, 3082 observations.
4 factors have been omited from the data

Warning in data_cor(da, lab = TRUE):
```

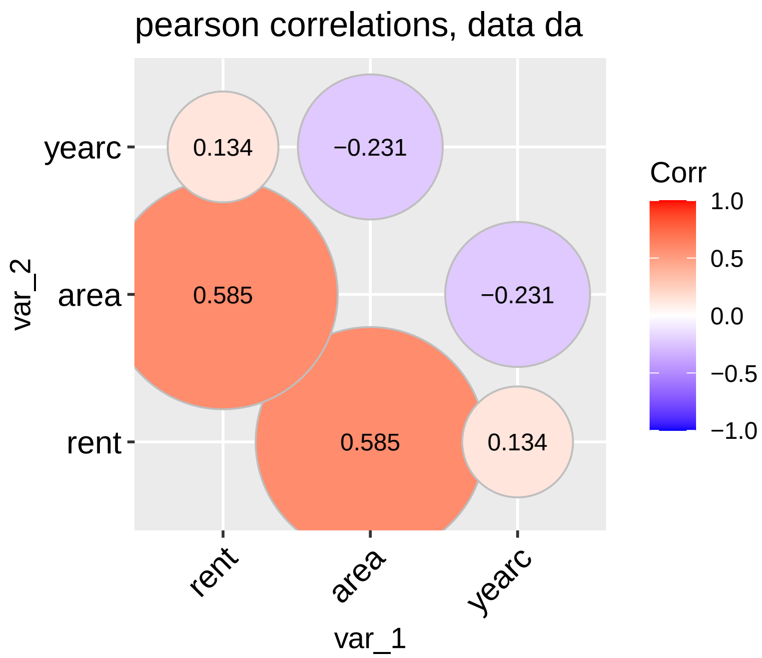


A different type of plot can be produce if we use;

```
data_cor(da, method="circle", circle.size = 40)
```

```
 100 % of data are saved,
that is, 3082 observations.
4 factors have been omited from the data

Warning in data_cor(da, method = "circle", circle.size = 40):
```



To get the variables with higher, say, than \$0.4 \$ correlation values use;

```
Tabcor <- data_cor(da, plot=FALSE)
```

```
 100 % of data are saved,
that is, 3082 observations.
4 factors have been omited from the data

Warning in data_cor(da, plot = FALSE):
```

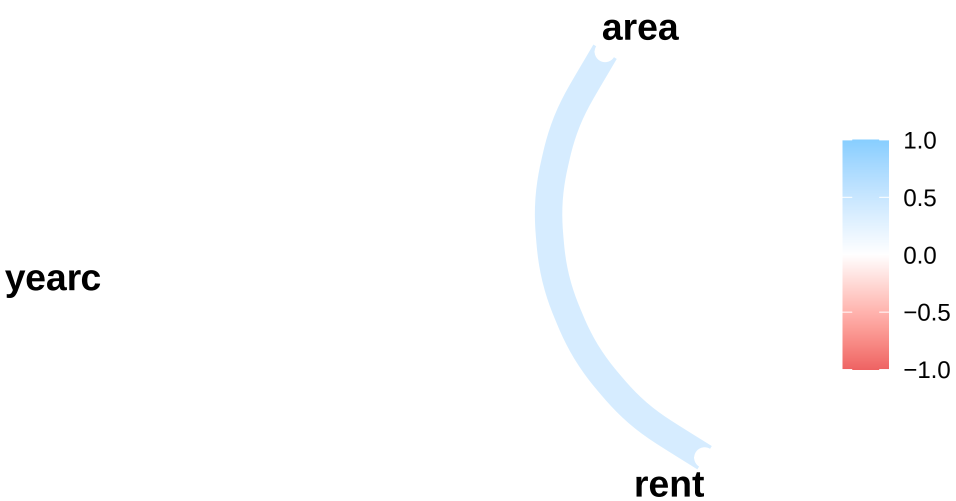
```
high_val(Tabcor, val=0.4)
```

```
     name1  name2   corrs
[1,] "rent" "area" "0.585"
```

We can plot the path of those variables using the package `corr`;

```
library(corrr)
network_plot(Tabcor)
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
i The deprecated feature was likely used in the corrr package.
  Please report the issue at <https://github.com/tidymodels/corrr/issues>.
```



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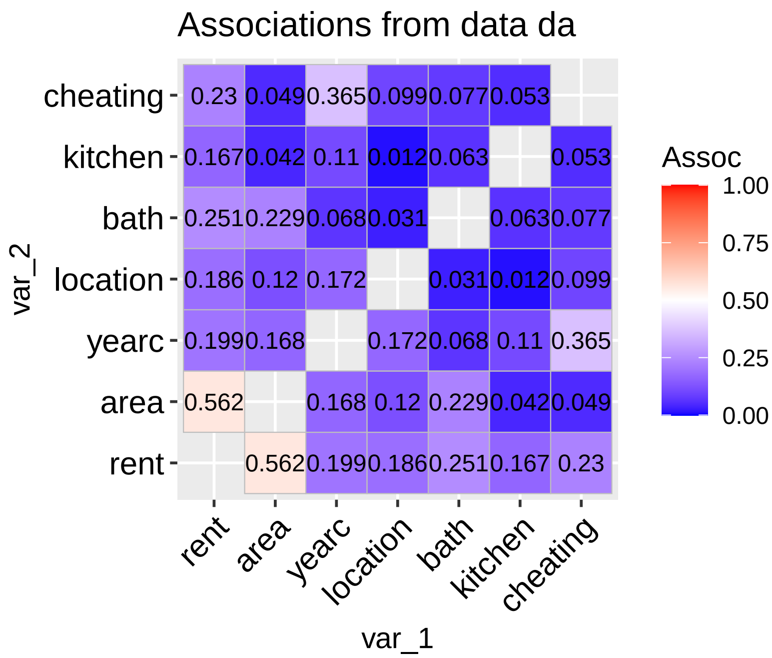
`data_association()`

The function `data_association()` is taking a `data.frame` object and plot the pair-wise association of all its variables. The pair-wise association for two continuous variables is given by default by the absolute value of the Spearman's correlation coefficient. For two categorical variables by the (adjusted for bias) Cramers' `v`-coefficient. For a continuous against a categorical variables by the $\sqrt{R}^2$ obtained by regressing the continuous variable against the categorical variable.

```
data_association(da, lab=TRUE)
```

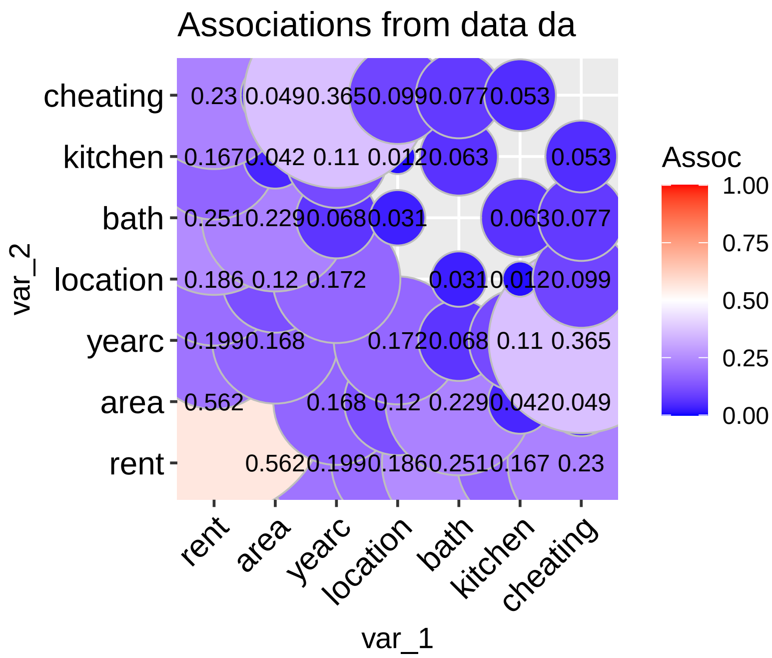
```
 100 % of data are saved,
that is, 3082 observations.
```



A different type of plot can be produce if we use;

```
data_association(da, method="circle", circle.size = 40)
```

```
 100 % of data are saved,
that is, 3082 observations.
```



To get the variables with higher, say, than \$0.4 \$ association values use;

```
Tabcor <- data_cor(da, plot=FALSE)
```

```
 100 % of data are saved,
that is, 3082 observations.
4 factors have been omited from the data

Warning in data_cor(da, plot = FALSE):
```

```
high_val(Tabcor, val=0.4)
```

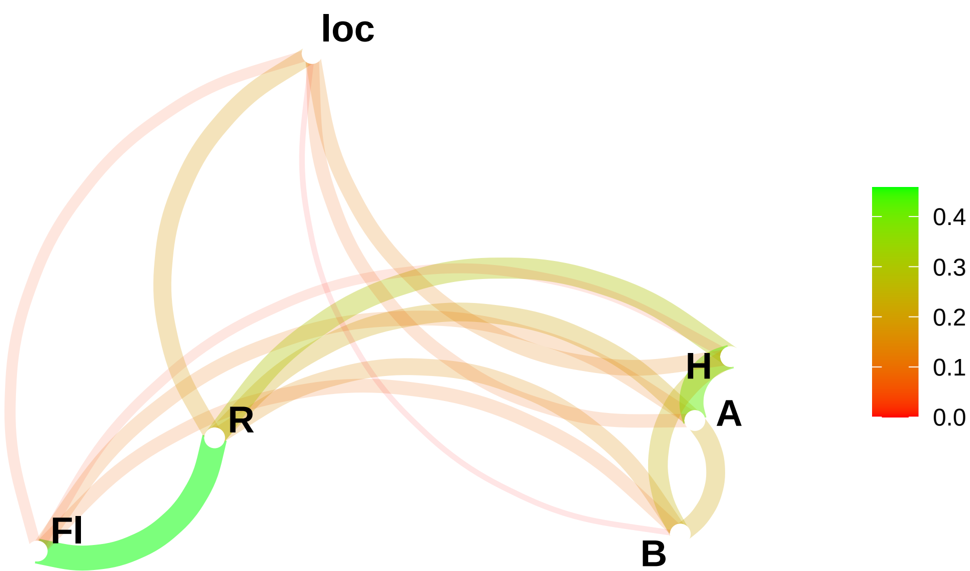
```
     name1  name2  corrs
[1,] "rent" "area" "0.585"
```

A different plot can be acheived using the funtion `network_plot()` of package `corr`;

```
pp=gamlss.prepdata:::data_association(rent[,-c(4,5,8)], plot=F)
```

```
 100 % of data are saved,
that is, 1969 observations.
```

```r
library(corrr)
network_plot(pp, min_cor = 0, colors = c("red", "green"), legend = "range")
```



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data_void()

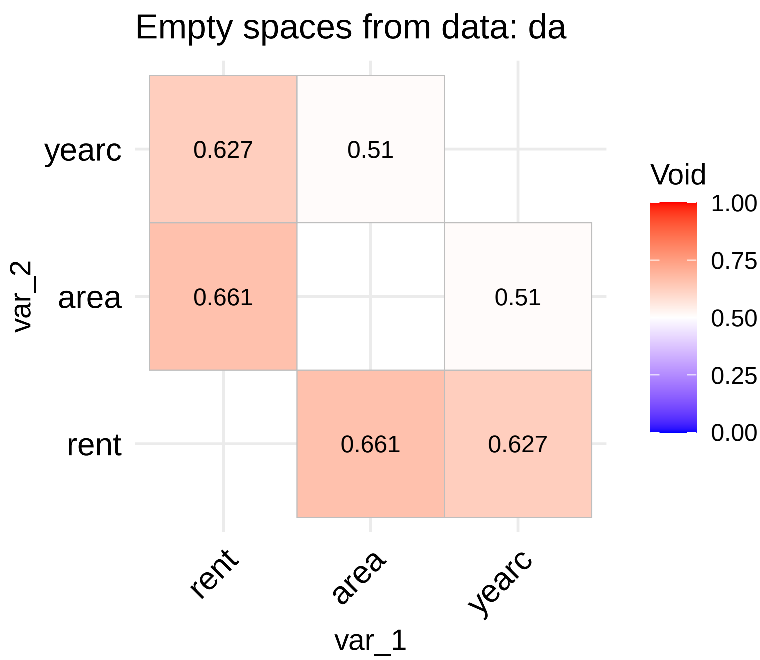
> ⚠ Warning
>
> This function is new. Its theoretical foundation are not proven yet. The function needs testing and therefore it should be used with causion.

The idea behind the functions `void()` and its equivalent `data.frame` version `data_void()` is to be able to identify whether the data in the direction of two continuous variables say x_i and x_j have a lot of empty spaces. The reason is that empty spaces effect prediction since interpolation at empty spaces is dengerous. The function `data_void()` is taking a `data.frame` object and plot the percentage of empty spaces for all pair-wise continuous variables. The function used the `foreach()` function of the package **foreach** to allow parallel processing.

```r
registerDoParallel(cores = 9)
data_void(da)
```

```
 100 % of data are saved,
that is, 3082 observations.
4 factors have been omited from the data
```

```
Warning in data_void(da):
```

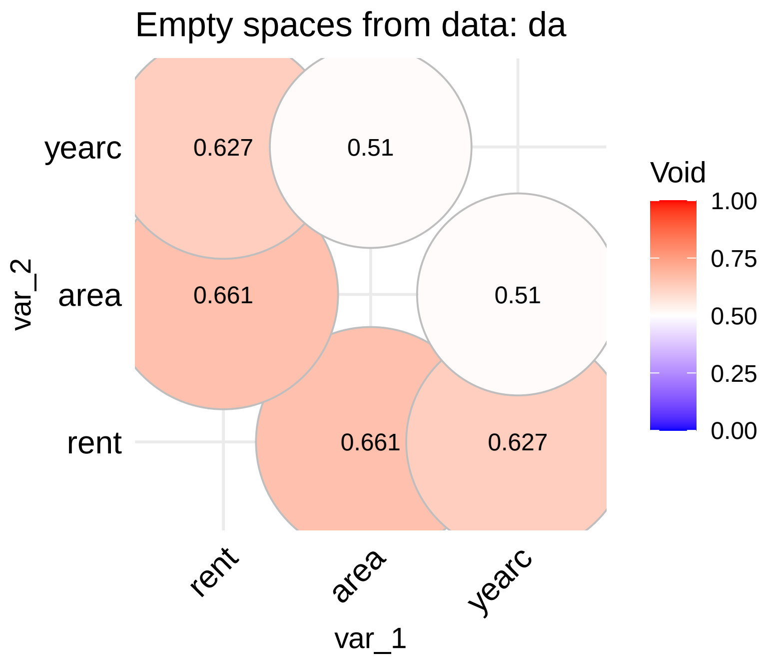


A different type of plot can be produce if we use;

```
data_void(da, method="circle", circle.size = 40)
```

```
 100 % of data are saved,
that is, 3082 observations.
4 factors have been omited from the data
```

```
Warning in data_void(da, method = "circle", circle.size = 40):
```




```
stopImplicitCluster()
```

To get the variables with highter than \$0.4 \$ values use;

```
Tabvoid <- data_void(da, plot=FALSE)
```

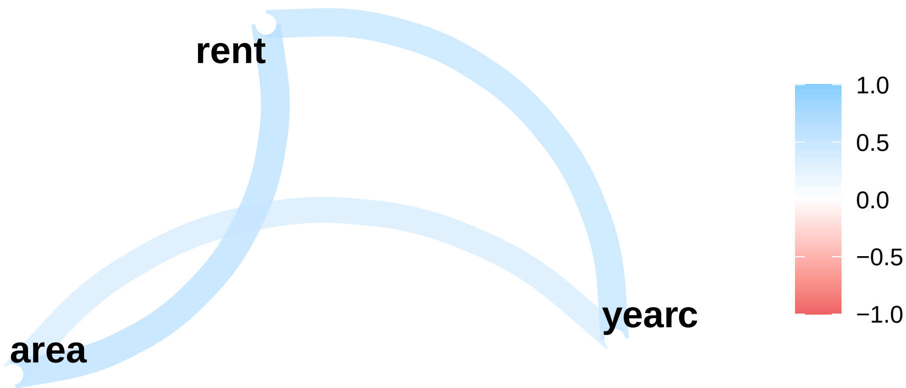
```
 100 % of data are saved,
that is, 3082 observations.
4 factors have been omited from the data

Warning in data_void(da, plot = FALSE):
```

```
high_val(Tabvoid, val=0.4)
```

```
     name1  name2   corrs
[1,] "rent" "area"  "0.661"
[2,] "rent" "yearc" "0.627"
[3,] "area" "yearc" "0.51"
```

To plot their paths use;

```r
library(corrr)
network_plot(Tabvoid)
```



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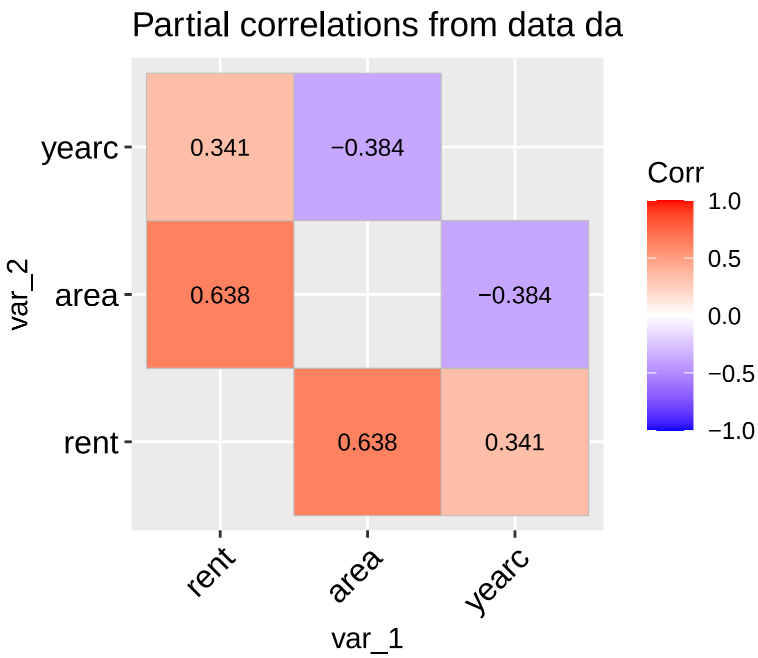
data_pcor()

The function `data_copr()` is taking a `data.frame` object and plot the partial correlation coefficients of all its continuous variables.

```r
data_pcor(da, lab=TRUE)
```

```
 100 % of data are saved,
that is, 3082 observations.
4 factors have been omited from the data
```

```
Warning in data_pcor(da, lab = TRUE):
```



To get the variables with highter than \$0.4 \$ partial correlation values use;

```
Tabpcor <- data_pcor(da, plot=FALSE)
```

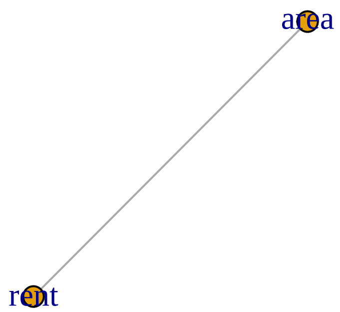
```
 100 % of data are saved,
that is, 3082 observations.
4 factors have been omited from the data

Warning in data_pcor(da, plot = FALSE):
```

```
high_val(Tabpcor, val=0.4)
```

```
     name1  name2   corrs
[1,] "rent" "area" "0.638"
```

```
high_val(Tabpcor, val=0.4, plot=T)
```



For plotting you can use;

```
library(corrr)
network_plot(Tabpcor)
```

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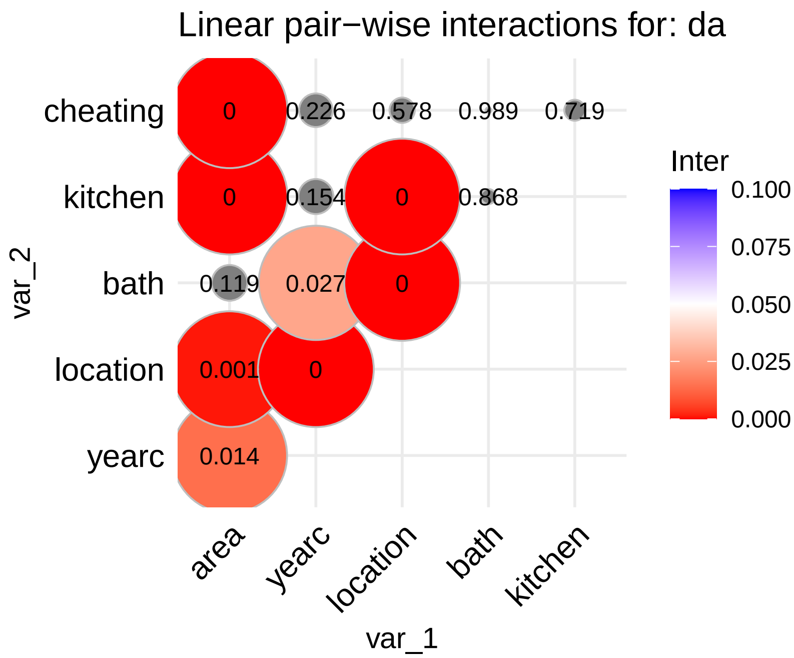
data_inter()

The function `data_inter()` takes a `data.frame`, fits all pair-wise interactions of the explanatory variables against the response (using a normal model) and produce a graph displaying their significance levels. The idea behind this is to identify possible first order interactions at an early stage of the analysis.

```
da |> gamlss.prepdata:::data_inter(response= rent)
```

```
 100 % of data are saved,
that is, 3082 observations.
```




```
tinter <-gamlss.prepdata:::data_inter(da,  response= rent, plot=FALSE)
```

```
 100 % of data are saved,
that is, 3082 observations.
```

```
tinter
```

```
         area yearc location  bath kitchen cheating
area       NA 0.014    0.001 0.119   0.000    0.000
yearc      NA    NA    0.000 0.027   0.154    0.226
location   NA    NA       NA 0.000   0.000    0.578
bath       NA    NA       NA    NA   0.868    0.989
kitchen    NA    NA       NA    NA      NA    0.719
cheating   NA    NA       NA    NA      NA       NA
```

To get the variables with lower than \$0.05 \$ significant interactions use;

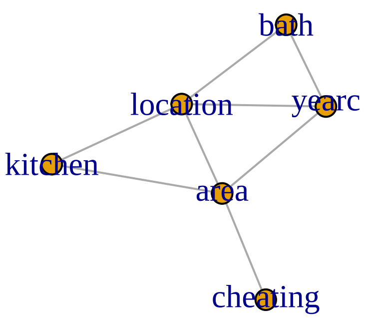
```
Tabinter <- gamlss.prepdata:::data_inter(da, response= rent, plot=FALSE)
```

```
 100 % of data are saved,
that is, 3082 observations.
```

```r
low_val(Tabinter)
```

```
     name1      name2      value  
[1,] "area"     "yearc"    "0.014"
[2,] "area"     "location" "0.001"
[3,] "yearc"    "location" "0"    
[4,] "yearc"    "bath"     "0.027"
[5,] "location" "bath"     "0"    
[6,] "area"     "kitchen"  "0"    
[7,] "location" "kitchen"  "0"    
[8,] "area"     "cheating" "0"    
```

```r
low_val(Tabinter, plot=T)
```



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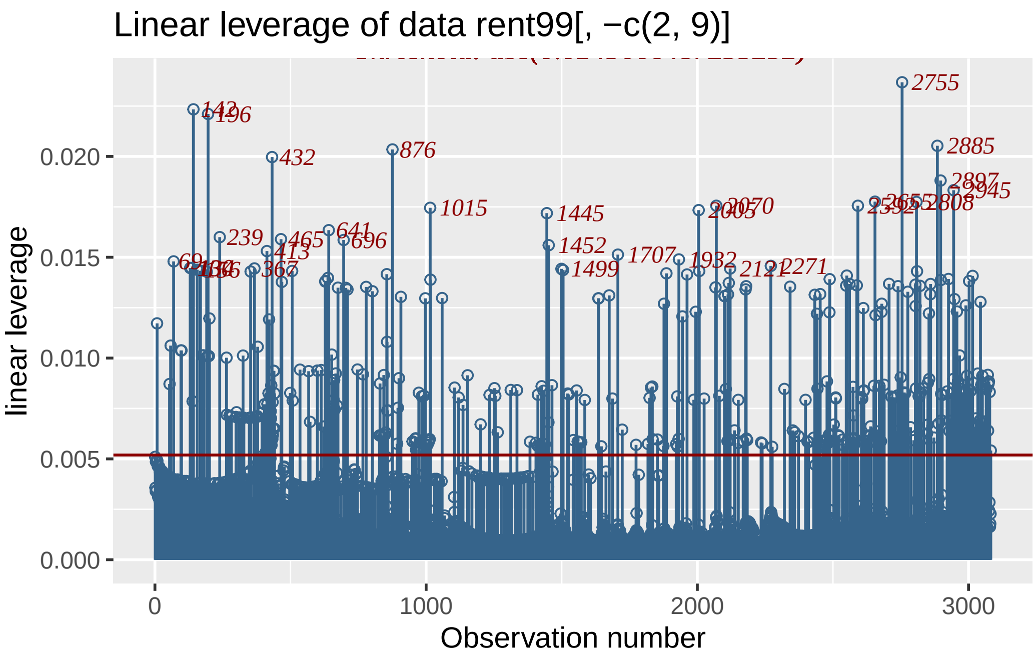
data_leverage()

The function `data_leverage()` uses the linear model methodology to identify posible unusual observations within the explanatory variables as a group (not indivisually). It fit a linear (normal) model with response, the response of the data, and all explanatory variables in the data, as x's. It then calculates the leverage points and plots them. A leverage is a number between zero and 1. Large leverage correspond to extremes in the x's.

```r
rent99[, -c(2,9)] |> data_leverage( response=rent)
```

```
 100 % of data are saved, 
that is, 3082 observations.
```



> **Note**
>
> The horizontal line is the plot is at point $2 \times (r/n)$, which is the threshold suggested in the literature; values beyond this point could be identified as extremes. It looks that the point $2 \times (r/n)$ is too low (at least for our data). Instead the plot identify only observation which are in the one per cent upper quantile. That is a quantile value of the leverage at value 0.99.

Section show how the information of the function `data_leverage()` can be combined with the information given by `data_outliers()` in order to confirm hight outliers in the data.

> **Important**
>
> What happen if multiple response are in the data?

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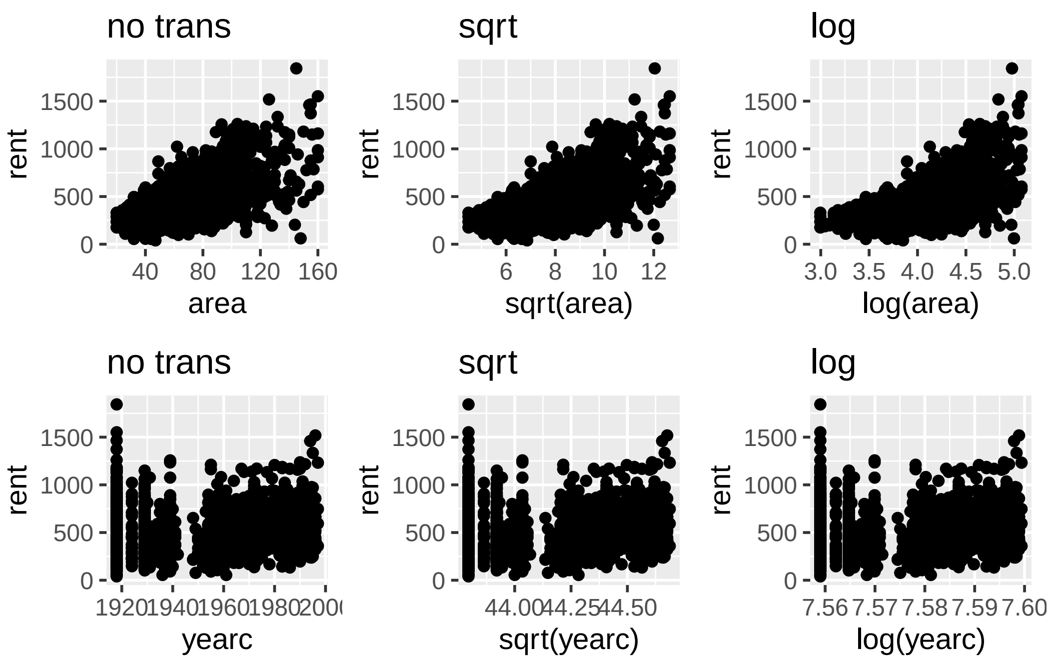
`data_Ptrans_plot()`

The essence of this plot is to visually decide whether certain values of λ in a power transformation are appropriate or not. Section describes the power transformation, $T = X^{\lambda}$ for $\lambda > 0$, in more details. For each continuous explanatory variable X, the function show the response against X, $(\lambda = 1)$, the response against the square root of X, $(\lambda = 1/2)$ and the response against the log of X $(\lambda \to 0)$. The user then can decide whether any of those transformation are appropriate. Note that the square root and log transformations are appropriate for righ skew variables.

```
gamlss.prepdata:::data_Ptrans_plot(da, rent)
```

```
 100 % of data are saved,
that is, 3082 observations.
```



It look that no transformation is needed for the continuous explanatory variables `area` of `yearc`.

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Feature functions

The features functions are functions to help with the continuous explanatory x-variables; We divide them here in five sections; Functions for

- identify `outliers`;
- identify hight `correlation` between variables;
- `scaling` continuous variables;
- `transforming` continuous variables and

- function appropriate for `dates` and `time` (helpful for time series data analysis).

The available functions are;

Table 3: The outliers, scaling, transformation and date and time functions

Functions	Usage
`y_outliers()`	identify possible outliers for one continuous variable
`y_outliers_both()`	identify possible outliers using both z-scores and the quantile rule methodology
`y_outliers_loop()`	identify outliers (using z-scores) by fitting several times the chossen `family` eliminating each time observations identifyied as outliers in the prievious fits
`y_outliers_by()`	identify possible outliers (using z-scores) at partitions of the data by a factor
`y_outliers_z()`	identify possible outliers (using z-scores) for different combination of `loop` and `by`
`data_outliers()`	identify possible outliers for all continuous x-variable (see also `data_zscores` in Section)
`data_leverage()`	this is a repeat of the Section function
`data_index_cor()`	this function gives an index of continuous variables with high correlation with others continuous variables
`data_index_association()`	this function gives an index of variables with high association with others continuous or factor variables
`data_scale()`	scales all continuous x-variables to zero mean and one s.d. or to the range [0,1]
`xy_Ptrans()`	looking for appropriate power transformation for response against one of the x-variable
`data_Ptrans()`	looking for appropriate power transformation for response against all x-variable
`data_Ptrans_plot()`	plotting all x's against the response using identity, square root and log transformations
`time_dt2dh()`	Separates a date and time variable into two variables
`time_hour2num()`	takes hours ans translate to numeric

Outliers

Outliers in the response variable, within a distributional regression framework like GAMLSS, are better handled by selecting an appropriate distribution for the response. For example, an outlier under the assumption of a normal distribution for the response may no longer be considered an outlier if a distribution from the t-family is assumed. Considerations of contanimnated responses data are not considered here. The function `data_response()`, discussed in Section and uses the z-scores methodology, provides some guidelines for appropriate action to identify ouliers in the response. This section focuses on outliers in the explanatory variables.

Outliers in the continuous explanatory variables can potentially affect curve fitting, whether using linear or smooth non-parametric terms. In such cases, removing outliers from the x-variables can make the model more robust. Outliers are observations that deviate significantly from the rest of the data, but the concept also depends on the dimension being considered. An single observation value might be an outlier on an explanatory variable. A pair of observations could be an outlier when examined within a pair-wise relationships. Extreme observations identification on continuous variables, is based on how far the observation lies from the majority of the data. Pairwise plots of continuous variables could identify outliers in the two-dimensions. Leverage points are useful to identify outliers in r dimensions where r is the size of the explanatory variables.

> **Note**
>
> At this preliminary stage of the analysis (no model has been fitted yet), it is difficult to identify outliers among `factors`. This is because outliers in factors typically do not appear unusual unless are examined in combination with rest of factors or variables. Identifying `factor` outliers is a task better suited for the modelling stage of the analysis.

The function `y_outliers()` in Section , is applied to continuous variables only, and it uses two different methodologies (option `type`);

- the z-scores, `type="zscorses"` (the default) and
- the quantile rule, `type="quantile"`.

Both methodologies are trying to remove extreme skewness in the variables before looking for outliers by using the (internal) function `y_Ptrans()`. The function is applied to any variable with positive values. The assumption, is that positive variables are more likely to be right skewed and therefor a power transformation will correct that. The function `y_Ptrans()` attempts to find the power λ, of a transformation $T = X^{\lambda}$, that minimizes the Jarque-Bera test statistic—a measure of deviation from normality. The Jarque-Bera test statistic is always non-negative and a value way from zero indicates strong evidence against normality. The goal of this methodology is to reduce extreme skewness in the data before attempt to identify the outliers. If the data are not so skewed the power transformation will genraly do no harm.

> ℹ Note
>
> At the momment we allow the power parameter of the tranformation λ to vary at the range 0 to 1.5. This range cover only right skewed behaviour. More work has to be done to establishes suitable range.

Now given that the positive variables are transformed (using the power transformation) and that all the rest of variables, containg negative values, remain untransformed, the z-scores methodology fits a parametertic distribution to the variable and computes the residuals of the fitted model (z-scores). The default distribution is the four parameter `SHASHo` distribution. The z-scores methodology identify outliers if a spesific residual (z-score) has an absolute value bigger that a predetermining `value` ay K. The quantile rule methodology works similar. It identify an observation if it is further than $K \times MAD$ way from the median. The value of K is set by the option `value` of the function `y_outliers()`. If the `value` is missing it is calculated using the formulas $\Phi^{-1}(1/(10 \times N))$, where N is the number of observations. The two methodologies are appropriate for identifying outliers in one dimension. To identify outliers in p dimensions the function `data_leverage()`, see Section , shouls be used.

> ℹ Note
>
> Several of the graphical functions described in Section are good for identify outliers visually. In fact we recommend that the function `data_outliers()` and its related function `data_outliers_both()`, `data_outliers_by()` and `data_outliers_z()` should be used in combination with functions like `data_plot()` or `data_zscores()` or `y_dots()` of the package `gamlss.gplots`.

`y_outliers()`

The function `y_outliers()` identify outliers in one x variable using the two methodologies described above, The

- `zscores`: wherte a distribution is fitted to the data (default `SHASHo`), the residuals (z-scores) are taken and observations with large residuals are identified as outliers.
- `quantile`: This is more realing with the classical methodology based on sample quantiles. An observation is identified if it far from the median. How far usually is taken as K time the Mean Absolute Deviation (MAD).

Hewe is an example using the zscores;

```
y_outliers(da$rent)
```

```
the x was tansformed using the power 0.2412793
```

```
named integer(0)
```

and here using the `quantile`;

```r
y_outliers(da$rent, type="quantile")
```

```
the x was tansformed using the power 0.2412793
```

```
[1]  38 426
```

Note that the function `y_outliers()` is used by `data_outliers()` to identify outliers in all continuous variables in the data. There are several variation of the basic function `y_outliers()` described next.

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`y_outliers_both()`

The function `y_outliers_both()` uses both `zscores` and `quantile` methodology and reports the observations appearing in the intersection of both sets, `method=intersection` (the default), or the union of both, `method=union`.

```r
y_outliers_both(da$rent, method="union")
```

```
the x was tansformed using the power 0.2412793
the x was tansformed using the power 0.2412793
```

```
[1]  38 426
```

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`y_outliers_by()`

The function `y_outliers_by()` splits the data according to given factor, option `by` before it performes the detection of outliers. It is useful if the given factor split the data in a meaningful way. Continuous variable can be split in advange to create a factors using the **R** function `cut()`.

```r
y_outliers_by(da$rent, by=da$location)
```

```
$`1`
[1] 1747

$`2`
[1] 243

$`3`
numeric(0)
```

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`y_outliers_loop()`

The function `y_outliers_loop()` tries to identify more data than the `y_outliers()` function by using the folwing extra loop; It starts by fitting the distribution to the data which help to identify from the z-scores certain obsevations as outliers. Thaose outliers are 'then removed and the procees starts again. It stops whete there are no more observations removed. This loop allows more obsrvations to be identify and it is design to help when banch of data points masks other point to be identified as outliers.

```r
gamlss.prepdata:::y_outliers_loop(da$rent)
```

```
the x was tansformed using the power 0.2412793

integer(0)
```

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`y_outliers_z()`

The function `y_outliers_z()` tries to put the functions `y_outliers_loop()` and `y_outliers_by()` together. It works only for the z-scores methodology but its option allow different combinations of the algorithm. For example the options;

- `by`: allows to fit for levels of a factor;
- `value`: allows different values in th edetection of z-scores outliers;
- `family`; allows different families of distribution to be fitted (instead of the `SHASHo`)

- `transform` allows to switch off the transformation option
- `loop`; allows looping the algorithm, `TRUE` or `FALSE`.

```
gamlss.prepdata:::y_outliers_z(da$rent, by=da$loc)
```

```
the x was tansformed using the power 0.2412793
```

```
$`1`
integer(0)

$`2`
[1] 38

$`3`
integer(0)
```

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`data_outliers()`

The function `data_outliers()` uses the *z-scores* technique described above in order to detect outliers in the continuous variables in the data. It fits a `SHASHo` distribution to the continuous variable in the data and uses the z-scores (quantile residuals) to identify (one dimension) outliers for those continuous variables.

```
data_outliers(da)
```

```
the x was tansformed using the power 0.2412793
the x was tansformed using the power 0.3872234
the x was tansformed using the power 1.499942
```

```
$rent
named integer(0)

$area
named integer(0)

$yearc
named integer(0)
```

As it happen no individual variable outliers were highlighted in the `rent` data using the z-scores methodology. In general we recommend the function `data_outliers()` to be used in combination of the graphical functions `data_plot()` and `data_zscores()`

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`data_leverage()` (repeat)

The function `data_leverage()`, fisrt describe in Section , can be used to detect extremes in the continuous variables in the data by fiting a linear model with all the continuous variables in the data and then useing the higher leverage points to identify (multi-dimension) outliers for the observations for all continuous variables. By default, it identifies one percent of the observations with high leverage in the data. Here we use the function as a way to identify the obsrvation (not plottoing).

```
data_leverage(da, response=rent, plot=FALSE)
```

```
 100 % of data are saved,
that is, 3082 observations.
```

```
 [1]   69  130  134  142  156  196  239  367  413  432  465  641  696  876 1015
[16] 1445 1452 1499 1707 1932 2005 2070 2121 2271 2592 2655 2755 2808 2885 2897
[31] 2945
```

> **Note**
>
> The function`data_leverage()` always identifies the top one percent of observations with the highest leverage, which may or may not be outliers. These high-leverage points can have a strong influence on model estimates. To better assess their impact, the list of high-leverage observations returned by `data_leverage()` should be compared with the list of outliers identified by the `data_outliers()` function. Observations that appear in both lists are particularly noteworthy, as they may be influential outliers that warrant further investigation.

Next we are trying the process of identifying if data with high leverage are also coincide with outliers identified using the function `data_outliers()` function. The function `intersect()` will flash out the intersection of the two set of observations. First use `data_outliers()`

```
ul <- unlist(data_outliers(da))
```

```
the x was tansformed using the power 0.2412793
the x was tansformed using the power 0.3872234
the x was tansformed using the power 1.499942
```

then `data_leverage()`;

```
ll<-data_leverage(da, response=rent, plot=FALSE)
```

```
 100 % of data are saved,
that is, 3082 observations.
```

and finally we inspect the intersection of the two sets using `intersect()`;

```
intersect(ll,ul)
```

```
integer(0)
```

Here we have zero outliers but in general we would expect to find common observations for further checking.

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Hight correlations

The idea here is to identify variables with high correlation or association with the possibility to eliminated them from the data. The first function is appropriate for continuous variables while the second for both continuous and factors. Note that the method used to identify the variables is identical to the method used in the package `caret` see page 5 of Kuhn and Max (2008).

`data_index_cor()`

The function `data_index_cor()`, It takes a `data.frame` as input calculates correlation coefficients of the continuous variables in the data and creates an index for variables which could be possible to eliminate from the further analysis.

```
gamlss.prepdata:::data_index_cor(rent99[,-1])
```

```
4 factors have been omited from the data
```

```
Warning in data_cor(data, type = type, plot = FALSE, percentage = percentage, :
```

```
numeric(0)
```

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data_index_association()

The function `data_index_association()`, It takes a `data.frame` as input calculates the association coefficients of all the variables in the data and creates an index for variables which could be possible to eliminate from the further analysis.

```
gamlss.prepdata:::data_index_association(rent99[,-1])
```

```
numeric(0)
```

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Scaling

Scaling is another form of transformation for the continuous explanatory variables in the data which brings the explanatory variables in the same scale. It is a form of **standardization**. Standardization means bringing all continuous explanatory variables to similar range of values. For some machine learning techniques, i.e., principal component regression or neural networks standardization is mandatory in other like LASSO is recommended. There are two types of standardisation

i) *scaling to normality* and

ii) *scaling to a range from zero to one.*

Scaling to normality it means that the variable should have a mean zero and a standard deviation one. Note that, this is equivalent of fitting a Normal distribution to the variables and then taking the z-scores (residuals) of the fit as the transformed variable. The problem with this type of scaling is that skewness and kurtosis in the standardised data persist. One could go further and fit a four parameter distribution instead of a normal where the two extra parameters could account for skewness and kurtosis. The function `data_scale()` described in Section it gives this option with te argument `family` in which can used to specify a different distribution.

data_scale()

The function `data_scale()` perform scaling to all continuous variables an a data set. Note that factors in the data set are left untouched because when fitted within a model they will be transform to dummy variable which take values 0 or 1 (a kind of standardization). The response is left also untouched because it is assumed that an appropriate distribution will be fitted. The response variable has to be declared using the argument `response.` First we use `data_scale()` to scale to normality.

```
data_scale(da, response=rent)
```

```
    area    yearc location     bath  kitchen cheating
     132       68        3        2        2        2
 100 % of data are saved,
that is, 3082 observations.
```

```
          rent        area       yearc location bath kitchen cheating
1     109.94872 -1.74454868 -1.71736817        2    0       0        0
2     243.28204 -1.66021955 -1.71736817        2    0       0        1
3     261.64102 -1.57589041 -1.71736817        1    0       0        1
4     106.41026 -1.57589041 -1.71736817        2    0       0        0
5     133.38461 -1.57589041 -1.71736817        2    0       0        1
6     339.02563 -1.57589041 -1.71736817        2    0       0        1
7     215.23077 -1.53372585 -1.71736817        1    0       0        0
8     323.23077 -1.53372585 -1.71736817        1    0       1        1
9     216.30769 -1.49156128 -1.71736817        1    0       0        0
10    245.28204 -1.44939671 -1.71736817        2    0       0        0
11    285.38461 -1.40723215 -1.71736817        2    0       0        0
12    238.30769 -1.36506758 -1.71736817        1    0       0        1
13    374.46155 -1.36506758 -1.71736817        2    0       0        1
14    137.94872 -1.32290301 -1.71736817        2    0       0        1
15    188.35896 -1.23857388 -1.71736817        1    0       0        0
16    261.94870 -1.23857388 -1.71736817        1    0       0        0
17    363.38461 -1.19640931 -1.71736817        2    0       0        1
18    280.05127 -1.19640931 -1.71736817        2    0       0        0
19    182.71794 -1.15424475 -1.71736817        1    0       0        1
20     56.66667 -1.15424475 -1.71736817        2    0       0        0
21    147.74359 -1.15424475 -1.71736817        2    0       0        0
22    358.82053 -1.15424475 -1.71736817        2    0       0        1
23    331.12820 -1.06991561 -1.71736817        1    0       0        1
24    170.25641 -1.02775105 -1.71736817        1    0       0        0
25    173.17949 -1.02775105 -1.71736817        2    0       0        1
```

26	402.66669	-1.02775105	-1.71736817	1	0	0	1
27	401.94870	-1.02775105	-1.71736817	1	0	0	0
28	289.84616	-1.02775105	-1.71736817	2	0	0	1
29	210.76923	-0.98558648	-1.71736817	1	0	0	0
30	164.15385	-0.94342192	-1.71736817	1	0	0	1
31	381.89743	-0.94342192	-1.71736817	2	0	0	1
32	278.82053	-0.94342192	-1.71736817	2	0	0	0
33	243.02563	-0.94342192	-1.71736817	2	0	0	1
34	168.87180	-0.94342192	-1.71736817	2	0	0	0
35	178.61537	-0.94342192	-1.71736817	2	0	0	0
36	369.64102	-0.90125735	-1.71736817	2	0	0	1
37	94.05128	-0.90125735	-1.71736817	1	0	0	1
38	40.51282	-0.85909278	-1.71736817	2	0	0	1
39	186.71796	-0.81692822	-1.71736817	1	0	0	0
40	268.15387	-0.81692822	-1.71736817	1	0	0	1
41	212.71794	-0.81692822	-1.71736817	2	0	0	1
42	239.02565	-0.81692822	-1.71736817	2	0	0	1
43	355.64102	-0.81692822	-1.71736817	2	0	0	0
44	203.64102	-0.77476365	-1.71736817	2	0	0	0
45	437.43588	-0.77476365	-1.71736817	1	0	0	1
46	439.38461	-0.77476365	-1.71736817	1	0	0	1
47	252.82051	-0.73259908	-1.71736817	1	0	0	1
48	201.23076	-0.73259908	-1.71736817	1	0	0	0
49	353.79489	-0.73259908	-1.71736817	2	0	0	0
50	346.10257	-0.73259908	-1.71736817	1	0	0	0
51	221.28206	-0.73259908	-1.71736817	1	0	0	1
52	409.38461	-0.73259908	-1.71736817	2	0	0	0
53	184.76923	-0.69043452	-1.71736817	2	0	0	1
54	507.94873	-0.64826995	-1.71736817	1	1	0	1
55	282.56412	-0.64826995	-1.71736817	1	0	0	1
56	464.05130	-0.64826995	-1.71736817	1	0	0	1
57	278.71796	-0.64826995	-1.71736817	2	0	0	0
58	459.64102	-0.64826995	-1.71736817	1	0	1	1
59	216.05127	-0.64826995	-1.71736817	1	0	0	1
60	294.51282	-0.56394082	-1.71736817	1	0	0	0
61	599.94873	-0.56394082	-1.71736817	2	0	0	1
62	550.82050	-0.56394082	-1.71736817	2	0	0	1
63	311.94870	-0.56394082	-1.71736817	2	0	0	0
64	328.71796	-0.52177625	-1.71736817	2	0	0	0
65	459.74359	-0.52177625	-1.71736817	2	0	0	1
66	342.71796	-0.52177625	-1.71736817	1	0	0	1
67	355.28204	-0.52177625	-1.71736817	1	0	0	1
68	455.79486	-0.52177625	-1.71736817	1	0	0	0

69	375.23077	-0.52177625	-1.71736817	3	0	0	1
70	303.89743	-0.52177625	-1.71736817	2	0	0	1
71	408.87180	-0.52177625	-1.71736817	1	0	0	1
72	432.71796	-0.52177625	-1.71736817	1	0	0	1
73	540.41028	-0.47961168	-1.71736817	2	0	0	1
74	268.10257	-0.47961168	-1.71736817	1	0	0	1
75	404.30771	-0.47961168	-1.71736817	1	0	0	1
76	599.02563	-0.47961168	-1.71736817	1	0	0	1
77	258.15384	-0.47961168	-1.71736817	2	0	0	0
78	498.35898	-0.47961168	-1.71736817	2	0	0	1
79	430.41025	-0.47961168	-1.71736817	2	0	0	0
80	245.23077	-0.47961168	-1.71736817	1	0	0	1
81	630.92310	-0.47961168	-1.71736817	1	0	0	1
82	350.61539	-0.47961168	-1.71736817	1	0	0	1
83	145.28204	-0.47961168	-1.71736817	1	0	0	0
84	384.00000	-0.43744712	-1.71736817	2	0	0	1
85	204.05128	-0.43744712	-1.71736817	1	0	0	1
86	217.43590	-0.43744712	-1.71736817	1	0	0	1
87	229.53847	-0.43744712	-1.71736817	2	0	0	0
88	182.71794	-0.43744712	-1.71736817	2	0	0	1
89	369.23077	-0.39528255	-1.71736817	1	0	0	0
90	365.38461	-0.39528255	-1.71736817	1	0	0	0
91	405.94870	-0.39528255	-1.71736817	1	0	0	1
92	513.94873	-0.39528255	-1.71736817	2	0	0	1
93	134.97437	-0.39528255	-1.71736817	1	0	0	0
94	265.33334	-0.39528255	-1.71736817	2	0	0	0
95	508.82053	-0.39528255	-1.71736817	2	0	0	1
96	425.23077	-0.35311798	-1.71736817	1	0	1	1
97	218.30769	-0.35311798	-1.71736817	1	0	0	0
98	499.74359	-0.35311798	-1.71736817	1	0	1	1
99	226.76924	-0.35311798	-1.71736817	2	0	0	0
100	246.61539	-0.35311798	-1.71736817	2	0	0	0
101	443.38461	-0.35311798	-1.71736817	2	0	0	1
102	654.92303	-0.31095342	-1.71736817	2	0	0	1
103	501.58972	-0.31095342	-1.71736817	2	0	0	1
104	514.30768	-0.31095342	-1.71736817	2	0	0	1
105	216.00000	-0.31095342	-1.71736817	2	0	0	0
106	495.58975	-0.31095342	-1.71736817	1	0	0	1
107	284.92307	-0.31095342	-1.71736817	1	0	0	0
108	697.28204	-0.31095342	-1.71736817	2	0	0	1
109	518.30768	-0.31095342	-1.71736817	2	0	0	0
110	309.58975	-0.31095342	-1.71736817	1	0	0	0
111	361.89743	-0.31095342	-1.71736817	1	0	0	1

112	351.33331	-0.31095342	-1.71736817	2	0	0	0
113	292.46152	-0.31095342	-1.71736817	1	0	0	1
114	280.76923	-0.31095342	-1.71736817	2	0	0	0
115	259.53845	-0.31095342	-1.71736817	1	0	0	0
116	174.25641	-0.31095342	-1.71736817	1	0	0	1
117	254.25641	-0.31095342	-1.71736817	1	0	0	0
118	391.07690	-0.26878885	-1.71736817	1	0	0	0
119	542.15381	-0.26878885	-1.71736817	2	0	0	1
120	279.84616	-0.22662428	-1.71736817	2	0	0	0
121	252.20512	-0.22662428	-1.71736817	1	0	0	0
122	282.10254	-0.22662428	-1.71736817	1	0	0	1
123	248.25641	-0.22662428	-1.71736817	1	0	0	0
124	343.74359	-0.22662428	-1.71736817	2	0	0	1
125	590.61536	-0.22662428	-1.71736817	1	0	0	1
126	195.48718	-0.22662428	-1.71736817	1	0	0	0
127	261.64102	-0.18445972	-1.71736817	2	0	0	0
128	523.07690	-0.18445972	-1.71736817	1	0	0	1
129	97.53846	-0.18445972	-1.71736817	1	0	0	0
130	664.05127	-0.14229515	-1.71736817	3	0	0	1
131	589.28204	-0.14229515	-1.71736817	1	0	0	1
132	508.51282	-0.14229515	-1.71736817	1	0	0	1
133	274.82053	-0.14229515	-1.71736817	2	0	0	1
134	720.30768	-0.14229515	-1.71736817	3	0	0	1
135	391.28204	-0.14229515	-1.71736817	1	0	0	1
136	400.76923	-0.14229515	-1.71736817	1	0	0	1
137	442.87180	-0.14229515	-1.71736817	1	0	0	1
138	298.30771	-0.10013058	-1.71736817	1	0	0	1
139	227.43590	-0.10013058	-1.71736817	2	1	0	1
140	453.94873	-0.10013058	-1.71736817	2	0	0	1
141	535.17950	-0.10013058	-1.71736817	1	0	0	1
142	800.76923	-0.10013058	-1.71736817	3	0	1	1
143	452.05127	-0.10013058	-1.71736817	1	0	0	1
144	462.56412	-0.10013058	-1.71736817	2	0	0	0
145	334.10257	-0.10013058	-1.71736817	2	0	0	0
146	308.51282	-0.10013058	-1.71736817	2	0	0	1
147	348.66669	-0.10013058	-1.71736817	2	0	0	1
148	459.74359	-0.10013058	-1.71736817	2	0	0	1
149	463.07693	-0.10013058	-1.71736817	2	0	0	1
150	593.48718	-0.10013058	-1.71736817	1	0	0	1
151	423.89743	-0.05796602	-1.71736817	1	0	0	1
152	210.92307	-0.05796602	-1.71736817	2	0	0	0
153	274.71796	-0.01580145	-1.71736817	2	0	0	0
154	172.15385	-0.01580145	-1.71736817	2	0	0	0

155	641.69232	-0.01580145	-1.71736817	2	0	0	1
156	508.20514	-0.01580145	-1.71736817	3	0	0	1
157	597.12823	-0.01580145	-1.71736817	2	0	0	1
158	414.56412	0.02636312	-1.71736817	2	0	0	1
159	507.38461	0.02636312	-1.71736817	2	0	0	1
160	168.61537	0.02636312	-1.71736817	1	0	0	0
161	314.00000	0.02636312	-1.71736817	1	0	0	0
162	274.41025	0.02636312	-1.71736817	1	0	0	1
163	476.35898	0.02636312	-1.71736817	1	0	0	1
164	505.23077	0.02636312	-1.71736817	2	0	0	1
165	718.46155	0.06852768	-1.71736817	2	0	0	1
166	395.53845	0.06852768	-1.71736817	1	0	0	1
167	412.41025	0.06852768	-1.71736817	1	0	0	0
168	588.51282	0.06852768	-1.71736817	3	0	0	1
169	178.97437	0.06852768	-1.71736817	1	0	0	1
170	342.76926	0.06852768	-1.71736817	1	0	0	0
171	644.10254	0.11069225	-1.71736817	1	0	0	1
172	512.00000	0.11069225	-1.71736817	2	0	0	1
173	423.48718	0.11069225	-1.71736817	2	0	0	1
174	429.79486	0.11069225	-1.71736817	2	0	0	0
175	339.89743	0.11069225	-1.71736817	2	0	0	0
176	335.07693	0.11069225	-1.71736817	2	0	1	1
177	496.92307	0.11069225	-1.71736817	2	0	0	1
178	280.30768	0.11069225	-1.71736817	2	0	0	0
179	365.33334	0.11069225	-1.71736817	1	0	0	1
180	750.76923	0.11069225	-1.71736817	2	0	1	1
181	190.97435	0.11069225	-1.71736817	1	0	0	0
182	190.71794	0.11069225	-1.71736817	2	0	0	0
183	451.58972	0.11069225	-1.71736817	2	0	0	1
184	607.79486	0.11069225	-1.71736817	1	0	0	1
185	250.10257	0.11069225	-1.71736817	2	0	0	1
186	208.46153	0.11069225	-1.71736817	2	0	0	0
187	245.23077	0.11069225	-1.71736817	2	0	0	0
188	183.58974	0.11069225	-1.71736817	2	0	0	0
189	616.00000	0.11069225	-1.71736817	2	0	0	1
190	436.05127	0.15285682	-1.71736817	2	0	0	1
191	593.48718	0.15285682	-1.71736817	3	0	0	1
192	494.82053	0.15285682	-1.71736817	1	0	0	1
193	618.51282	0.19502138	-1.71736817	2	0	0	1
194	691.23077	0.19502138	-1.71736817	2	0	1	1
195	756.00000	0.19502138	-1.71736817	2	0	1	1
196	318.56412	0.19502138	-1.71736817	3	0	1	1
197	443.43591	0.19502138	-1.71736817	1	0	0	1

198	694.71790	0.19502138	-1.71736817	2	0	0	1
199	732.25641	0.19502138	-1.71736817	2	0	1	1
200	335.53845	0.23718595	-1.71736817	2	0	0	1
201	443.28207	0.23718595	-1.71736817	1	0	1	0
202	553.74359	0.23718595	-1.71736817	1	0	0	1
203	391.17947	0.23718595	-1.71736817	1	0	0	1
204	390.25641	0.23718595	-1.71736817	1	0	0	1
205	263.74359	0.27935051	-1.71736817	1	0	0	1
206	380.76923	0.27935051	-1.71736817	2	0	0	1
207	379.23077	0.27935051	-1.71736817	2	0	0	1
208	698.30768	0.32151508	-1.71736817	1	0	0	1
209	381.33331	0.32151508	-1.71736817	2	0	0	0
210	663.38458	0.32151508	-1.71736817	2	0	0	1
211	595.43591	0.32151508	-1.71736817	2	0	0	1
212	366.66666	0.32151508	-1.71736817	2	0	0	0
213	247.79488	0.32151508	-1.71736817	2	0	0	1
214	735.69232	0.32151508	-1.71736817	2	0	0	1
215	307.64105	0.36367965	-1.71736817	2	0	0	0
216	371.23077	0.36367965	-1.71736817	1	0	0	1
217	619.53845	0.36367965	-1.71736817	1	0	0	1
218	365.23077	0.36367965	-1.71736817	1	0	0	0
219	213.23076	0.36367965	-1.71736817	2	0	0	0
220	632.87177	0.36367965	-1.71736817	1	0	0	1
221	371.28204	0.36367965	-1.71736817	1	0	0	0
222	308.46155	0.36367965	-1.71736817	2	0	0	0
223	270.30768	0.40584421	-1.71736817	1	0	0	1
224	763.02563	0.40584421	-1.71736817	2	0	0	0
225	461.48718	0.40584421	-1.71736817	2	0	0	1
226	293.64102	0.40584421	-1.71736817	2	0	0	0
227	446.87180	0.40584421	-1.71736817	1	0	0	1
228	497.33331	0.40584421	-1.71736817	1	0	0	1
229	293.79489	0.40584421	-1.71736817	2	0	0	0
230	243.89745	0.44800878	-1.71736817	2	0	0	0
231	550.87177	0.44800878	-1.71736817	1	0	0	1
232	388.51282	0.44800878	-1.71736817	1	0	0	1
233	338.05130	0.44800878	-1.71736817	1	0	0	0
234	505.43588	0.44800878	-1.71736817	1	0	0	1
235	563.48718	0.49017335	-1.71736817	2	0	0	0
236	241.84616	0.49017335	-1.71736817	2	0	0	0
237	444.56412	0.49017335	-1.71736817	2	0	0	1
238	312.15384	0.49017335	-1.71736817	1	0	0	0
239	495.94870	0.53233791	-1.71736817	3	0	0	0
240	438.15387	0.53233791	-1.71736817	2	0	0	0

241	300.15384	0.53233791	-1.71736817	1	0	0	0
242	619.89746	0.53233791	-1.71736817	2	0	0	1
243	525.23077	0.53233791	-1.71736817	1	0	0	1
244	376.56409	0.53233791	-1.71736817	1	0	0	1
245	268.30771	0.53233791	-1.71736817	2	0	0	0
246	655.74359	0.53233791	-1.71736817	2	0	0	1
247	544.15381	0.53233791	-1.71736817	2	0	0	0
248	312.56412	0.53233791	-1.71736817	1	0	0	1
249	833.33331	0.57450248	-1.71736817	2	0	0	1
250	317.28204	0.57450248	-1.71736817	2	0	0	0
251	806.61542	0.57450248	-1.71736817	2	0	0	1
252	275.69229	0.57450248	-1.71736817	2	0	0	0
253	292.25641	0.61666705	-1.71736817	1	0	0	0
254	230.71794	0.61666705	-1.71736817	2	0	0	1
255	519.74359	0.61666705	-1.71736817	1	0	0	1
256	415.84616	0.65883161	-1.71736817	1	0	0	1
257	602.35895	0.65883161	-1.71736817	2	0	0	1
258	712.61536	0.65883161	-1.71736817	2	0	0	0
259	675.64105	0.70099618	-1.71736817	2	0	0	1
260	418.76923	0.70099618	-1.71736817	2	0	0	0
261	524.46155	0.70099618	-1.71736817	1	0	0	0
262	412.56412	0.70099618	-1.71736817	1	0	0	1
263	700.82050	0.74316075	-1.71736817	2	0	0	1
264	830.87177	0.74316075	-1.71736817	2	0	1	1
265	709.74359	0.74316075	-1.71736817	2	1	0	1
266	350.20514	0.74316075	-1.71736817	1	0	0	0
267	358.97437	0.74316075	-1.71736817	2	0	0	1
268	678.82050	0.74316075	-1.71736817	1	0	0	1
269	283.02567	0.78532531	-1.71736817	2	0	0	1
270	755.64105	0.78532531	-1.71736817	2	0	0	1
271	546.97437	0.78532531	-1.71736817	1	0	0	1
272	343.69232	0.82748988	-1.71736817	2	0	0	0
273	504.10257	0.82748988	-1.71736817	2	0	0	1
274	355.58975	0.82748988	-1.71736817	2	0	0	0
275	950.97437	0.82748988	-1.71736817	2	1	0	1
276	300.71796	0.82748988	-1.71736817	2	0	0	0
277	884.20508	0.82748988	-1.71736817	1	0	0	1
278	309.38461	0.82748988	-1.71736817	2	0	0	0
279	373.84616	0.86965445	-1.71736817	2	0	0	0
280	177.33333	0.86965445	-1.71736817	2	0	0	0
281	470.20514	0.95398358	-1.71736817	2	0	0	0
282	728.82050	0.95398358	-1.71736817	2	1	0	1
283	515.07697	0.95398358	-1.71736817	2	0	0	1

284	637.79486	0.95398358	-1.71736817	1	0	0	1
285	330.61539	0.95398358	-1.71736817	2	0	0	0
286	673.64099	0.95398358	-1.71736817	1	0	0	1
287	345.07693	0.95398358	-1.71736817	2	0	0	0
288	546.51282	0.95398358	-1.71736817	2	0	0	0
289	602.97437	0.95398358	-1.71736817	2	0	0	0
290	488.97437	0.95398358	-1.71736817	2	0	0	0
291	305.38461	0.95398358	-1.71736817	1	0	0	1
292	331.17947	0.95398358	-1.71736817	2	0	0	1
293	458.05130	0.95398358	-1.71736817	2	0	0	1
294	436.35898	0.99614815	-1.71736817	1	0	0	0
295	844.30768	0.99614815	-1.71736817	2	0	0	1
296	283.17950	1.03831271	-1.71736817	1	0	0	1
297	790.92310	1.03831271	-1.71736817	2	0	0	1
298	449.89743	1.03831271	-1.71736817	1	0	0	1
299	533.23077	1.03831271	-1.71736817	2	1	0	1
300	796.10260	1.03831271	-1.71736817	1	0	0	1
301	567.64105	1.03831271	-1.71736817	1	1	0	1
302	588.97437	1.03831271	-1.71736817	1	0	0	1
303	536.76923	1.03831271	-1.71736817	2	0	0	1
304	425.58975	1.08047728	-1.71736817	1	0	0	0
305	558.15387	1.08047728	-1.71736817	2	0	0	1
306	402.87180	1.08047728	-1.71736817	2	0	0	0
307	220.41025	1.08047728	-1.71736817	2	0	0	0
308	255.48718	1.08047728	-1.71736817	1	0	0	0
309	435.69229	1.08047728	-1.71736817	2	0	0	0
310	404.30771	1.12264185	-1.71736817	2	0	0	0
311	584.82050	1.12264185	-1.71736817	2	1	0	1
312	567.07697	1.12264185	-1.71736817	2	0	0	1
313	283.23077	1.12264185	-1.71736817	1	0	0	0
314	258.92307	1.16480641	-1.71736817	1	0	0	1
315	239.38461	1.16480641	-1.71736817	1	0	0	0
316	960.20514	1.16480641	-1.71736817	2	0	0	1
317	597.79486	1.16480641	-1.71736817	2	0	0	0
318	898.92310	1.16480641	-1.71736817	2	0	0	1
319	400.20514	1.16480641	-1.71736817	2	0	0	1
320	440.10257	1.16480641	-1.71736817	1	0	0	1
321	1083.79480	1.20697098	-1.71736817	2	1	0	1
322	705.28210	1.20697098	-1.71736817	1	0	0	1
323	785.43591	1.20697098	-1.71736817	2	0	0	1
324	449.17950	1.20697098	-1.71736817	2	0	0	0
325	1020.10254	1.20697098	-1.71736817	2	0	1	1
326	319.02563	1.24913555	-1.71736817	1	0	0	1

327	496.51282	1.24913555	-1.71736817	2	0	0	0
328	325.07693	1.24913555	-1.71736817	2	1	0	1
329	497.48715	1.29130011	-1.71736817	1	0	0	1
330	470.56409	1.29130011	-1.71736817	1	0	0	1
331	414.76923	1.33346468	-1.71736817	2	0	0	1
332	476.97433	1.33346468	-1.71736817	1	0	0	1
333	391.84613	1.33346468	-1.71736817	2	0	0	1
334	445.58975	1.37562925	-1.71736817	2	0	0	0
335	397.12820	1.37562925	-1.71736817	2	0	0	1
336	794.10254	1.37562925	-1.71736817	2	0	0	1
337	860.71796	1.37562925	-1.71736817	1	0	0	1
338	670.76923	1.37562925	-1.71736817	2	0	0	1
339	562.30768	1.37562925	-1.71736817	1	0	0	1
340	776.35901	1.37562925	-1.71736817	2	0	0	1
341	322.30768	1.37562925	-1.71736817	2	0	0	0
342	460.76923	1.37562925	-1.71736817	1	0	0	0
343	358.82053	1.37562925	-1.71736817	1	0	0	1
344	459.43591	1.41779381	-1.71736817	2	0	0	0
345	466.20511	1.45995838	-1.71736817	2	0	0	0
346	419.64102	1.45995838	-1.71736817	1	0	0	1
347	384.46155	1.45995838	-1.71736817	1	0	0	0
348	875.02570	1.45995838	-1.71736817	2	1	0	1
349	865.64105	1.50212295	-1.71736817	2	1	0	1
350	1260.51282	1.54428751	-1.71736817	1	0	0	1
351	332.05127	1.54428751	-1.71736817	1	0	0	0
352	314.05130	1.54428751	-1.71736817	1	0	0	1
353	521.43591	1.54428751	-1.71736817	3	0	0	1
354	1181.58984	1.58645208	-1.71736817	2	1	0	1
355	514.30768	1.58645208	-1.71736817	1	0	0	1
356	601.33331	1.58645208	-1.71736817	2	0	0	1
357	354.41025	1.58645208	-1.71736817	2	0	0	1
358	501.94870	1.62861664	-1.71736817	2	0	0	1
359	925.28210	1.67078121	-1.71736817	2	0	0	1
360	633.69226	1.71294578	-1.71736817	2	0	0	1
361	916.71796	1.71294578	-1.71736817	1	0	0	1
362	285.02563	1.71294578	-1.71736817	1	0	0	1
363	595.89746	1.75511034	-1.71736817	2	0	0	1
364	539.23077	1.75511034	-1.71736817	1	0	0	0
365	531.38458	1.79727491	-1.71736817	2	0	0	1
366	407.43588	1.79727491	-1.71736817	2	0	0	1
367	656.87183	1.79727491	-1.71736817	3	0	0	1
368	126.00000	1.79727491	-1.71736817	2	0	0	0
369	522.61536	1.79727491	-1.71736817	2	0	0	1

370	463.38461	1.79727491	-1.71736817	2	1	0	1
371	564.20508	1.79727491	-1.71736817	1	0	0	1
372	820.61536	1.79727491	-1.71736817	2	0	0	1
373	307.17947	1.79727491	-1.71736817	2	0	0	0
374	300.66666	1.79727491	-1.71736817	2	0	0	1
375	883.74359	1.79727491	-1.71736817	2	0	0	1
376	246.92308	1.79727491	-1.71736817	1	0	0	0
377	833.23077	1.79727491	-1.71736817	2	1	0	1
378	641.53845	1.88160404	-1.71736817	2	1	0	1
379	1072.30774	1.88160404	-1.71736817	2	0	1	1
380	697.84619	1.88160404	-1.71736817	2	0	0	1
381	522.00000	1.88160404	-1.71736817	2	0	0	1
382	606.76923	1.92376861	-1.71736817	2	0	0	0
383	707.43591	1.96593318	-1.71736817	1	0	0	1
384	559.02563	1.96593318	-1.71736817	2	0	0	1
385	514.10254	2.00809774	-1.71736817	2	0	0	1
386	589.23077	2.00809774	-1.71736817	2	0	0	1
387	521.07690	2.00809774	-1.71736817	1	0	0	0
388	633.89740	2.09242688	-1.71736817	2	0	0	1
389	415.02563	2.13459144	-1.71736817	1	0	0	0
390	333.43591	2.13459144	-1.71736817	1	0	0	1
391	671.64099	2.13459144	-1.71736817	1	0	0	1
392	286.05127	2.13459144	-1.71736817	1	0	0	1
393	733.89740	2.21892058	-1.71736817	2	0	0	1
394	292.15384	2.21892058	-1.71736817	2	0	0	0
395	1060.56421	2.21892058	-1.71736817	2	1	0	1
396	762.20514	2.21892058	-1.71736817	2	0	0	1
397	474.25641	2.21892058	-1.71736817	2	0	0	1
398	474.00000	2.21892058	-1.71736817	2	0	0	1
399	422.82053	2.21892058	-1.71736817	1	0	0	1
400	553.07690	2.21892058	-1.71736817	1	0	0	1
401	540.15387	2.30324971	-1.71736817	1	0	0	1
402	808.30768	2.34541428	-1.71736817	2	0	0	1
403	272.82053	2.34541428	-1.71736817	2	0	0	0
404	1136.46155	2.38757884	-1.71736817	1	1	0	1
405	512.66669	2.42974341	-1.71736817	2	0	0	0
406	657.58978	2.42974341	-1.71736817	2	0	0	1
407	966.10260	2.42974341	-1.71736817	2	0	0	0
408	584.35895	2.51407254	-1.71736817	2	0	0	1
409	195.17949	2.55623711	-1.71736817	2	0	0	0
410	713.38458	2.59840168	-1.71736817	2	0	0	1
411	882.00000	2.64056624	-1.71736817	1	0	0	1
412	518.46155	2.64056624	-1.71736817	1	0	0	1

413	841.64099	2.64056624	-1.71736817	3	0	0	1
414	507.64105	2.68273081	-1.71736817	2	0	0	0
415	445.53845	2.76705994	-1.71736817	2	1	0	1
416	415.74359	2.80922451	-1.71736817	2	0	0	1
417	998.46155	2.89355364	-1.71736817	2	0	0	1
418	478.82053	2.89355364	-1.71736817	2	0	0	1
419	884.82050	2.93571821	-1.71736817	1	1	0	1
420	1038.82043	2.93571821	-1.71736817	2	0	0	1
421	1174.15393	2.93571821	-1.71736817	2	0	1	1
422	372.00000	2.97788277	-1.71736817	2	0	0	1
423	671.64099	3.06221191	-1.71736817	2	0	0	1
424	1125.48718	3.06221191	-1.71736817	2	1	0	1
425	204.05128	3.23087017	-1.71736817	1	0	0	0
426	1843.38464	3.27303474	-1.71736817	2	0	0	1
427	629.38464	3.35736387	-1.71736817	1	0	0	0
428	61.53846	3.39952844	-1.71736817	2	0	0	0
429	1180.51282	3.48385757	-1.71736817	2	1	0	1
430	776.15387	3.56818671	-1.71736817	2	0	0	1
431	1372.51282	3.69468041	-1.71736817	2	0	0	1
432	877.53845	3.69468041	-1.71736817	3	1	0	1
433	1465.43591	3.69468041	-1.71736817	2	0	0	1
434	1150.15393	3.73684497	-1.71736817	2	0	0	0
435	785.74359	3.77900954	-1.71736817	2	0	0	1
436	985.89746	3.90550324	-1.71736817	2	0	0	1
437	911.23077	3.90550324	-1.71736817	2	0	0	1
438	1161.33337	3.90550324	-1.71736817	2	1	0	1
439	1551.17957	3.90550324	-1.71736817	2	1	0	1
440	604.35895	3.90550324	-1.71736817	2	0	0	1
441	575.74359	3.90550324	-1.71736817	2	0	0	0
442	146.87180	-1.40723215	-1.44837298	1	0	0	1
443	190.35898	-1.28073845	-1.44837298	1	0	0	1
444	152.35898	-0.98558648	-1.44837298	1	0	0	0
445	229.23077	-0.85909278	-1.44837298	2	0	0	1
446	227.07692	-0.85909278	-1.44837298	1	0	0	1
447	223.33333	-0.85909278	-1.44837298	2	0	0	1
448	291.02563	-0.73259908	-1.44837298	1	0	0	1
449	237.58974	-0.64826995	-1.44837298	1	0	0	1
450	339.79486	-0.52177625	-1.44837298	1	0	0	1
451	294.10257	-0.47961168	-1.44837298	1	0	0	1
452	362.10254	-0.43744712	-1.44837298	1	0	0	1
453	244.46155	-0.43744712	-1.44837298	1	0	0	0
454	277.53848	-0.39528255	-1.44837298	2	0	0	1
455	711.23077	-0.35311798	-1.44837298	2	0	0	1

456	216.97437	-0.18445972	-1.44837298	1	0	0	1
457	457.02563	-0.18445972	-1.44837298	2	0	0	1
458	547.84619	-0.14229515	-1.44837298	2	0	0	1
459	559.12823	-0.14229515	-1.44837298	2	0	0	1
460	337.69232	-0.01580145	-1.44837298	2	0	0	0
461	446.30768	0.11069225	-1.44837298	2	0	0	1
462	833.12817	0.27935051	-1.44837298	2	0	0	1
463	675.48718	0.36367965	-1.44837298	2	0	0	1
464	274.46155	0.36367965	-1.44837298	1	0	0	0
465	373.58975	0.49017335	-1.44837298	3	0	0	0
466	876.15387	0.53233791	-1.44837298	2	0	0	1
467	462.66669	0.65883161	-1.44837298	1	0	0	1
468	904.56409	0.82748988	-1.44837298	3	0	0	1
469	701.89740	0.95398358	-1.44837298	2	0	0	1
470	575.94873	1.20697098	-1.44837298	1	0	0	1
471	373.12820	1.29130011	-1.44837298	1	0	0	1
472	307.84616	1.37562925	-1.44837298	2	0	0	1
473	487.07690	1.45995838	-1.44837298	1	0	0	1
474	333.89743	1.62861664	-1.44837298	1	0	0	0
475	547.58978	1.71294578	-1.44837298	2	0	0	1
476	344.61539	1.75511034	-1.44837298	1	0	0	0
477	446.30768	1.79727491	-1.44837298	1	0	0	0
478	808.41028	2.13459144	-1.44837298	2	0	0	1
479	1021.38458	2.21892058	-1.44837298	2	0	0	1
480	447.74359	2.21892058	-1.44837298	2	0	0	1
481	283.94873	-1.66021955	-1.22421032	2	0	0	1
482	231.84616	-1.57589041	-1.22421032	1	0	0	1
483	342.05127	-1.53372585	-1.22421032	2	0	0	1
484	308.66669	-1.53372585	-1.22421032	1	0	0	1
485	202.25641	-1.49156128	-1.22421032	2	0	0	1
486	299.38461	-1.40723215	-1.22421032	1	0	0	1
487	302.92310	-1.36506758	-1.22421032	2	0	0	1
488	383.23077	-1.36506758	-1.22421032	2	0	0	1
489	319.69232	-1.36506758	-1.22421032	2	0	0	1
490	373.64102	-1.28073845	-1.22421032	2	0	0	1
491	278.82053	-1.23857388	-1.22421032	2	0	0	1
492	415.69229	-1.23857388	-1.22421032	1	0	0	1
493	271.53845	-1.23857388	-1.22421032	1	0	0	1
494	361.89743	-1.15424475	-1.22421032	1	0	0	1
495	413.94873	-1.15424475	-1.22421032	2	0	0	1
496	368.35898	-1.15424475	-1.22421032	2	0	0	1
497	202.00000	-0.94342192	-1.22421032	2	0	0	0
498	366.56409	-0.94342192	-1.22421032	1	0	0	1

499	332.87180	-0.94342192	-1.22421032	1	1	0	1
500	222.92308	-0.85909278	-1.22421032	1	0	0	1
501	295.38461	-0.85909278	-1.22421032	1	0	0	1
502	243.38461	-0.81692822	-1.22421032	1	0	0	1
503	397.38461	-0.73259908	-1.22421032	1	0	0	1
504	184.82051	-0.73259908	-1.22421032	2	0	0	0
505	494.10257	-0.73259908	-1.22421032	2	0	0	0
506	405.07693	-0.73259908	-1.22421032	3	0	0	1
507	177.43590	-0.73259908	-1.22421032	2	0	0	0
508	452.25641	-0.69043452	-1.22421032	1	1	0	1
509	437.48715	-0.64826995	-1.22421032	1	0	0	1
510	224.92308	-0.64826995	-1.22421032	1	0	0	1
511	468.92310	-0.64826995	-1.22421032	1	0	0	1
512	566.66669	-0.64826995	-1.22421032	2	0	0	1
513	281.38461	-0.56394082	-1.22421032	2	0	0	1
514	190.00000	-0.56394082	-1.22421032	1	0	0	1
515	499.07693	-0.56394082	-1.22421032	1	0	0	0
516	351.17947	-0.56394082	-1.22421032	2	0	0	1
517	455.48718	-0.52177625	-1.22421032	2	0	0	1
518	390.35898	-0.52177625	-1.22421032	1	0	0	1
519	239.28206	-0.47961168	-1.22421032	1	0	0	1
520	526.92310	-0.47961168	-1.22421032	2	0	0	1
521	148.25641	-0.47961168	-1.22421032	1	0	0	0
522	197.69231	-0.47961168	-1.22421032	1	0	0	0
523	332.10254	-0.43744712	-1.22421032	1	0	0	1
524	157.23077	-0.39528255	-1.22421032	2	0	0	0
525	121.64102	-0.35311798	-1.22421032	1	0	0	1
526	174.82051	-0.35311798	-1.22421032	1	0	0	1
527	502.71796	-0.31095342	-1.22421032	1	0	0	1
528	305.94870	-0.31095342	-1.22421032	2	0	0	0
529	301.23077	-0.31095342	-1.22421032	2	0	0	1
530	497.33331	-0.31095342	-1.22421032	1	0	0	1
531	473.69232	-0.31095342	-1.22421032	2	0	0	1
532	137.02565	-0.26878885	-1.22421032	1	0	0	0
533	345.33334	-0.22662428	-1.22421032	2	0	0	1
534	250.10257	-0.22662428	-1.22421032	1	0	0	0
535	568.61542	-0.22662428	-1.22421032	1	0	1	1
536	562.41022	-0.22662428	-1.22421032	2	0	0	1
537	245.89743	-0.18445972	-1.22421032	1	0	0	0
538	483.02567	-0.18445972	-1.22421032	1	0	0	0
539	597.23077	-0.18445972	-1.22421032	2	0	0	1
540	269.94873	-0.18445972	-1.22421032	2	0	0	1
541	256.66666	-0.14229515	-1.22421032	1	0	0	1

542	557.94873	-0.14229515	-1.22421032	1	0	0	1
543	242.56410	-0.14229515	-1.22421032	1	0	0	0
544	266.25641	-0.10013058	-1.22421032	1	0	0	1
545	319.48718	-0.10013058	-1.22421032	1	0	0	1
546	635.53851	-0.10013058	-1.22421032	1	0	0	1
547	512.97437	-0.10013058	-1.22421032	2	0	0	0
548	591.64099	-0.10013058	-1.22421032	1	0	0	1
549	454.25641	-0.10013058	-1.22421032	2	0	0	1
550	258.51282	-0.10013058	-1.22421032	2	0	0	0
551	701.89740	-0.10013058	-1.22421032	2	0	0	1
552	601.17950	-0.10013058	-1.22421032	1	0	0	0
553	252.61539	-0.05796602	-1.22421032	2	0	0	1
554	624.35895	-0.05796602	-1.22421032	2	0	0	1
555	402.56412	-0.01580145	-1.22421032	2	0	0	0
556	622.82050	-0.01580145	-1.22421032	2	0	0	1
557	306.92307	0.02636312	-1.22421032	2	0	0	1
558	599.17950	0.02636312	-1.22421032	2	0	0	1
559	593.28204	0.06852768	-1.22421032	2	0	0	1
560	384.92307	0.11069225	-1.22421032	1	0	0	1
561	388.25641	0.11069225	-1.22421032	2	0	0	0
562	372.97437	0.11069225	-1.22421032	1	0	0	1
563	605.74359	0.11069225	-1.22421032	1	0	0	1
564	102.71795	0.11069225	-1.22421032	1	0	0	0
565	528.71796	0.15285682	-1.22421032	1	0	0	1
566	303.69232	0.23718595	-1.22421032	1	0	0	1
567	418.61539	0.27935051	-1.22421032	2	0	1	1
568	597.74359	0.27935051	-1.22421032	1	0	0	1
569	414.97437	0.32151508	-1.22421032	1	0	0	1
570	773.33331	0.32151508	-1.22421032	2	0	0	1
571	320.25641	0.36367965	-1.22421032	1	0	0	1
572	547.02563	0.36367965	-1.22421032	1	1	0	1
573	401.17947	0.36367965	-1.22421032	1	0	0	1
574	282.71796	0.36367965	-1.22421032	1	0	0	0
575	460.30768	0.36367965	-1.22421032	2	0	0	1
576	339.02563	0.36367965	-1.22421032	1	0	0	1
577	347.64105	0.36367965	-1.22421032	2	0	0	0
578	412.46152	0.36367965	-1.22421032	1	0	0	1
579	491.28204	0.40584421	-1.22421032	2	0	0	1
580	404.71796	0.44800878	-1.22421032	1	0	0	1
581	260.30771	0.44800878	-1.22421032	2	0	0	0
582	759.07690	0.44800878	-1.22421032	1	0	0	1
583	440.30768	0.44800878	-1.22421032	1	0	0	1
584	717.94873	0.44800878	-1.22421032	2	0	0	1

585	403.58975	0.49017335	-1.22421032	1	0	0	1
586	271.12820	0.49017335	-1.22421032	1	0	0	0
587	435.12820	0.53233791	-1.22421032	1	0	0	1
588	492.35895	0.53233791	-1.22421032	2	0	0	1
589	705.79492	0.53233791	-1.22421032	2	0	0	1
590	209.12820	0.53233791	-1.22421032	1	0	0	0
591	391.94870	0.53233791	-1.22421032	1	0	0	1
592	251.07692	0.53233791	-1.22421032	1	0	0	0
593	409.17950	0.53233791	-1.22421032	1	0	0	1
594	230.92307	0.53233791	-1.22421032	2	0	0	0
595	357.23074	0.57450248	-1.22421032	1	0	0	0
596	461.84613	0.57450248	-1.22421032	2	0	0	1
597	385.74359	0.61666705	-1.22421032	2	0	0	1
598	419.43591	0.61666705	-1.22421032	1	0	0	1
599	374.82053	0.65883161	-1.22421032	1	1	0	0
600	379.84616	0.65883161	-1.22421032	1	0	0	1
601	465.07693	0.65883161	-1.22421032	2	0	0	1
602	652.10254	0.70099618	-1.22421032	2	0	0	1
603	370.00000	0.74316075	-1.22421032	2	0	0	1
604	567.43591	0.74316075	-1.22421032	2	0	0	1
605	179.89743	0.74316075	-1.22421032	1	0	0	0
606	346.87180	0.78532531	-1.22421032	2	0	0	1
607	141.23076	0.78532531	-1.22421032	1	0	0	0
608	357.89746	0.82748988	-1.22421032	2	0	0	0
609	380.66666	0.86965445	-1.22421032	2	0	0	0
610	291.58972	0.86965445	-1.22421032	1	0	0	1
611	280.35898	1.03831271	-1.22421032	1	0	0	0
612	333.48718	1.08047728	-1.22421032	1	0	0	0
613	218.05128	1.12264185	-1.22421032	1	0	1	1
614	417.43588	1.16480641	-1.22421032	1	0	0	1
615	914.61536	1.20697098	-1.22421032	2	0	0	1
616	755.64105	1.20697098	-1.22421032	1	0	0	0
617	470.61539	1.20697098	-1.22421032	1	0	0	1
618	498.82053	1.29130011	-1.22421032	2	0	0	1
619	1061.58984	1.29130011	-1.22421032	2	0	0	1
620	794.66663	1.29130011	-1.22421032	2	0	0	1
621	956.87183	1.37562925	-1.22421032	2	0	0	1
622	543.84613	1.37562925	-1.22421032	1	0	0	1
623	517.02563	1.41779381	-1.22421032	1	1	0	1
624	556.41028	1.41779381	-1.22421032	2	0	0	1
625	304.51282	1.41779381	-1.22421032	2	0	0	1
626	336.00000	1.45995838	-1.22421032	2	0	0	1
627	322.41025	1.50212295	-1.22421032	1	0	0	0

628	545.84619	1.54428751	-1.22421032	3	0	0	1
629	502.97437	1.58645208	-1.22421032	3	0	0	1
630	558.97437	1.58645208	-1.22421032	1	0	0	1
631	418.92310	1.67078121	-1.22421032	2	0	0	1
632	361.02563	1.75511034	-1.22421032	2	0	0	1
633	741.28204	1.75511034	-1.22421032	1	0	0	1
634	558.71796	1.75511034	-1.22421032	2	0	0	1
635	589.38464	1.79727491	-1.22421032	1	0	0	1
636	638.61542	1.79727491	-1.22421032	2	0	0	0
637	995.48718	1.79727491	-1.22421032	2	0	0	1
638	709.84613	1.79727491	-1.22421032	3	0	0	1
639	1008.30768	1.79727491	-1.22421032	2	0	0	1
640	212.15385	1.79727491	-1.22421032	2	0	0	1
641	205.23077	1.79727491	-1.22421032	3	0	0	0
642	711.07690	1.92376861	-1.22421032	2	0	0	1
643	578.35901	1.96593318	-1.22421032	1	0	0	1
644	659.28204	2.00809774	-1.22421032	2	0	0	1
645	820.71796	2.21892058	-1.22421032	2	1	0	1
646	544.82050	2.21892058	-1.22421032	1	0	0	1
647	661.28204	2.34541428	-1.22421032	1	0	0	1
648	693.33331	2.34541428	-1.22421032	2	0	0	1
649	496.71793	2.34541428	-1.22421032	1	0	0	1
650	712.35895	2.42974341	-1.22421032	2	0	0	1
651	875.23077	2.55623711	-1.22421032	1	1	0	1
652	654.82050	2.64056624	-1.22421032	1	1	0	0
653	968.82050	2.64056624	-1.22421032	2	0	0	1
654	885.69232	2.72489538	-1.22421032	2	0	0	1
655	523.74359	2.89355364	-1.22421032	1	0	0	0
656	549.17950	3.02004734	-1.22421032	1	0	0	1
657	1149.53845	3.06221191	-1.22421032	2	1	0	1
658	721.94873	3.10437647	-1.22421032	1	0	0	1
659	655.43591	3.27303474	-1.22421032	2	1	0	1
660	562.20514	3.27303474	-1.22421032	1	0	0	1
661	516.66669	3.69468041	-1.22421032	1	1	0	1
662	155.58974	-0.47961168	-1.17937779	2	0	0	0
663	273.43591	-0.39528255	-1.17937779	1	1	0	1
664	500.35898	0.11069225	-1.17937779	2	0	0	1
665	378.10257	0.53233791	-1.17937779	1	0	0	1
666	816.00000	0.74316075	-1.17937779	2	0	0	1
667	809.53845	0.82748988	-1.17937779	1	0	1	1
668	275.38461	0.91181901	-1.17937779	1	0	0	1
669	428.00000	1.45995838	-1.17937779	1	0	0	1
670	512.35895	1.50212295	-1.17937779	1	0	0	1

671	134.35898	-1.32290301	-1.13454525	1	0	0	0
672	474.97437	-0.60610538	-1.13454525	1	1	0	1
673	565.89746	-0.14229515	-1.13454525	2	0	0	0
674	157.74359	0.11069225	-1.13454525	1	0	0	0
675	1075.94873	1.96593318	-1.13454525	2	1	1	1
676	272.76926	-0.85909278	-1.08971272	1	0	0	1
677	262.61539	-0.81692822	-1.08971272	1	0	0	1
678	258.41025	-0.69043452	-1.08971272	1	0	0	1
679	351.94870	-0.64826995	-1.08971272	1	0	0	1
680	278.10257	-0.47961168	-1.08971272	1	0	0	1
681	269.79486	-0.94342192	-1.04488019	1	0	0	1
682	417.69232	-0.35311798	-1.04488019	1	0	0	1
683	377.58975	-0.56394082	-1.00004766	2	0	0	1
684	342.82053	-0.52177625	-1.00004766	2	0	0	1
685	292.30768	-0.39528255	-1.00004766	1	0	0	0
686	223.64102	-0.14229515	-1.00004766	2	0	0	1
687	589.79486	0.23718595	-1.00004766	2	0	0	1
688	354.56412	0.86965445	-1.00004766	2	0	0	1
689	275.53845	0.86965445	-1.00004766	2	0	0	1
690	185.48718	-1.19640931	-0.95521513	1	0	0	1
691	306.25641	-1.15424475	-0.95521513	1	0	0	1
692	382.51285	-0.14229515	-0.95521513	1	0	0	1
693	799.48718	0.32151508	-0.95521513	1	0	0	1
694	398.20514	0.61666705	-0.95521513	1	0	0	1
695	713.28204	0.65883161	-0.95521513	2	0	0	1
696	377.53848	0.70099618	-0.95521513	3	0	0	0
697	510.41025	1.37562925	-0.95521513	2	0	0	1
698	55.23077	-0.98558648	-0.91038260	2	0	0	0
699	254.35898	-0.94342192	-0.91038260	1	0	0	1
700	304.10257	-0.52177625	-0.91038260	1	0	0	1
701	248.35896	-0.43744712	-0.91038260	1	0	0	1
702	516.00000	-0.39528255	-0.91038260	2	0	0	0
703	640.76923	-0.18445972	-0.91038260	2	0	0	1
704	459.58975	-0.10013058	-0.91038260	3	0	0	1
705	341.84613	-0.01580145	-0.91038260	2	0	0	0
706	584.00000	0.40584421	-0.91038260	1	0	0	1
707	407.07690	0.44800878	-0.91038260	2	0	0	1
708	319.94873	0.44800878	-0.91038260	2	0	0	1
709	392.56412	0.82748988	-0.91038260	2	0	0	1
710	368.15387	1.20697098	-0.91038260	3	0	0	1
711	119.58974	-1.02775105	-0.86555006	2	0	0	0
712	361.58972	-0.94342192	-0.86555006	1	0	0	1
713	360.30768	-0.60610538	-0.86555006	2	0	0	1

714	271.89743	-0.56394082	-0.86555006	1	0	0	1
715	406.61539	-0.39528255	-0.86555006	2	0	0	1
716	381.17947	-0.31095342	-0.86555006	1	0	0	1
717	577.17950	-0.26878885	-0.86555006	2	0	0	0
718	297.07690	-0.14229515	-0.86555006	2	0	0	1
719	626.05133	-0.05796602	-0.86555006	2	0	0	1
720	529.69232	0.36367965	-0.86555006	2	0	0	1
721	543.74359	0.36367965	-0.86555006	1	0	0	1
722	542.76923	0.65883161	-0.86555006	2	0	0	1
723	583.58972	0.78532531	-0.86555006	1	0	0	0
724	137.17949	-1.32290301	-0.82071753	1	0	0	0
725	302.15384	-1.15424475	-0.82071753	2	0	0	1
726	160.35898	-1.06991561	-0.82071753	1	0	0	1
727	203.94873	-0.85909278	-0.82071753	2	0	0	0
728	366.35898	-0.39528255	-0.82071753	1	0	0	1
729	283.94873	-0.35311798	-0.82071753	1	0	0	1
730	111.23077	-0.18445972	-0.82071753	1	0	0	0
731	303.69232	-0.18445972	-0.82071753	1	0	0	1
732	579.74359	0.27935051	-0.82071753	2	0	0	1
733	240.25641	-1.66021955	-0.77588500	1	0	0	1
734	293.23077	-1.57589041	-0.77588500	1	0	0	1
735	168.35896	-1.57589041	-0.77588500	1	0	0	0
736	270.92307	-1.53372585	-0.77588500	2	0	0	0
737	307.69232	-1.44939671	-0.77588500	1	0	0	1
738	175.23077	-1.44939671	-0.77588500	2	0	0	1
739	273.17950	-1.44939671	-0.77588500	1	0	0	1
740	120.97436	-1.36506758	-0.77588500	1	0	0	0
741	114.46154	-1.36506758	-0.77588500	1	0	0	0
742	299.17950	-1.36506758	-0.77588500	2	0	0	1
743	375.53845	-1.36506758	-0.77588500	1	0	0	1
744	205.28204	-1.32290301	-0.77588500	1	0	0	1
745	92.10257	-1.19640931	-0.77588500	1	0	0	0
746	250.56410	-1.19640931	-0.77588500	1	0	0	1
747	204.20514	-1.06991561	-0.77588500	1	0	1	1
748	347.33331	-1.06991561	-0.77588500	2	0	0	1
749	99.23077	-1.06991561	-0.77588500	2	0	0	1
750	387.12820	-1.02775105	-0.77588500	2	0	0	1
751	366.05127	-1.02775105	-0.77588500	1	0	0	1
752	478.82053	-0.98558648	-0.77588500	1	0	0	1
753	172.97435	-0.94342192	-0.77588500	2	0	0	1
754	203.74358	-0.94342192	-0.77588500	1	0	0	1
755	271.28204	-0.94342192	-0.77588500	1	0	0	1
756	347.38461	-0.81692822	-0.77588500	1	0	0	1

757	329.84616	-0.81692822	-0.77588500	2	0	0	1
758	279.02563	-0.81692822	-0.77588500	1	0	0	1
759	423.58975	-0.73259908	-0.77588500	1	0	0	1
760	522.05127	-0.64826995	-0.77588500	1	0	0	1
761	342.92310	-0.60610538	-0.77588500	2	0	0	0
762	408.92310	-0.60610538	-0.77588500	2	0	0	1
763	535.28210	-0.60610538	-0.77588500	1	0	0	1
764	539.17950	-0.56394082	-0.77588500	2	0	0	1
765	193.33333	-0.52177625	-0.77588500	2	0	0	1
766	402.35895	-0.52177625	-0.77588500	1	0	0	1
767	250.30769	-0.52177625	-0.77588500	2	0	1	1
768	269.23077	-0.52177625	-0.77588500	2	0	0	1
769	630.76923	-0.52177625	-0.77588500	2	0	0	1
770	507.79489	-0.47961168	-0.77588500	1	0	0	1
771	180.30769	-0.47961168	-0.77588500	1	0	0	1
772	340.10257	-0.43744712	-0.77588500	1	0	0	1
773	301.64102	-0.43744712	-0.77588500	1	0	0	0
774	382.25641	-0.39528255	-0.77588500	1	0	0	1
775	456.71793	-0.39528255	-0.77588500	2	0	0	1
776	435.84616	-0.39528255	-0.77588500	2	0	0	1
777	337.02563	-0.39528255	-0.77588500	2	0	0	0
778	246.05127	-0.35311798	-0.77588500	1	0	0	0
779	645.58978	-0.35311798	-0.77588500	3	0	0	1
780	509.12820	-0.31095342	-0.77588500	2	0	0	1
781	168.25641	-0.31095342	-0.77588500	1	0	0	1
782	594.41022	-0.31095342	-0.77588500	1	0	0	1
783	274.41025	-0.31095342	-0.77588500	2	0	0	1
784	122.82051	-0.31095342	-0.77588500	1	0	0	0
785	533.12817	-0.31095342	-0.77588500	2	0	0	1
786	118.61539	-0.31095342	-0.77588500	1	0	0	0
787	297.74359	-0.26878885	-0.77588500	1	0	0	1
788	324.92307	-0.22662428	-0.77588500	2	0	0	1
789	325.12820	-0.22662428	-0.77588500	1	0	0	1
790	263.48718	-0.22662428	-0.77588500	2	0	0	1
791	583.17944	-0.18445972	-0.77588500	2	0	0	1
792	317.48715	-0.18445972	-0.77588500	1	0	0	1
793	314.66666	-0.18445972	-0.77588500	2	0	0	1
794	506.56409	-0.10013058	-0.77588500	2	0	0	0
795	542.30768	-0.10013058	-0.77588500	2	0	0	1
796	354.20514	-0.10013058	-0.77588500	2	0	0	0
797	283.17950	-0.01580145	-0.77588500	2	0	0	1
798	319.38461	-0.01580145	-0.77588500	1	0	0	1
799	271.17947	-0.01580145	-0.77588500	2	0	0	1

800	485.89743	-0.01580145	-0.77588500	2	0	0	1
801	637.79486	-0.01580145	-0.77588500	1	0	0	0
802	557.07697	0.02636312	-0.77588500	3	0	0	1
803	324.41025	0.02636312	-0.77588500	2	0	0	0
804	377.58975	0.02636312	-0.77588500	1	0	0	1
805	186.82051	0.06852768	-0.77588500	2	0	0	1
806	247.94872	0.06852768	-0.77588500	1	0	0	1
807	620.30768	0.06852768	-0.77588500	2	0	0	0
808	418.61539	0.11069225	-0.77588500	1	0	0	1
809	626.56415	0.11069225	-0.77588500	2	0	0	1
810	467.02563	0.15285682	-0.77588500	2	0	0	1
811	607.28204	0.23718595	-0.77588500	2	0	0	1
812	613.48718	0.27935051	-0.77588500	2	0	0	0
813	271.07690	0.27935051	-0.77588500	1	0	0	1
814	537.28204	0.27935051	-0.77588500	1	0	0	1
815	489.58975	0.36367965	-0.77588500	1	0	0	1
816	218.56410	0.36367965	-0.77588500	1	0	0	1
817	501.12820	0.40584421	-0.77588500	1	0	0	1
818	189.07693	0.44800878	-0.77588500	1	0	0	1
819	517.53845	0.44800878	-0.77588500	1	0	0	1
820	371.58972	0.44800878	-0.77588500	2	0	0	1
821	285.53845	0.49017335	-0.77588500	1	0	0	1
822	856.05133	0.53233791	-0.77588500	2	0	0	0
823	371.89743	0.57450248	-0.77588500	2	0	0	1
824	453.28207	0.61666705	-0.77588500	1	0	0	1
825	377.43588	0.61666705	-0.77588500	1	0	0	1
826	337.53848	0.61666705	-0.77588500	1	0	0	1
827	398.35898	0.70099618	-0.77588500	1	0	0	1
828	461.02563	0.70099618	-0.77588500	1	1	0	1
829	264.97437	0.70099618	-0.77588500	1	0	0	1
830	890.97437	0.74316075	-0.77588500	1	0	1	1
831	365.53845	0.74316075	-0.77588500	2	0	0	0
832	375.53845	0.78532531	-0.77588500	1	0	0	1
833	137.74359	0.78532531	-0.77588500	2	0	0	1
834	467.38461	0.86965445	-0.77588500	1	1	0	1
835	438.20514	0.95398358	-0.77588500	2	0	0	1
836	299.58975	0.99614815	-0.77588500	1	0	0	0
837	506.76923	1.03831271	-0.77588500	1	0	0	1
838	1256.51282	1.08047728	-0.77588500	2	0	0	1
839	323.17950	1.08047728	-0.77588500	1	0	0	0
840	679.33331	1.12264185	-0.77588500	2	1	0	1
841	793.23077	1.16480641	-0.77588500	1	0	0	1
842	432.00000	1.20697098	-0.77588500	1	0	0	0

843	584.56409	1.24913555	-0.77588500	2	0	0	1
844	661.84613	1.33346468	-0.77588500	2	0	1	1
845	348.05130	1.37562925	-0.77588500	1	0	0	1
846	787.43591	1.37562925	-0.77588500	2	0	0	1
847	745.74359	1.50212295	-0.77588500	1	0	0	1
848	436.97433	1.54428751	-0.77588500	1	1	0	1
849	642.61536	1.62861664	-0.77588500	2	0	0	1
850	419.48718	1.71294578	-0.77588500	2	0	0	1
851	1075.64099	1.79727491	-0.77588500	1	1	0	1
852	845.33337	1.79727491	-0.77588500	2	0	0	1
853	575.74359	2.17675601	-0.77588500	2	0	0	0
854	570.61536	2.21892058	-0.77588500	1	0	0	1
855	646.82056	2.21892058	-0.77588500	3	0	0	1
856	1234.71790	2.72489538	-0.77588500	2	0	1	1
857	404.46155	2.93571821	-0.77588500	1	1	0	1
858	462.61539	3.06221191	-0.77588500	2	1	0	1
859	273.69232	-1.06991561	-0.73105247	1	0	0	1
860	142.82051	-0.69043452	-0.73105247	2	0	0	0
861	300.71796	-0.43744712	-0.73105247	1	0	0	1
862	307.23074	-0.35311798	-0.73105247	1	0	0	1
863	571.12817	0.11069225	-0.73105247	2	0	0	1
864	362.61539	0.32151508	-0.73105247	1	0	0	1
865	745.79492	1.20697098	-0.73105247	2	0	0	1
866	485.12820	-0.94342192	-0.68621994	1	0	0	1
867	304.25641	-0.94342192	-0.68621994	1	0	0	1
868	374.76923	0.53233791	-0.68621994	1	0	0	1
869	503.53848	0.61666705	-0.68621994	1	0	0	1
870	367.84616	0.61666705	-0.68621994	1	0	0	1
871	612.25641	1.08047728	-0.68621994	1	0	0	1
872	449.69232	1.20697098	-0.68621994	1	0	0	1
873	269.48718	-0.35311798	-0.64138741	1	0	0	1
874	221.02563	-0.90125735	-0.37239222	1	0	0	1
875	214.56410	-0.77476365	-0.37239222	1	0	0	1
876	652.56409	0.27935051	-0.37239222	3	0	1	1
877	77.07693	-1.19640931	-0.32755968	1	0	0	0
878	538.56409	1.50212295	-0.32755968	1	0	0	1
879	285.58975	-1.66021955	-0.28272715	1	0	0	1
880	256.71796	-1.36506758	-0.28272715	1	0	0	1
881	445.33334	-0.81692822	-0.28272715	2	0	0	1
882	421.79486	-0.77476365	-0.28272715	1	0	0	1
883	256.41025	-0.56394082	-0.28272715	2	0	0	0
884	289.69232	-0.31095342	-0.28272715	1	0	0	0
885	419.74359	-0.18445972	-0.28272715	1	0	0	1

886	445.84616	0.78532531	-0.28272715	1	0	0	0
887	336.92307	0.95398358	-0.28272715	1	0	0	0
888	232.20512	-1.11208018	-0.23789462	2	0	0	1
889	387.89746	-0.18445972	-0.23789462	1	0	0	1
890	484.41025	-0.05796602	-0.23789462	1	0	0	1
891	302.25641	0.11069225	-0.23789462	1	0	0	0
892	346.76923	0.19502138	-0.23789462	2	0	0	1
893	442.56412	0.86965445	-0.23789462	1	1	0	1
894	262.56412	-1.49156128	-0.19306209	2	0	0	1
895	320.56409	-1.40723215	-0.19306209	1	0	0	1
896	372.66669	-1.15424475	-0.19306209	2	1	0	1
897	345.38461	-1.06991561	-0.19306209	1	0	0	1
898	202.00000	-1.06991561	-0.19306209	1	0	0	0
899	330.05127	-1.02775105	-0.19306209	1	0	0	1
900	442.05127	-0.90125735	-0.19306209	2	0	1	1
901	306.25641	-0.85909278	-0.19306209	1	0	0	1
902	152.46153	-0.35311798	-0.19306209	1	0	0	0
903	148.51282	-0.31095342	-0.19306209	1	0	0	1
904	266.25641	-0.31095342	-0.19306209	2	0	0	0
905	293.53848	-0.14229515	-0.19306209	1	0	0	1
906	350.05127	-0.05796602	-0.19306209	2	0	0	1
907	200.41025	0.02636312	-0.19306209	3	0	0	1
908	281.64102	-1.74454868	-0.14822956	2	0	0	1
909	344.51282	-1.36506758	-0.14822956	1	0	0	1
910	325.28204	-1.28073845	-0.14822956	2	0	0	1
911	332.76926	-1.28073845	-0.14822956	1	0	0	1
912	378.71796	-0.81692822	-0.14822956	1	0	0	1
913	130.82051	-0.77476365	-0.14822956	1	0	0	1
914	434.41025	-0.39528255	-0.14822956	2	0	0	1
915	360.82050	-0.22662428	-0.14822956	2	0	0	0
916	319.28204	0.19502138	-0.14822956	2	0	0	0
917	173.12820	0.23718595	-0.14822956	2	0	0	1
918	448.05130	0.61666705	-0.14822956	1	0	0	1
919	475.69229	0.78532531	-0.14822956	1	0	0	1
920	444.41025	1.08047728	-0.14822956	1	0	0	1
921	721.23077	1.62861664	-0.14822956	1	0	0	1
922	229.02565	-1.28073845	-0.10339703	1	0	0	1
923	203.43590	-1.02775105	-0.10339703	1	0	0	1
924	132.61539	-0.90125735	-0.10339703	1	0	0	0
925	211.02563	-0.73259908	-0.10339703	2	0	0	1
926	249.74359	-0.69043452	-0.10339703	1	0	0	1
927	309.84616	-0.47961168	-0.10339703	2	0	0	1
928	164.20514	-0.43744712	-0.10339703	1	0	0	1

929	283.89743	-0.39528255	-0.10339703	1	0	0	0
930	423.94873	-0.35311798	-0.10339703	1	0	0	1
931	427.64105	-0.31095342	-0.10339703	2	0	0	0
932	236.05127	-0.31095342	-0.10339703	1	0	0	1
933	417.58975	-0.31095342	-0.10339703	1	0	0	1
934	448.46155	-0.26878885	-0.10339703	1	0	0	1
935	485.07693	-0.26878885	-0.10339703	1	0	0	1
936	441.64102	-0.26878885	-0.10339703	1	0	0	1
937	348.15387	-0.22662428	-0.10339703	2	0	0	1
938	340.82050	-0.18445972	-0.10339703	1	0	0	1
939	245.28204	-0.10013058	-0.10339703	2	0	0	1
940	384.82053	-0.10013058	-0.10339703	2	0	0	1
941	463.69232	-0.05796602	-0.10339703	1	0	0	1
942	416.87180	0.02636312	-0.10339703	1	0	0	1
943	531.38458	0.19502138	-0.10339703	2	0	0	1
944	589.79486	0.19502138	-0.10339703	2	0	0	1
945	303.23077	0.19502138	-0.10339703	1	0	0	0
946	270.00000	0.23718595	-0.10339703	2	0	0	1
947	394.76923	0.23718595	-0.10339703	2	0	0	0
948	434.41025	0.36367965	-0.10339703	2	0	0	1
949	728.82050	0.44800878	-0.10339703	1	0	0	1
950	818.97437	0.53233791	-0.10339703	2	1	0	1
951	374.82053	0.53233791	-0.10339703	1	0	0	1
952	429.53845	0.61666705	-0.10339703	1	0	0	1
953	464.46155	0.61666705	-0.10339703	1	0	0	1
954	456.35898	0.65883161	-0.10339703	1	0	0	1
955	440.46155	0.78532531	-0.10339703	1	0	0	1
956	897.07697	0.91181901	-0.10339703	1	0	0	1
957	811.53845	1.37562925	-0.10339703	2	0	0	1
958	568.46155	1.45995838	-0.10339703	1	1	0	1
959	577.74359	1.62861664	-0.10339703	1	0	0	1
960	460.51282	1.79727491	-0.10339703	1	1	0	1
961	552.46155	1.83943948	-0.10339703	1	1	0	1
962	582.00000	1.83943948	-0.10339703	1	1	0	1
963	442.92310	3.48385757	-0.10339703	2	0	0	1
964	143.43590	-1.49156128	-0.05856449	1	0	0	1
965	205.23077	-1.32290301	-0.05856449	2	0	0	1
966	251.58975	-1.23857388	-0.05856449	1	0	0	0
967	202.92308	-1.11208018	-0.05856449	1	0	0	1
968	346.05127	-1.02775105	-0.05856449	2	0	0	1
969	91.02564	-0.98558648	-0.05856449	1	0	0	1
970	540.30768	-0.81692822	-0.05856449	1	0	0	1
971	184.15385	-0.69043452	-0.05856449	1	0	0	1

972	161.38463	-0.56394082	-0.05856449	1	0	0	0
973	417.69232	-0.52177625	-0.05856449	1	0	1	1
974	292.15384	-0.52177625	-0.05856449	1	0	0	0
975	391.38461	-0.47961168	-0.05856449	1	0	0	1
976	388.66669	-0.43744712	-0.05856449	1	0	0	1
977	285.84616	-0.39528255	-0.05856449	1	0	0	1
978	311.69229	-0.35311798	-0.05856449	1	0	0	0
979	498.00000	-0.31095342	-0.05856449	1	0	0	1
980	259.23077	-0.22662428	-0.05856449	1	0	0	1
981	270.30768	0.02636312	-0.05856449	1	0	0	1
982	271.02563	0.06852768	-0.05856449	2	0	0	1
983	547.28204	0.06852768	-0.05856449	1	0	0	1
984	532.76923	0.11069225	-0.05856449	1	0	1	1
985	476.10257	0.19502138	-0.05856449	1	0	0	1
986	390.76923	0.23718595	-0.05856449	1	0	0	1
987	311.74359	0.27935051	-0.05856449	1	0	0	1
988	659.94873	0.32151508	-0.05856449	2	0	0	1
989	662.35895	0.32151508	-0.05856449	2	0	0	1
990	510.00000	0.36367965	-0.05856449	2	0	0	1
991	544.56409	0.36367965	-0.05856449	1	0	0	1
992	623.12817	0.44800878	-0.05856449	1	0	1	1
993	605.38464	0.44800878	-0.05856449	2	0	0	1
994	718.10260	0.44800878	-0.05856449	1	0	0	1
995	491.64102	0.49017335	-0.05856449	2	0	0	1
996	304.82053	0.53233791	-0.05856449	1	0	0	1
997	712.46155	0.57450248	-0.05856449	3	0	0	1
998	409.89743	0.86965445	-0.05856449	2	0	0	1
999	450.51282	0.95398358	-0.05856449	2	0	0	1
1000	479.48718	1.03831271	-0.05856449	1	1	0	1
1001	421.33331	1.03831271	-0.05856449	1	1	0	1
1002	415.74359	1.08047728	-0.05856449	1	1	0	1
1003	405.89743	1.08047728	-0.05856449	1	1	0	1
1004	624.15381	1.08047728	-0.05856449	1	0	0	1
1005	410.20514	1.08047728	-0.05856449	1	1	0	1
1006	583.48718	1.16480641	-0.05856449	2	0	0	1
1007	818.92310	1.29130011	-0.05856449	1	0	0	1
1008	749.53845	1.29130011	-0.05856449	1	0	0	1
1009	557.02563	1.50212295	-0.05856449	1	1	0	1
1010	472.05127	1.71294578	-0.05856449	1	1	0	1
1011	453.58975	1.71294578	-0.05856449	1	0	0	1
1012	455.43588	1.71294578	-0.05856449	1	1	0	1
1013	469.07693	1.79727491	-0.05856449	1	1	0	1
1014	691.79486	2.00809774	-0.05856449	2	0	0	1

1015	1163.48718	2.00809774	-0.05856449	3	1	0	1
1016	1210.92310	2.00809774	-0.05856449	3	0	0	1
1017	175.64102	-1.99753608	-0.01373196	1	0	0	1
1018	234.25641	-1.70238411	-0.01373196	1	0	0	1
1019	294.20514	-1.61805498	-0.01373196	2	0	0	1
1020	208.10255	-1.57589041	-0.01373196	2	0	0	1
1021	222.56410	-1.57589041	-0.01373196	1	0	0	1
1022	289.84616	-1.57589041	-0.01373196	1	0	0	1
1023	253.53845	-1.44939671	-0.01373196	2	0	0	1
1024	228.56410	-1.36506758	-0.01373196	1	0	0	1
1025	455.12820	-1.23857388	-0.01373196	1	0	0	1
1026	321.84613	-1.23857388	-0.01373196	1	0	0	1
1027	377.38461	-1.11208018	-0.01373196	2	0	0	1
1028	273.53848	-1.06991561	-0.01373196	1	0	0	1
1029	322.10254	-1.02775105	-0.01373196	2	0	0	1
1030	257.02563	-0.94342192	-0.01373196	1	0	0	1
1031	131.89745	-0.81692822	-0.01373196	1	0	0	0
1032	340.71796	-0.73259908	-0.01373196	1	0	0	0
1033	483.94873	-0.69043452	-0.01373196	2	0	0	1
1034	336.97433	-0.64826995	-0.01373196	1	0	0	0
1035	462.25641	-0.60610538	-0.01373196	1	0	0	1
1036	344.05130	-0.56394082	-0.01373196	1	0	0	1
1037	349.43591	-0.52177625	-0.01373196	2	0	0	1
1038	178.82053	-0.47961168	-0.01373196	1	0	0	0
1039	547.02563	-0.47961168	-0.01373196	2	0	0	0
1040	183.58974	-0.43744712	-0.01373196	2	0	0	1
1041	322.41025	-0.39528255	-0.01373196	2	0	0	1
1042	295.94870	-0.39528255	-0.01373196	1	0	0	1
1043	493.84616	-0.35311798	-0.01373196	1	0	0	1
1044	427.02563	-0.31095342	-0.01373196	1	0	0	1
1045	380.41025	-0.31095342	-0.01373196	2	0	0	1
1046	219.64102	-0.31095342	-0.01373196	1	0	0	0
1047	257.64102	-0.31095342	-0.01373196	1	0	0	0
1048	617.84619	-0.22662428	-0.01373196	1	0	0	0
1049	380.82050	-0.18445972	-0.01373196	2	0	0	1
1050	644.20508	-0.10013058	-0.01373196	2	0	0	1
1051	753.28204	-0.10013058	-0.01373196	1	0	0	1
1052	173.12820	-0.10013058	-0.01373196	2	0	0	1
1053	512.97437	-0.10013058	-0.01373196	2	0	0	1
1054	439.79486	0.02636312	-0.01373196	1	0	0	1
1055	389.64102	0.06852768	-0.01373196	2	0	0	1
1056	244.00000	0.11069225	-0.01373196	2	0	0	1
1057	197.94872	0.11069225	-0.01373196	1	0	0	0

1058	362.46152	0.15285682	-0.01373196	2	0	0	1
1059	231.07692	0.15285682	-0.01373196	3	0	0	1
1060	304.82053	0.32151508	-0.01373196	2	0	0	1
1061	633.69226	0.36367965	-0.01373196	1	0	0	1
1062	479.07693	0.40584421	-0.01373196	2	0	0	1
1063	393.53848	0.49017335	-0.01373196	1	0	0	1
1064	487.23074	0.49017335	-0.01373196	1	0	0	1
1065	410.10257	0.70099618	-0.01373196	2	0	0	1
1066	664.30768	0.95398358	-0.01373196	1	0	0	1
1067	417.84616	1.71294578	-0.01373196	2	0	0	1
1068	310.20514	-1.87104238	0.03110057	2	0	0	1
1069	218.05128	-1.66021955	0.03110057	1	0	0	1
1070	264.97437	-1.61805498	0.03110057	1	0	0	1
1071	252.71794	-1.40723215	0.03110057	1	0	0	1
1072	428.05130	-1.40723215	0.03110057	2	0	0	1
1073	263.12820	-1.40723215	0.03110057	1	0	0	1
1074	369.84616	-1.36506758	0.03110057	2	0	0	1
1075	243.84616	-1.28073845	0.03110057	2	0	0	1
1076	314.35898	-1.19640931	0.03110057	1	0	0	1
1077	233.74358	-1.06991561	0.03110057	1	0	0	1
1078	386.00000	-0.81692822	0.03110057	1	0	0	1
1079	144.56410	-0.73259908	0.03110057	2	0	0	1
1080	412.61539	-0.69043452	0.03110057	2	0	0	1
1081	224.51282	-0.64826995	0.03110057	1	0	0	1
1082	362.05127	-0.56394082	0.03110057	1	0	0	1
1083	476.20511	-0.35311798	0.03110057	2	0	0	1
1084	330.20514	-0.31095342	0.03110057	1	0	0	1
1085	350.76923	-0.14229515	0.03110057	1	0	0	1
1086	676.87183	-0.14229515	0.03110057	1	0	0	1
1087	399.28204	-0.10013058	0.03110057	1	0	0	1
1088	390.35898	-0.10013058	0.03110057	1	0	0	1
1089	256.61539	-0.10013058	0.03110057	1	0	0	1
1090	221.89745	0.02636312	0.03110057	1	0	0	1
1091	395.38461	0.06852768	0.03110057	1	0	0	1
1092	636.41028	0.27935051	0.03110057	1	0	0	1
1093	643.33331	0.40584421	0.03110057	2	0	0	1
1094	646.00000	0.44800878	0.03110057	1	0	0	1
1095	348.76923	0.49017335	0.03110057	1	0	0	1
1096	426.46152	0.49017335	0.03110057	1	0	0	1
1097	417.33331	0.70099618	0.03110057	1	0	0	1
1098	395.74359	0.78532531	0.03110057	1	0	0	1
1099	449.69232	0.86965445	0.03110057	1	0	0	1
1100	418.10257	1.20697098	0.03110057	1	0	0	1

1101	544.15381	1.41779381	0.03110057	1	0	0	1
1102	686.82056	1.54428751	0.03110057	2	0	0	1
1103	553.53845	2.47190798	0.03110057	1	0	0	1
1104	289.43591	-1.99753608	0.05351684	2	0	0	1
1105	299.48718	-1.78671325	0.05351684	2	1	0	1
1106	365.12820	-1.74454868	0.05351684	1	0	0	1
1107	295.17947	-1.70238411	0.05351684	1	0	0	1
1108	188.41025	-1.66021955	0.05351684	1	0	0	1
1109	336.97433	-1.66021955	0.05351684	1	0	0	1
1110	296.35898	-1.66021955	0.05351684	2	0	0	1
1111	294.87180	-1.66021955	0.05351684	1	0	0	1
1112	412.71796	-1.61805498	0.05351684	2	0	0	1
1113	296.66666	-1.61805498	0.05351684	1	0	0	1
1114	286.51282	-1.61805498	0.05351684	2	0	0	1
1115	247.17949	-1.57589041	0.05351684	2	0	0	1
1116	245.38461	-1.57589041	0.05351684	1	0	0	1
1117	324.41025	-1.57589041	0.05351684	2	0	0	1
1118	242.05128	-1.57589041	0.05351684	2	0	0	1
1119	208.41025	-1.53372585	0.05351684	1	0	0	1
1120	429.79486	-1.53372585	0.05351684	2	1	0	1
1121	315.28204	-1.49156128	0.05351684	2	0	0	1
1122	406.51282	-1.49156128	0.05351684	2	0	0	1
1123	260.05130	-1.49156128	0.05351684	2	0	0	1
1124	493.69232	-1.49156128	0.05351684	1	0	0	1
1125	253.38461	-1.44939671	0.05351684	2	0	0	1
1126	235.94872	-1.44939671	0.05351684	1	0	0	1
1127	343.53848	-1.40723215	0.05351684	1	0	0	1
1128	160.35898	-1.40723215	0.05351684	2	0	0	1
1129	381.94870	-1.36506758	0.05351684	2	0	0	1
1130	394.30771	-1.36506758	0.05351684	1	0	0	0
1131	187.38461	-1.36506758	0.05351684	2	0	0	1
1132	367.28204	-1.36506758	0.05351684	1	0	0	1
1133	334.76923	-1.36506758	0.05351684	2	0	0	1
1134	322.00000	-1.36506758	0.05351684	2	0	0	1
1135	178.41025	-1.32290301	0.05351684	2	0	0	0
1136	289.48718	-1.32290301	0.05351684	2	1	0	1
1137	331.48718	-1.28073845	0.05351684	1	0	0	1
1138	256.92307	-1.28073845	0.05351684	2	0	0	1
1139	469.28204	-1.28073845	0.05351684	1	0	0	1
1140	351.64102	-1.28073845	0.05351684	2	0	0	1
1141	243.33333	-1.28073845	0.05351684	1	0	0	1
1142	256.56409	-1.28073845	0.05351684	1	0	0	1
1143	378.30771	-1.23857388	0.05351684	1	0	0	1

1144	105.74359	-1.23857388	0.05351684	1	0	0	1
1145	284.71796	-1.23857388	0.05351684	2	0	0	1
1146	335.79486	-1.23857388	0.05351684	1	0	0	1
1147	247.74359	-1.19640931	0.05351684	2	0	0	1
1148	270.41025	-1.19640931	0.05351684	2	0	0	1
1149	306.30768	-1.19640931	0.05351684	2	0	0	1
1150	331.53845	-1.19640931	0.05351684	2	0	0	1
1151	456.25641	-1.19640931	0.05351684	2	0	0	1
1152	178.30769	-1.19640931	0.05351684	1	0	0	1
1153	431.53845	-1.19640931	0.05351684	2	0	1	1
1154	283.84616	-1.15424475	0.05351684	1	0	0	1
1155	443.69232	-1.15424475	0.05351684	1	0	0	1
1156	320.35898	-1.15424475	0.05351684	1	0	0	1
1157	224.15385	-1.15424475	0.05351684	1	0	0	1
1158	270.05127	-1.11208018	0.05351684	2	0	0	0
1159	356.61539	-1.06991561	0.05351684	2	0	0	1
1160	300.97437	-1.06991561	0.05351684	1	0	0	1
1161	392.05127	-1.02775105	0.05351684	1	0	0	1
1162	187.33333	-1.02775105	0.05351684	2	0	0	1
1163	416.76923	-1.02775105	0.05351684	2	0	0	1
1164	474.92307	-0.98558648	0.05351684	1	0	0	1
1165	439.89743	-0.98558648	0.05351684	2	0	0	1
1166	230.35898	-0.98558648	0.05351684	1	0	0	1
1167	425.17947	-0.98558648	0.05351684	2	0	0	1
1168	214.51282	-0.94342192	0.05351684	1	0	0	1
1169	149.74359	-0.94342192	0.05351684	2	0	0	0
1170	451.17947	-0.94342192	0.05351684	2	0	0	1
1171	276.15384	-0.90125735	0.05351684	2	0	0	1
1172	382.30768	-0.90125735	0.05351684	1	0	0	1
1173	299.43591	-0.90125735	0.05351684	1	0	0	1
1174	313.33334	-0.85909278	0.05351684	1	0	0	1
1175	210.61539	-0.85909278	0.05351684	1	0	0	1
1176	398.20514	-0.81692822	0.05351684	2	0	0	1
1177	237.07692	-0.81692822	0.05351684	1	0	0	1
1178	395.64102	-0.81692822	0.05351684	1	0	0	1
1179	334.51282	-0.81692822	0.05351684	2	0	0	1
1180	436.56409	-0.81692822	0.05351684	2	0	0	1
1181	339.43591	-0.77476365	0.05351684	2	0	0	1
1182	298.20514	-0.77476365	0.05351684	2	0	0	1
1183	284.82053	-0.77476365	0.05351684	1	0	0	1
1184	217.69231	-0.77476365	0.05351684	2	0	0	1
1185	318.46155	-0.77476365	0.05351684	2	0	0	0
1186	491.07690	-0.77476365	0.05351684	1	0	0	1

1187	362.41025	-0.77476365	0.05351684	1	0	0	1
1188	452.00000	-0.73259908	0.05351684	2	0	0	1
1189	373.84616	-0.73259908	0.05351684	2	0	0	1
1190	349.64102	-0.73259908	0.05351684	1	0	0	1
1191	218.46153	-0.73259908	0.05351684	2	0	0	1
1192	260.46155	-0.73259908	0.05351684	2	0	0	1
1193	265.89743	-0.73259908	0.05351684	2	0	0	1
1194	465.38461	-0.73259908	0.05351684	2	0	0	1
1195	425.38461	-0.73259908	0.05351684	2	0	0	1
1196	312.76926	-0.73259908	0.05351684	2	0	0	0
1197	276.35898	-0.73259908	0.05351684	2	0	0	1
1198	274.20514	-0.69043452	0.05351684	1	0	0	0
1199	337.74359	-0.64826995	0.05351684	2	0	0	1
1200	398.71796	-0.64826995	0.05351684	2	0	0	1
1201	428.30771	-0.64826995	0.05351684	2	1	0	1
1202	308.82053	-0.64826995	0.05351684	1	0	0	1
1203	275.43588	-0.64826995	0.05351684	2	0	0	1
1204	252.56410	-0.64826995	0.05351684	1	0	0	1
1205	267.79489	-0.64826995	0.05351684	2	0	0	1
1206	235.69231	-0.60610538	0.05351684	1	0	0	1
1207	177.64102	-0.60610538	0.05351684	1	0	0	0
1208	280.97437	-0.60610538	0.05351684	2	0	0	0
1209	480.87180	-0.60610538	0.05351684	2	0	0	1
1210	629.33331	-0.60610538	0.05351684	2	0	0	1
1211	532.76923	-0.56394082	0.05351684	2	0	0	1
1212	588.05127	-0.56394082	0.05351684	2	0	0	1
1213	381.94870	-0.56394082	0.05351684	2	0	0	0
1214	460.05127	-0.56394082	0.05351684	2	0	0	1
1215	468.56412	-0.56394082	0.05351684	2	0	0	1
1216	225.28204	-0.56394082	0.05351684	2	0	0	1
1217	381.17947	-0.56394082	0.05351684	2	0	0	1
1218	187.94872	-0.56394082	0.05351684	2	0	0	1
1219	551.53845	-0.52177625	0.05351684	2	0	0	1
1220	407.94873	-0.52177625	0.05351684	1	0	0	1
1221	478.66669	-0.52177625	0.05351684	1	0	0	1
1222	484.46155	-0.52177625	0.05351684	1	0	0	1
1223	380.61539	-0.52177625	0.05351684	2	0	0	1
1224	286.46152	-0.52177625	0.05351684	1	0	0	1
1225	241.58975	-0.52177625	0.05351684	2	0	0	0
1226	507.12820	-0.47961168	0.05351684	1	0	0	0
1227	482.41025	-0.47961168	0.05351684	2	0	0	1
1228	382.87180	-0.47961168	0.05351684	2	0	0	1
1229	374.71796	-0.47961168	0.05351684	2	0	0	1

1230	509.43591	-0.47961168	0.05351684	1	0	0	1
1231	283.69232	-0.47961168	0.05351684	2	0	0	1
1232	420.97437	-0.47961168	0.05351684	2	0	0	1
1233	390.00000	-0.47961168	0.05351684	1	0	0	1
1234	405.58975	-0.43744712	0.05351684	1	0	1	1
1235	385.12820	-0.43744712	0.05351684	2	0	0	0
1236	477.43588	-0.43744712	0.05351684	2	0	0	1
1237	200.15384	-0.43744712	0.05351684	2	0	0	1
1238	302.82053	-0.43744712	0.05351684	1	0	0	0
1239	574.05127	-0.39528255	0.05351684	2	0	0	1
1240	301.17947	-0.39528255	0.05351684	1	0	0	0
1241	294.05130	-0.39528255	0.05351684	2	0	0	1
1242	344.00000	-0.39528255	0.05351684	1	0	0	1
1243	542.66663	-0.39528255	0.05351684	2	0	0	1
1244	578.66669	-0.39528255	0.05351684	2	0	0	1
1245	439.53845	-0.39528255	0.05351684	2	0	0	1
1246	201.48718	-0.35311798	0.05351684	1	0	0	1
1247	367.53848	-0.35311798	0.05351684	2	0	0	1
1248	314.66666	-0.35311798	0.05351684	1	0	0	0
1249	433.28207	-0.31095342	0.05351684	2	0	0	1
1250	384.51282	-0.31095342	0.05351684	2	0	0	1
1251	266.41025	-0.31095342	0.05351684	2	0	0	1
1252	434.00000	-0.31095342	0.05351684	2	0	1	1
1253	366.00000	-0.31095342	0.05351684	2	0	0	1
1254	299.69232	-0.31095342	0.05351684	1	0	0	0
1255	231.33334	-0.31095342	0.05351684	1	0	1	1
1256	370.71796	-0.31095342	0.05351684	1	0	0	1
1257	562.61536	-0.31095342	0.05351684	2	0	0	1
1258	398.41028	-0.31095342	0.05351684	2	0	0	1
1259	425.02563	-0.31095342	0.05351684	2	0	0	1
1260	427.53848	-0.31095342	0.05351684	2	0	0	0
1261	361.12820	-0.31095342	0.05351684	2	0	0	1
1262	478.15387	-0.31095342	0.05351684	1	0	0	1
1263	303.64102	-0.31095342	0.05351684	2	0	0	1
1264	370.97437	-0.26878885	0.05351684	2	1	0	1
1265	322.41025	-0.26878885	0.05351684	2	0	0	1
1266	358.66669	-0.26878885	0.05351684	1	0	0	1
1267	604.76923	-0.26878885	0.05351684	2	0	0	1
1268	366.92307	-0.26878885	0.05351684	2	0	0	1
1269	426.92307	-0.26878885	0.05351684	2	0	0	1
1270	325.79486	-0.26878885	0.05351684	2	0	0	1
1271	580.76923	-0.22662428	0.05351684	1	0	0	1
1272	365.79486	-0.22662428	0.05351684	2	0	0	1

1273	649.17950	-0.22662428	0.05351684	2	0	0	1
1274	463.28207	-0.22662428	0.05351684	2	0	0	1
1275	617.02563	-0.22662428	0.05351684	2	0	0	1
1276	528.25641	-0.22662428	0.05351684	1	0	0	1
1277	594.00000	-0.22662428	0.05351684	2	0	0	0
1278	472.87180	-0.22662428	0.05351684	1	0	0	1
1279	572.76923	-0.22662428	0.05351684	1	0	0	1
1280	666.66669	-0.18445972	0.05351684	2	0	0	1
1281	380.15384	-0.18445972	0.05351684	1	0	0	1
1282	363.48718	-0.18445972	0.05351684	2	0	0	1
1283	501.58972	-0.18445972	0.05351684	2	0	0	1
1284	325.84616	-0.18445972	0.05351684	2	0	0	1
1285	221.02563	-0.14229515	0.05351684	1	0	0	1
1286	377.33331	-0.14229515	0.05351684	2	0	0	1
1287	311.48718	-0.14229515	0.05351684	2	0	0	0
1288	248.61537	-0.14229515	0.05351684	1	0	0	1
1289	360.46155	-0.14229515	0.05351684	1	0	0	1
1290	464.66666	-0.10013058	0.05351684	1	0	0	1
1291	590.41028	-0.10013058	0.05351684	2	0	0	1
1292	468.00000	-0.10013058	0.05351684	2	0	0	1
1293	211.58975	-0.10013058	0.05351684	1	0	0	0
1294	292.46152	-0.10013058	0.05351684	2	0	0	1
1295	200.05128	-0.10013058	0.05351684	2	0	0	0
1296	359.43591	-0.10013058	0.05351684	1	0	0	1
1297	398.87180	-0.10013058	0.05351684	1	0	0	1
1298	367.74359	-0.10013058	0.05351684	1	0	0	1
1299	227.53847	-0.10013058	0.05351684	2	0	0	1
1300	317.48715	-0.10013058	0.05351684	1	0	0	1
1301	520.66669	-0.10013058	0.05351684	2	0	0	1
1302	490.66666	-0.05796602	0.05351684	2	0	0	0
1303	383.23077	-0.05796602	0.05351684	1	0	0	1
1304	551.33331	-0.05796602	0.05351684	1	0	0	1
1305	354.82053	-0.05796602	0.05351684	2	0	0	1
1306	407.07690	-0.01580145	0.05351684	2	0	0	1
1307	267.43588	-0.01580145	0.05351684	2	0	0	0
1308	527.33337	-0.01580145	0.05351684	2	0	0	1
1309	295.69229	-0.01580145	0.05351684	2	0	0	1
1310	471.48718	-0.01580145	0.05351684	2	0	0	1
1311	490.25641	-0.01580145	0.05351684	1	0	0	1
1312	794.15381	0.02636312	0.05351684	2	0	1	1
1313	588.71796	0.02636312	0.05351684	2	0	0	1
1314	204.66667	0.02636312	0.05351684	1	0	0	1
1315	406.76923	0.02636312	0.05351684	2	0	0	0

1316	362.20511	0.02636312	0.05351684	2	0	0	1
1317	487.58975	0.02636312	0.05351684	2	0	0	1
1318	313.17950	0.02636312	0.05351684	2	0	0	1
1319	544.15381	0.02636312	0.05351684	1	0	0	1
1320	513.48718	0.02636312	0.05351684	2	0	0	1
1321	551.07690	0.02636312	0.05351684	2	0	0	1
1322	501.94870	0.06852768	0.05351684	1	0	0	1
1323	297.69232	0.06852768	0.05351684	2	0	0	1
1324	188.51282	0.11069225	0.05351684	1	0	0	1
1325	542.20514	0.11069225	0.05351684	2	0	0	1
1326	387.74359	0.11069225	0.05351684	1	0	0	1
1327	472.56412	0.11069225	0.05351684	2	0	0	1
1328	351.43588	0.11069225	0.05351684	2	0	0	1
1329	368.51282	0.11069225	0.05351684	2	0	0	1
1330	151.07692	0.11069225	0.05351684	2	0	0	0
1331	654.10254	0.11069225	0.05351684	2	0	0	1
1332	349.12820	0.11069225	0.05351684	1	0	0	1
1333	532.46155	0.11069225	0.05351684	1	0	0	1
1334	545.28210	0.11069225	0.05351684	1	0	0	1
1335	696.41028	0.11069225	0.05351684	2	0	1	1
1336	541.28204	0.11069225	0.05351684	1	0	0	1
1337	510.87180	0.11069225	0.05351684	2	0	0	1
1338	383.58975	0.11069225	0.05351684	1	0	0	0
1339	519.07690	0.11069225	0.05351684	1	0	0	1
1340	391.02563	0.15285682	0.05351684	2	0	0	0
1341	344.15384	0.15285682	0.05351684	1	0	0	1
1342	143.84616	0.15285682	0.05351684	2	0	0	1
1343	237.94872	0.19502138	0.05351684	2	0	0	0
1344	506.82050	0.19502138	0.05351684	1	0	0	1
1345	360.71796	0.19502138	0.05351684	2	0	0	1
1346	501.64102	0.23718595	0.05351684	2	0	0	1
1347	384.10257	0.23718595	0.05351684	2	0	0	0
1348	424.46155	0.23718595	0.05351684	1	0	0	1
1349	414.92307	0.27935051	0.05351684	1	0	0	1
1350	509.94873	0.27935051	0.05351684	2	0	0	0
1351	655.33337	0.27935051	0.05351684	2	0	0	1
1352	367.58975	0.27935051	0.05351684	2	0	0	1
1353	410.51282	0.32151508	0.05351684	2	0	0	1
1354	419.17950	0.32151508	0.05351684	2	0	0	1
1355	419.17950	0.32151508	0.05351684	1	0	0	1
1356	279.02563	0.32151508	0.05351684	1	0	0	1
1357	494.20514	0.32151508	0.05351684	2	0	0	1
1358	335.89743	0.32151508	0.05351684	2	0	0	1

1359	462.76926	0.32151508	0.05351684	2	0	0	1
1360	416.41025	0.32151508	0.05351684	1	0	0	1
1361	314.87180	0.32151508	0.05351684	2	0	0	0
1362	497.48715	0.32151508	0.05351684	2	0	0	1
1363	593.79486	0.32151508	0.05351684	1	0	0	1
1364	619.79486	0.32151508	0.05351684	2	0	0	1
1365	357.38461	0.32151508	0.05351684	2	0	0	1
1366	704.05127	0.32151508	0.05351684	2	0	0	1
1367	446.66666	0.32151508	0.05351684	1	0	0	1
1368	333.89743	0.32151508	0.05351684	1	0	0	1
1369	537.17950	0.36367965	0.05351684	2	0	0	1
1370	428.00000	0.36367965	0.05351684	2	0	0	1
1371	626.10260	0.40584421	0.05351684	2	0	0	1
1372	421.33331	0.40584421	0.05351684	2	0	0	1
1373	388.25641	0.40584421	0.05351684	2	0	0	1
1374	664.56409	0.40584421	0.05351684	2	0	0	1
1375	271.12820	0.44800878	0.05351684	2	0	0	0
1376	373.17950	0.44800878	0.05351684	1	0	0	1
1377	446.97433	0.44800878	0.05351684	2	0	0	0
1378	578.76923	0.44800878	0.05351684	2	0	0	1
1379	541.28204	0.44800878	0.05351684	2	0	0	1
1380	403.94873	0.44800878	0.05351684	1	0	0	1
1381	525.23077	0.49017335	0.05351684	2	0	0	1
1382	558.51282	0.49017335	0.05351684	1	0	0	1
1383	572.66663	0.49017335	0.05351684	2	1	0	1
1384	509.12820	0.49017335	0.05351684	2	0	0	1
1385	297.38461	0.49017335	0.05351684	1	0	0	1
1386	639.89746	0.49017335	0.05351684	1	0	0	1
1387	389.28204	0.49017335	0.05351684	2	0	0	1
1388	600.10254	0.49017335	0.05351684	1	0	0	1
1389	313.17950	0.53233791	0.05351684	2	0	0	1
1390	411.64102	0.53233791	0.05351684	2	0	0	1
1391	724.51282	0.53233791	0.05351684	2	0	0	1
1392	581.17950	0.53233791	0.05351684	2	0	0	1
1393	272.97437	0.53233791	0.05351684	1	0	0	0
1394	691.23077	0.53233791	0.05351684	2	0	0	1
1395	620.35895	0.53233791	0.05351684	2	0	0	1
1396	641.12817	0.57450248	0.05351684	2	0	0	1
1397	525.53851	0.57450248	0.05351684	1	0	0	1
1398	399.89743	0.61666705	0.05351684	2	0	0	1
1399	365.28204	0.61666705	0.05351684	1	0	0	1
1400	481.07690	0.65883161	0.05351684	2	0	0	1
1401	494.82053	0.65883161	0.05351684	1	0	0	1

1402	467.89746	0.65883161	0.05351684	1	0	0	1
1403	272.66669	0.65883161	0.05351684	1	0	0	1
1404	805.28210	0.65883161	0.05351684	2	0	0	1
1405	677.89746	0.70099618	0.05351684	2	1	0	1
1406	462.20511	0.70099618	0.05351684	1	0	0	1
1407	673.69226	0.70099618	0.05351684	2	0	0	1
1408	498.15387	0.70099618	0.05351684	1	0	0	1
1409	477.94873	0.74316075	0.05351684	1	0	0	1
1410	560.51282	0.74316075	0.05351684	2	0	0	1
1411	458.25641	0.74316075	0.05351684	1	1	0	1
1412	631.17950	0.74316075	0.05351684	1	0	1	1
1413	502.41025	0.74316075	0.05351684	2	0	0	1
1414	503.02567	0.74316075	0.05351684	2	0	0	1
1415	606.71796	0.74316075	0.05351684	2	0	0	1
1416	501.89743	0.78532531	0.05351684	1	0	0	1
1417	587.12823	0.82748988	0.05351684	2	0	0	1
1418	647.58978	0.82748988	0.05351684	2	0	0	1
1419	328.82053	0.82748988	0.05351684	2	0	0	1
1420	461.12820	0.86965445	0.05351684	2	0	0	1
1421	306.46152	0.86965445	0.05351684	2	0	0	0
1422	310.35898	0.86965445	0.05351684	1	1	0	1
1423	591.43591	0.86965445	0.05351684	1	0	0	1
1424	377.69232	0.86965445	0.05351684	1	0	0	1
1425	336.00000	0.91181901	0.05351684	2	0	0	0
1426	480.05127	0.95398358	0.05351684	2	0	1	1
1427	761.69232	0.95398358	0.05351684	2	0	0	1
1428	411.48718	1.03831271	0.05351684	1	0	0	1
1429	396.20511	1.03831271	0.05351684	2	0	0	1
1430	343.64102	1.03831271	0.05351684	2	0	0	1
1431	721.17950	1.03831271	0.05351684	2	0	0	1
1432	635.48718	1.08047728	0.05351684	2	0	0	1
1433	810.82050	1.08047728	0.05351684	1	0	1	1
1434	597.94873	1.16480641	0.05351684	1	0	0	1
1435	420.00000	1.16480641	0.05351684	1	0	0	1
1436	376.56409	1.24913555	0.05351684	2	0	0	0
1437	704.10254	1.29130011	0.05351684	2	0	0	1
1438	439.48718	1.33346468	0.05351684	1	1	0	1
1439	786.61542	1.33346468	0.05351684	1	0	0	1
1440	455.33334	1.37562925	0.05351684	2	0	0	1
1441	515.53845	1.54428751	0.05351684	2	0	0	1
1442	438.30771	1.58645208	0.05351684	2	0	0	0
1443	543.69226	1.67078121	0.05351684	2	0	0	1
1444	534.87177	1.71294578	0.05351684	1	0	0	1

1445	465.17947	1.79727491	0.05351684	3	0	0	0
1446	747.28204	1.79727491	0.05351684	1	0	0	1
1447	514.05127	2.09242688	0.05351684	2	0	0	1
1448	555.17950	2.09242688	0.05351684	2	0	0	1
1449	703.12817	2.21892058	0.05351684	2	0	0	1
1450	727.33337	2.64056624	0.05351684	2	0	0	1
1451	559.43591	2.64056624	0.05351684	1	1	0	1
1452	1009.02563	3.06221191	0.05351684	3	0	0	1
1453	197.74359	-1.91320695	0.07593310	1	0	0	1
1454	265.28204	-1.66021955	0.07593310	1	0	0	1
1455	277.48715	-1.57589041	0.07593310	2	0	0	1
1456	191.43590	-1.57589041	0.07593310	1	0	0	1
1457	356.25641	-1.36506758	0.07593310	2	0	0	1
1458	266.15384	-1.23857388	0.07593310	1	0	0	1
1459	356.10257	-1.23857388	0.07593310	1	0	0	1
1460	226.71796	-1.23857388	0.07593310	1	0	0	1
1461	285.69229	-1.19640931	0.07593310	1	0	0	1
1462	287.02563	-1.19640931	0.07593310	1	0	0	1
1463	371.64102	-1.15424475	0.07593310	2	0	0	1
1464	235.17949	-1.15424475	0.07593310	1	0	1	1
1465	366.30768	-1.15424475	0.07593310	1	0	0	1
1466	165.07692	-1.06991561	0.07593310	2	0	0	0
1467	338.66669	-1.06991561	0.07593310	1	0	0	1
1468	197.79488	-0.94342192	0.07593310	1	0	0	1
1469	381.79486	-0.73259908	0.07593310	2	0	0	1
1470	379.94873	-0.73259908	0.07593310	2	0	0	1
1471	263.84616	-0.69043452	0.07593310	1	0	0	1
1472	383.38461	-0.64826995	0.07593310	1	0	0	1
1473	306.92307	-0.64826995	0.07593310	1	0	0	1
1474	354.05130	-0.52177625	0.07593310	1	0	0	1
1475	401.23077	-0.47961168	0.07593310	2	0	0	1
1476	412.56412	-0.43744712	0.07593310	1	0	0	1
1477	570.87177	-0.43744712	0.07593310	1	0	0	1
1478	481.07690	-0.39528255	0.07593310	1	0	0	1
1479	352.56412	-0.35311798	0.07593310	2	0	0	1
1480	497.48715	-0.35311798	0.07593310	2	0	0	1
1481	384.46155	-0.31095342	0.07593310	1	0	0	1
1482	480.66666	-0.18445972	0.07593310	1	0	0	1
1483	439.74359	-0.14229515	0.07593310	1	0	0	1
1484	593.74359	-0.01580145	0.07593310	2	0	0	1
1485	534.20508	0.19502138	0.07593310	2	0	0	1
1486	487.33331	0.19502138	0.07593310	1	0	0	1
1487	578.61542	0.32151508	0.07593310	2	0	0	1

1488	398.82053	0.49017335	0.07593310	1	0	0	1
1489	233.28204	0.53233791	0.07593310	1	0	0	1
1490	698.05127	0.53233791	0.07593310	2	0	0	1
1491	700.35895	0.53233791	0.07593310	1	0	0	1
1492	498.71796	0.70099618	0.07593310	1	0	0	1
1493	390.41025	0.74316075	0.07593310	2	0	0	1
1494	899.69232	0.99614815	0.07593310	1	0	0	1
1495	451.84613	1.03831271	0.07593310	2	0	0	1
1496	213.74358	-1.95537151	0.12076563	2	0	0	1
1497	276.71793	-1.74454868	0.12076563	1	0	0	1
1498	354.51282	-1.66021955	0.12076563	2	0	0	1
1499	179.53847	-1.61805498	0.12076563	3	0	0	1
1500	245.89743	-1.57589041	0.12076563	2	0	0	1
1501	270.51282	-1.57589041	0.12076563	1	0	0	1
1502	335.79486	-1.57589041	0.12076563	2	0	0	1
1503	233.38461	-1.57589041	0.12076563	2	0	0	1
1504	396.30768	-1.57589041	0.12076563	2	0	0	1
1505	185.33333	-1.57589041	0.12076563	3	0	0	1
1506	229.02565	-1.53372585	0.12076563	2	0	0	1
1507	312.05127	-1.44939671	0.12076563	1	0	0	1
1508	308.00000	-1.36506758	0.12076563	2	0	0	1
1509	349.53845	-1.32290301	0.12076563	2	0	0	1
1510	356.61539	-1.32290301	0.12076563	2	0	0	1
1511	407.94873	-1.28073845	0.12076563	2	0	0	1
1512	284.30771	-1.23857388	0.12076563	2	0	0	1
1513	365.74359	-1.23857388	0.12076563	1	0	0	1
1514	322.15384	-1.23857388	0.12076563	1	0	0	1
1515	357.28204	-1.02775105	0.12076563	2	0	0	1
1516	326.87180	-0.94342192	0.12076563	2	0	0	1
1517	293.94873	-0.90125735	0.12076563	1	0	0	1
1518	258.25641	-0.85909278	0.12076563	1	0	0	1
1519	265.84616	-0.77476365	0.12076563	2	0	0	1
1520	307.64105	-0.73259908	0.12076563	2	0	0	1
1521	400.41025	-0.73259908	0.12076563	1	0	0	1
1522	344.76923	-0.64826995	0.12076563	1	0	1	1
1523	405.64102	-0.64826995	0.12076563	1	0	0	1
1524	250.61539	-0.64826995	0.12076563	1	0	0	1
1525	368.56412	-0.56394082	0.12076563	1	0	0	1
1526	231.28206	-0.56394082	0.12076563	1	0	1	1
1527	535.12823	-0.52177625	0.12076563	1	0	0	1
1528	438.76923	-0.52177625	0.12076563	1	0	0	1
1529	353.17950	-0.47961168	0.12076563	1	0	0	1
1530	507.28204	-0.47961168	0.12076563	2	0	0	1

1531	492.10254	-0.43744712	0.12076563	2	0	0	1
1532	385.94870	-0.43744712	0.12076563	2	0	0	1
1533	615.84619	-0.39528255	0.12076563	2	0	0	1
1534	236.76924	-0.39528255	0.12076563	2	0	0	1
1535	458.76923	-0.39528255	0.12076563	1	0	0	1
1536	518.20514	-0.31095342	0.12076563	2	0	0	1
1537	458.05130	-0.14229515	0.12076563	1	0	0	1
1538	338.66669	-0.14229515	0.12076563	2	0	0	1
1539	437.07690	-0.14229515	0.12076563	2	0	0	1
1540	418.10257	-0.10013058	0.12076563	1	0	0	1
1541	352.05127	-0.10013058	0.12076563	1	0	0	1
1542	265.17947	-0.10013058	0.12076563	1	0	0	0
1543	379.94873	-0.05796602	0.12076563	1	1	0	1
1544	589.12823	-0.05796602	0.12076563	1	0	0	1
1545	389.89743	-0.01580145	0.12076563	1	0	0	1
1546	448.00000	0.02636312	0.12076563	1	0	0	1
1547	738.46155	0.02636312	0.12076563	2	0	0	1
1548	467.84616	0.06852768	0.12076563	1	0	0	1
1549	393.33334	0.11069225	0.12076563	1	0	0	1
1550	594.15381	0.11069225	0.12076563	1	0	0	1
1551	407.89746	0.15285682	0.12076563	1	0	0	1
1552	563.69226	0.19502138	0.12076563	1	0	0	1
1553	504.71796	0.19502138	0.12076563	1	0	0	1
1554	551.84613	0.23718595	0.12076563	1	0	0	1
1555	670.87177	0.27935051	0.12076563	2	0	1	1
1556	365.94870	0.32151508	0.12076563	2	0	0	1
1557	568.97437	0.36367965	0.12076563	1	0	0	1
1558	671.07690	0.40584421	0.12076563	2	0	0	1
1559	698.25641	0.44800878	0.12076563	2	1	0	1
1560	571.89740	0.44800878	0.12076563	1	0	0	1
1561	437.28204	0.53233791	0.12076563	2	0	0	1
1562	481.23077	0.57450248	0.12076563	2	0	0	1
1563	461.33331	0.57450248	0.12076563	2	1	0	1
1564	707.12823	0.61666705	0.12076563	1	0	0	1
1565	450.76923	0.61666705	0.12076563	2	0	0	1
1566	380.61539	0.65883161	0.12076563	1	0	0	1
1567	518.15387	0.86965445	0.12076563	1	0	0	1
1568	718.76923	0.95398358	0.12076563	1	0	0	1
1569	281.84613	1.16480641	0.12076563	2	0	0	1
1570	498.20514	1.29130011	0.12076563	1	0	0	1
1571	758.51282	1.29130011	0.12076563	2	0	0	1
1572	729.64105	1.29130011	0.12076563	2	1	0	1
1573	781.12817	1.37562925	0.12076563	1	0	0	1

1574	421.17947	1.41779381	0.12076563	1	0	0	1
1575	640.05127	1.45995838	0.12076563	1	0	0	1
1576	1081.79492	1.50212295	0.12076563	1	0	0	1
1577	586.66669	1.62861664	0.12076563	1	0	0	1
1578	914.51282	1.67078121	0.12076563	2	0	0	1
1579	311.07690	-1.82887781	0.16559816	2	0	0	1
1580	161.79488	-1.82887781	0.16559816	2	0	0	1
1581	326.15384	-1.74454868	0.16559816	1	0	0	1
1582	229.64102	-1.70238411	0.16559816	2	0	0	1
1583	244.10257	-1.66021955	0.16559816	2	0	0	1
1584	273.74359	-1.57589041	0.16559816	1	0	0	1
1585	177.89743	-1.49156128	0.16559816	2	1	0	1
1586	258.71796	-1.49156128	0.16559816	1	0	0	1
1587	132.76923	-1.44939671	0.16559816	1	0	0	1
1588	262.56412	-1.40723215	0.16559816	1	0	0	1
1589	296.97433	-1.36506758	0.16559816	1	0	0	1
1590	237.12820	-1.32290301	0.16559816	1	0	0	1
1591	315.64102	-1.23857388	0.16559816	1	0	0	1
1592	397.69232	-1.19640931	0.16559816	1	0	0	1
1593	329.89743	-1.11208018	0.16559816	1	0	0	1
1594	286.10257	-0.94342192	0.16559816	2	0	0	1
1595	431.12820	-0.81692822	0.16559816	1	0	0	1
1596	277.02563	-0.81692822	0.16559816	1	0	0	1
1597	387.12820	-0.77476365	0.16559816	1	0	0	1
1598	372.61539	-0.73259908	0.16559816	2	0	0	1
1599	240.56410	-0.64826995	0.16559816	2	0	0	0
1600	270.61539	-0.64826995	0.16559816	2	0	0	1
1601	495.28204	-0.64826995	0.16559816	2	0	0	1
1602	304.20514	-0.64826995	0.16559816	1	0	0	1
1603	276.66666	-0.64826995	0.16559816	2	0	0	1
1604	339.48718	-0.60610538	0.16559816	2	0	0	1
1605	316.35898	-0.60610538	0.16559816	2	0	0	1
1606	183.23076	-0.56394082	0.16559816	1	0	0	0
1607	330.00000	-0.56394082	0.16559816	1	0	0	1
1608	321.02563	-0.56394082	0.16559816	1	0	0	1
1609	232.35898	-0.52177625	0.16559816	1	0	0	1
1610	646.87183	-0.43744712	0.16559816	1	0	0	1
1611	370.66666	-0.39528255	0.16559816	1	0	0	1
1612	303.79489	-0.35311798	0.16559816	1	0	0	1
1613	294.20514	-0.35311798	0.16559816	1	0	0	1
1614	303.64102	-0.35311798	0.16559816	1	0	0	1
1615	469.33334	-0.31095342	0.16559816	1	0	0	1
1616	384.05130	-0.26878885	0.16559816	1	0	0	1

1617	292.56412	-0.22662428	0.16559816	1	0	0	1
1618	365.69229	-0.18445972	0.16559816	1	0	0	1
1619	601.12817	-0.14229515	0.16559816	2	0	0	1
1620	348.87180	-0.14229515	0.16559816	1	0	0	1
1621	345.58975	-0.14229515	0.16559816	1	0	0	1
1622	384.30771	-0.14229515	0.16559816	1	0	0	1
1623	449.48718	-0.10013058	0.16559816	1	0	0	1
1624	397.64105	-0.10013058	0.16559816	1	0	0	1
1625	495.28204	-0.10013058	0.16559816	2	0	0	1
1626	482.66669	-0.10013058	0.16559816	1	0	0	1
1627	493.43591	-0.01580145	0.16559816	1	0	0	1
1628	556.20514	-0.01580145	0.16559816	2	0	0	1
1629	369.58975	0.02636312	0.16559816	1	0	0	1
1630	615.58978	0.02636312	0.16559816	1	0	0	1
1631	322.25641	0.02636312	0.16559816	1	0	0	1
1632	477.58975	0.23718595	0.16559816	1	0	0	1
1633	642.00000	0.23718595	0.16559816	1	0	0	1
1634	604.46149	0.27935051	0.16559816	1	0	0	1
1635	531.84613	0.27935051	0.16559816	3	0	0	1
1636	312.61539	0.32151508	0.16559816	3	0	0	1
1637	480.30768	0.32151508	0.16559816	2	0	0	1
1638	309.94873	0.36367965	0.16559816	1	0	0	1
1639	506.46152	0.40584421	0.16559816	1	0	0	1
1640	517.12823	0.49017335	0.16559816	1	0	0	1
1641	503.12820	0.53233791	0.16559816	1	0	0	1
1642	347.94873	0.53233791	0.16559816	1	0	0	1
1643	332.66669	0.53233791	0.16559816	2	0	0	1
1644	386.15384	0.61666705	0.16559816	1	0	0	1
1645	402.15384	0.65883161	0.16559816	1	0	0	1
1646	239.28206	0.78532531	0.16559816	1	1	0	1
1647	648.41028	0.86965445	0.16559816	1	0	0	1
1648	576.82056	0.91181901	0.16559816	2	0	0	1
1649	652.87177	0.95398358	0.16559816	1	0	0	1
1650	487.79489	0.99614815	0.16559816	1	0	0	1
1651	439.43591	1.24913555	0.16559816	2	0	0	1
1652	814.87177	1.37562925	0.16559816	2	0	0	1
1653	515.38464	1.79727491	0.16559816	1	0	0	1
1654	272.35895	-1.87104238	0.21043070	1	0	0	1
1655	218.10255	-1.78671325	0.21043070	1	0	0	1
1656	304.25641	-1.66021955	0.21043070	2	0	0	1
1657	324.61539	-1.61805498	0.21043070	1	0	0	1
1658	54.92308	-1.49156128	0.21043070	1	0	0	1
1659	324.25641	-1.49156128	0.21043070	1	0	0	1

1660	256.87180	-1.44939671	0.21043070	1	0	0	1
1661	368.56412	-1.40723215	0.21043070	2	0	0	1
1662	243.48717	-1.32290301	0.21043070	1	0	0	1
1663	252.51282	-1.19640931	0.21043070	1	0	0	0
1664	331.69229	-1.11208018	0.21043070	1	0	0	1
1665	462.00000	-0.73259908	0.21043070	2	0	0	1
1666	521.33331	-0.69043452	0.21043070	1	0	0	1
1667	595.94873	-0.64826995	0.21043070	1	0	0	1
1668	360.82050	-0.64826995	0.21043070	2	0	0	1
1669	274.56412	-0.60610538	0.21043070	2	0	0	1
1670	352.15384	-0.47961168	0.21043070	1	0	0	1
1671	243.84616	-0.39528255	0.21043070	2	0	0	1
1672	444.25641	-0.31095342	0.21043070	1	0	0	1
1673	347.79489	-0.31095342	0.21043070	1	0	0	1
1674	476.76923	-0.31095342	0.21043070	2	0	0	1
1675	497.12820	-0.26878885	0.21043070	3	0	0	1
1676	256.20514	-0.26878885	0.21043070	1	0	0	1
1677	355.74359	-0.18445972	0.21043070	1	0	0	1
1678	384.92307	-0.18445972	0.21043070	1	0	0	1
1679	590.61536	-0.14229515	0.21043070	1	0	0	1
1680	529.33331	-0.14229515	0.21043070	1	0	0	1
1681	303.38461	-0.10013058	0.21043070	1	0	0	1
1682	457.12820	-0.10013058	0.21043070	2	0	0	1
1683	621.17950	-0.01580145	0.21043070	2	0	0	1
1684	508.41028	0.02636312	0.21043070	1	0	0	1
1685	675.64105	0.02636312	0.21043070	1	0	0	1
1686	342.71796	0.02636312	0.21043070	1	0	0	1
1687	523.58972	0.02636312	0.21043070	1	0	1	1
1688	468.15387	0.06852768	0.21043070	1	0	0	1
1689	640.56409	0.11069225	0.21043070	1	0	0	1
1690	516.30768	0.11069225	0.21043070	1	0	0	1
1691	396.66666	0.11069225	0.21043070	2	0	0	1
1692	571.69232	0.23718595	0.21043070	1	0	0	1
1693	310.66666	0.32151508	0.21043070	1	0	0	1
1694	341.94870	0.40584421	0.21043070	1	0	0	1
1695	523.17950	0.44800878	0.21043070	1	0	0	1
1696	345.38461	0.44800878	0.21043070	1	0	0	1
1697	486.20511	0.65883161	0.21043070	1	0	0	1
1698	291.38461	0.65883161	0.21043070	1	0	0	1
1699	256.66666	0.70099618	0.21043070	1	0	0	1
1700	441.23077	0.95398358	0.21043070	1	0	0	1
1701	806.82056	0.95398358	0.21043070	1	0	0	1
1702	629.69232	0.95398358	0.21043070	1	0	0	1

1703	607.17950	1.08047728	0.21043070	2	0	0	1
1704	559.84613	1.37562925	0.21043070	1	0	0	1
1705	859.58972	1.54428751	0.21043070	1	0	0	1
1706	458.56412	1.58645208	0.21043070	1	0	0	1
1707	939.28204	2.76705994	0.21043070	3	0	0	1
1708	239.28206	-1.74454868	0.25526323	1	0	0	1
1709	211.84616	-1.74454868	0.25526323	1	0	0	1
1710	382.76926	-1.66021955	0.25526323	1	0	0	1
1711	295.12820	-1.44939671	0.25526323	1	0	0	1
1712	261.48718	-1.44939671	0.25526323	1	0	0	1
1713	243.79488	-1.40723215	0.25526323	1	0	0	1
1714	249.94872	-1.23857388	0.25526323	1	0	0	1
1715	276.35898	-1.19640931	0.25526323	2	0	0	1
1716	279.33334	-1.15424475	0.25526323	1	0	0	1
1717	237.89743	-1.06991561	0.25526323	1	0	0	1
1718	441.69229	-0.81692822	0.25526323	1	0	0	1
1719	532.46155	-0.81692822	0.25526323	2	0	0	1
1720	209.58975	-0.81692822	0.25526323	2	0	0	1
1721	495.33334	-0.73259908	0.25526323	2	0	0	1
1722	489.69232	-0.73259908	0.25526323	1	0	0	1
1723	320.00000	-0.69043452	0.25526323	1	1	0	1
1724	391.28204	-0.56394082	0.25526323	2	0	0	1
1725	313.43591	-0.56394082	0.25526323	1	0	0	1
1726	358.41028	-0.52177625	0.25526323	1	0	0	1
1727	460.51282	-0.52177625	0.25526323	1	0	0	1
1728	320.46155	-0.52177625	0.25526323	1	0	0	1
1729	348.00000	-0.47961168	0.25526323	1	0	0	1
1730	302.05127	-0.47961168	0.25526323	2	0	0	1
1731	366.15384	-0.47961168	0.25526323	1	0	0	1
1732	373.48718	-0.43744712	0.25526323	1	0	0	1
1733	379.79486	-0.39528255	0.25526323	1	0	0	1
1734	548.30768	-0.39528255	0.25526323	1	0	0	1
1735	422.76926	-0.39528255	0.25526323	1	0	0	1
1736	509.58975	-0.31095342	0.25526323	1	0	0	1
1737	482.82053	-0.22662428	0.25526323	1	0	0	1
1738	365.94870	-0.22662428	0.25526323	1	0	0	1
1739	376.97433	-0.22662428	0.25526323	2	0	0	1
1740	541.53845	-0.22662428	0.25526323	1	0	0	1
1741	345.12820	-0.22662428	0.25526323	1	0	0	1
1742	378.61539	-0.18445972	0.25526323	1	0	0	1
1743	330.30768	-0.14229515	0.25526323	1	0	0	1
1744	465.33334	-0.10013058	0.25526323	2	0	0	1
1745	374.46155	-0.10013058	0.25526323	1	0	0	1

1746	306.66666	-0.01580145	0.25526323	2	0	0	1
1747	355.84616	0.11069225	0.25526323	1	0	0	1
1748	366.61539	0.11069225	0.25526323	2	0	0	1
1749	518.61536	0.15285682	0.25526323	1	0	0	1
1750	449.64102	0.15285682	0.25526323	1	0	0	1
1751	370.71796	0.23718595	0.25526323	2	0	0	1
1752	465.17947	0.23718595	0.25526323	2	0	0	1
1753	289.33334	0.23718595	0.25526323	1	0	0	1
1754	364.92307	0.32151508	0.25526323	1	0	0	1
1755	592.41022	0.40584421	0.25526323	1	0	0	1
1756	570.51282	0.44800878	0.25526323	1	0	0	1
1757	710.15387	0.49017335	0.25526323	1	0	0	1
1758	530.46155	0.49017335	0.25526323	1	0	0	1
1759	431.17947	0.49017335	0.25526323	1	0	0	1
1760	661.28204	0.53233791	0.25526323	1	0	0	1
1761	451.02563	0.53233791	0.25526323	1	0	0	1
1762	424.15384	0.57450248	0.25526323	1	0	0	1
1763	624.92303	0.61666705	0.25526323	1	0	0	1
1764	457.84616	0.61666705	0.25526323	1	0	0	1
1765	766.10260	0.61666705	0.25526323	1	0	0	1
1766	384.92307	0.70099618	0.25526323	2	0	0	1
1767	408.61539	0.70099618	0.25526323	2	0	0	1
1768	465.53845	0.70099618	0.25526323	1	0	0	1
1769	498.97437	0.78532531	0.25526323	1	0	0	1
1770	447.69232	0.78532531	0.25526323	1	0	0	1
1771	621.94873	0.82748988	0.25526323	1	0	0	1
1772	592.87177	0.91181901	0.25526323	2	0	0	1
1773	561.07690	0.95398358	0.25526323	1	0	0	1
1774	625.28210	1.29130011	0.25526323	1	1	0	1
1775	495.28204	1.33346468	0.25526323	1	0	0	1
1776	642.71796	1.79727491	0.25526323	2	0	0	1
1777	255.12820	-1.66021955	0.30009576	1	0	0	1
1778	278.30771	-1.53372585	0.30009576	1	0	0	1
1779	284.30771	-1.44939671	0.30009576	1	0	0	1
1780	260.41025	-1.44939671	0.30009576	1	0	0	1
1781	288.92310	-1.06991561	0.30009576	2	0	0	1
1782	325.89743	-0.98558648	0.30009576	2	0	0	1
1783	265.58975	-0.77476365	0.30009576	1	0	0	1
1784	201.89745	-0.77476365	0.30009576	1	0	0	0
1785	278.10257	-0.73259908	0.30009576	1	0	0	1
1786	216.41026	-0.73259908	0.30009576	1	0	0	1
1787	295.58975	-0.60610538	0.30009576	1	0	0	1
1788	290.56409	-0.60610538	0.30009576	2	0	0	1

1789	378.05130	-0.47961168	0.30009576	1	0	0	1
1790	361.12820	-0.43744712	0.30009576	1	0	0	1
1791	466.10257	-0.43744712	0.30009576	1	0	0	1
1792	372.30768	-0.39528255	0.30009576	1	0	0	1
1793	436.92307	-0.39528255	0.30009576	1	0	0	1
1794	418.51282	-0.31095342	0.30009576	1	0	0	1
1795	604.25641	-0.31095342	0.30009576	2	0	0	1
1796	305.48718	-0.31095342	0.30009576	1	0	0	1
1797	580.46155	-0.22662428	0.30009576	1	0	0	1
1798	510.97437	-0.18445972	0.30009576	1	0	0	1
1799	338.00000	-0.14229515	0.30009576	2	0	0	1
1800	331.12820	-0.14229515	0.30009576	1	0	0	1
1801	430.30768	-0.14229515	0.30009576	1	0	0	1
1802	460.00000	-0.14229515	0.30009576	1	0	0	1
1803	460.30768	-0.10013058	0.30009576	1	0	0	1
1804	406.25641	-0.10013058	0.30009576	2	0	0	1
1805	403.64102	-0.10013058	0.30009576	1	0	0	1
1806	472.35895	-0.10013058	0.30009576	1	0	0	1
1807	377.74359	-0.05796602	0.30009576	2	0	0	1
1808	408.30771	-0.05796602	0.30009576	1	0	0	1
1809	353.74359	-0.05796602	0.30009576	1	0	0	1
1810	458.87180	-0.01580145	0.30009576	1	0	0	1
1811	419.69232	0.02636312	0.30009576	2	0	0	1
1812	564.15381	0.02636312	0.30009576	1	0	0	1
1813	428.05130	0.06852768	0.30009576	2	0	0	1
1814	439.33334	0.11069225	0.30009576	1	0	0	1
1815	427.43588	0.11069225	0.30009576	1	0	0	1
1816	277.17947	0.15285682	0.30009576	1	0	0	1
1817	578.71796	0.19502138	0.30009576	2	0	0	1
1818	359.12820	0.19502138	0.30009576	1	1	0	1
1819	760.76923	0.23718595	0.30009576	1	0	0	1
1820	461.28204	0.32151508	0.30009576	1	0	0	1
1821	282.56412	0.36367965	0.30009576	1	0	0	1
1822	470.30768	0.36367965	0.30009576	2	0	0	1
1823	405.43588	0.44800878	0.30009576	2	0	0	1
1824	626.15387	0.49017335	0.30009576	1	0	1	1
1825	415.17947	0.53233791	0.30009576	1	0	0	1
1826	626.15387	0.61666705	0.30009576	1	0	0	1
1827	391.43588	0.74316075	0.30009576	2	0	0	1
1828	587.28204	0.78532531	0.30009576	1	0	0	1
1829	725.12823	0.82748988	0.30009576	2	0	1	1
1830	457.48715	0.82748988	0.30009576	1	0	0	1
1831	579.64105	0.95398358	0.30009576	1	0	0	1

1832	543.69226	1.03831271	0.30009576	1	0	0	1
1833	358.66669	1.29130011	0.30009576	2	0	0	1
1834	682.82050	1.37562925	0.30009576	1	0	1	1
1835	483.74359	1.79727491	0.30009576	1	1	0	1
1836	207.43590	-1.49156128	0.34492829	1	0	0	1
1837	503.23077	-1.15424475	0.34492829	1	0	0	1
1838	367.12820	-1.15424475	0.34492829	1	0	0	1
1839	276.25641	-1.06991561	0.34492829	1	0	0	1
1840	360.46155	-1.02775105	0.34492829	1	0	0	1
1841	359.79486	-0.85909278	0.34492829	1	0	0	1
1842	351.64102	-0.85909278	0.34492829	1	0	0	1
1843	468.25641	-0.77476365	0.34492829	1	0	0	1
1844	476.35898	-0.73259908	0.34492829	1	0	0	1
1845	384.76923	-0.47961168	0.34492829	1	0	0	1
1846	336.71793	-0.47961168	0.34492829	2	0	0	1
1847	478.92310	-0.47961168	0.34492829	1	0	0	1
1848	338.15387	-0.43744712	0.34492829	2	0	0	1
1849	339.84616	-0.43744712	0.34492829	2	0	0	1
1850	273.69232	-0.39528255	0.34492829	1	0	0	1
1851	378.41028	-0.35311798	0.34492829	2	0	0	1
1852	407.02563	-0.31095342	0.34492829	1	0	0	1
1853	357.12820	-0.18445972	0.34492829	1	1	0	1
1854	320.46155	-0.14229515	0.34492829	1	0	0	1
1855	365.94870	-0.10013058	0.34492829	1	0	0	0
1856	597.33337	-0.05796602	0.34492829	1	0	0	1
1857	458.51282	-0.05796602	0.34492829	1	0	0	1
1858	205.33333	-0.01580145	0.34492829	1	0	0	0
1859	312.10254	-0.01580145	0.34492829	1	0	0	1
1860	430.46155	0.02636312	0.34492829	1	0	0	1
1861	425.38461	0.06852768	0.34492829	1	0	0	1
1862	594.97430	0.11069225	0.34492829	1	0	0	1
1863	519.79486	0.11069225	0.34492829	2	0	0	1
1864	303.53848	0.23718595	0.34492829	1	0	0	1
1865	497.64105	0.27935051	0.34492829	1	0	0	1
1866	689.89746	0.36367965	0.34492829	2	0	0	1
1867	387.28204	0.40584421	0.34492829	1	0	0	1
1868	500.97437	0.40584421	0.34492829	1	0	0	1
1869	372.00000	0.61666705	0.34492829	1	0	0	1
1870	739.64105	0.65883161	0.34492829	1	0	0	1
1871	181.94872	0.70099618	0.34492829	1	0	0	1
1872	275.58975	0.74316075	0.34492829	2	0	0	1
1873	560.92310	0.74316075	0.34492829	1	0	0	1
1874	753.69226	0.86965445	0.34492829	1	0	0	1

1875	578.66669	1.08047728	0.34492829	1	1	0	1
1876	544.61536	1.16480641	0.34492829	1	0	0	1
1877	1041.89746	2.05026231	0.34492829	2	1	1	1
1878	336.87180	-1.78671325	0.38976082	1	0	0	1
1879	276.71793	-1.66021955	0.38976082	1	0	0	1
1880	313.94873	-1.66021955	0.38976082	1	0	0	1
1881	253.48717	-1.57589041	0.38976082	1	0	0	1
1882	226.61539	-1.57589041	0.38976082	1	0	0	1
1883	323.69232	-1.53372585	0.38976082	1	0	0	1
1884	354.76923	-1.53372585	0.38976082	1	0	0	1
1885	225.43590	-1.49156128	0.38976082	1	0	0	1
1886	370.51282	-1.49156128	0.38976082	3	0	0	1
1887	333.53848	-1.40723215	0.38976082	1	0	0	1
1888	315.43588	-1.28073845	0.38976082	1	0	0	1
1889	315.28204	-1.23857388	0.38976082	1	0	0	1
1890	275.33334	-1.15424475	0.38976082	1	0	0	1
1891	339.74359	-1.02775105	0.38976082	2	0	0	1
1892	421.89743	-0.94342192	0.38976082	1	0	0	1
1893	212.25641	-0.90125735	0.38976082	1	0	0	1
1894	244.76923	-0.81692822	0.38976082	1	0	0	1
1895	300.66666	-0.81692822	0.38976082	1	0	0	1
1896	321.02563	-0.77476365	0.38976082	1	0	0	1
1897	218.71794	-0.73259908	0.38976082	2	0	0	1
1898	256.87180	-0.31095342	0.38976082	1	0	0	1
1899	396.92307	-0.31095342	0.38976082	1	0	0	1
1900	285.02563	-0.14229515	0.38976082	1	0	0	1
1901	471.64102	-0.14229515	0.38976082	1	0	0	1
1902	690.35895	-0.10013058	0.38976082	1	0	0	1
1903	385.48718	-0.10013058	0.38976082	2	0	0	1
1904	539.28204	-0.10013058	0.38976082	1	0	0	1
1905	344.92307	-0.05796602	0.38976082	2	0	0	1
1906	463.53848	-0.01580145	0.38976082	1	0	0	1
1907	373.84616	-0.01580145	0.38976082	1	0	0	1
1908	313.33334	0.02636312	0.38976082	1	0	0	1
1909	533.64099	0.02636312	0.38976082	1	0	0	1
1910	431.79486	0.02636312	0.38976082	1	0	0	1
1911	614.20508	0.06852768	0.38976082	1	0	0	1
1912	418.00000	0.06852768	0.38976082	1	0	0	1
1913	467.23074	0.06852768	0.38976082	1	0	0	1
1914	333.79489	0.11069225	0.38976082	1	0	0	1
1915	238.97437	0.11069225	0.38976082	1	0	0	1
1916	498.15387	0.11069225	0.38976082	2	0	0	1
1917	357.69232	0.11069225	0.38976082	1	0	0	1

1918	412.35895	0.11069225	0.38976082	1	0	0	1
1919	478.82053	0.15285682	0.38976082	2	0	0	1
1920	316.66666	0.15285682	0.38976082	1	0	0	1
1921	472.97437	0.19502138	0.38976082	1	0	0	1
1922	515.69232	0.19502138	0.38976082	1	1	0	1
1923	494.41025	0.23718595	0.38976082	1	0	0	1
1924	494.20514	0.36367965	0.38976082	1	0	0	1
1925	540.51282	0.53233791	0.38976082	2	0	0	1
1926	452.97437	0.70099618	0.38976082	1	0	1	1
1927	849.79486	0.70099618	0.38976082	1	1	0	1
1928	632.30768	0.74316075	0.38976082	2	0	0	1
1929	486.05127	0.78532531	0.38976082	1	0	0	1
1930	691.94873	0.78532531	0.38976082	1	0	0	1
1931	531.53845	1.79727491	0.38976082	1	1	0	1
1932	188.35896	-1.95537151	0.43459335	3	0	0	1
1933	291.84613	-1.61805498	0.43459335	1	0	0	1
1934	289.64102	-1.36506758	0.43459335	2	0	0	1
1935	359.38461	-1.32290301	0.43459335	1	0	0	1
1936	260.35898	-1.28073845	0.43459335	1	0	0	1
1937	322.87180	-1.28073845	0.43459335	2	0	0	1
1938	394.46155	-1.23857388	0.43459335	2	0	0	1
1939	280.05127	-1.15424475	0.43459335	1	0	0	1
1940	275.28204	-1.15424475	0.43459335	1	0	0	1
1941	305.12820	-0.98558648	0.43459335	1	0	0	1
1942	317.79489	-0.81692822	0.43459335	2	0	0	1
1943	455.89743	-0.64826995	0.43459335	1	0	0	1
1944	384.51282	-0.64826995	0.43459335	2	0	0	1
1945	244.00000	-0.56394082	0.43459335	2	0	1	0
1946	667.07697	-0.56394082	0.43459335	1	0	0	1
1947	329.28204	-0.56394082	0.43459335	1	0	0	1
1948	470.00000	-0.31095342	0.43459335	2	0	0	1
1949	388.35898	-0.14229515	0.43459335	2	0	0	1
1950	367.69232	-0.10013058	0.43459335	2	0	0	1
1951	525.84619	0.36367965	0.43459335	1	0	0	1
1952	650.56409	0.53233791	0.43459335	1	0	0	1
1953	527.28204	0.78532531	0.43459335	1	0	0	1
1954	180.71794	0.82748988	0.43459335	1	0	0	1
1955	808.30768	1.03831271	0.43459335	1	0	0	1
1956	774.51282	1.29130011	0.43459335	1	0	0	1
1957	624.10254	1.37562925	0.43459335	1	0	0	1
1958	237.07692	-1.99753608	0.47942589	1	0	0	1
1959	259.89743	-1.61805498	0.47942589	1	0	0	1
1960	268.66669	-1.49156128	0.47942589	1	0	0	1

1961	358.25641	-1.49156128	0.47942589	1	0	0	1
1962	345.89743	-1.44939671	0.47942589	3	0	0	1
1963	375.07693	-1.44939671	0.47942589	1	0	0	1
1964	296.56409	-1.36506758	0.47942589	1	0	0	1
1965	325.58975	-1.32290301	0.47942589	1	0	0	1
1966	270.92307	-1.28073845	0.47942589	1	0	0	1
1967	487.69232	-1.23857388	0.47942589	1	0	0	1
1968	406.87180	-1.15424475	0.47942589	1	0	0	1
1969	323.02567	-0.85909278	0.47942589	1	0	0	1
1970	398.92310	-0.81692822	0.47942589	1	0	0	1
1971	463.02567	-0.73259908	0.47942589	2	0	0	1
1972	394.61539	-0.73259908	0.47942589	1	0	0	1
1973	435.89743	-0.69043452	0.47942589	1	0	0	1
1974	440.71796	-0.56394082	0.47942589	2	0	0	1
1975	528.76923	-0.52177625	0.47942589	1	0	0	1
1976	339.23077	-0.47961168	0.47942589	1	0	0	1
1977	424.92307	-0.47961168	0.47942589	1	0	0	1
1978	441.17947	-0.47961168	0.47942589	1	0	0	1
1979	348.25641	-0.39528255	0.47942589	1	0	0	1
1980	362.35895	-0.31095342	0.47942589	1	0	0	1
1981	441.23077	-0.26878885	0.47942589	1	0	0	1
1982	669.17950	-0.22662428	0.47942589	1	0	0	1
1983	644.76923	-0.10013058	0.47942589	1	0	0	1
1984	417.89746	-0.10013058	0.47942589	1	0	0	1
1985	368.05130	-0.10013058	0.47942589	1	0	0	1
1986	590.76923	-0.05796602	0.47942589	1	0	0	1
1987	577.48718	-0.01580145	0.47942589	1	0	1	1
1988	522.66669	0.06852768	0.47942589	1	0	0	1
1989	612.10254	0.11069225	0.47942589	1	0	0	1
1990	358.56412	0.11069225	0.47942589	1	0	0	1
1991	495.53845	0.15285682	0.47942589	1	0	0	1
1992	435.33334	0.15285682	0.47942589	1	0	0	1
1993	397.48715	0.23718595	0.47942589	1	0	0	1
1994	530.20514	0.32151508	0.47942589	1	1	1	1
1995	423.02567	0.32151508	0.47942589	1	0	0	1
1996	395.28204	0.32151508	0.47942589	1	0	0	1
1997	409.94873	0.36367965	0.47942589	1	0	0	1
1998	613.43585	0.40584421	0.47942589	2	0	0	1
1999	419.79486	0.44800878	0.47942589	1	0	0	1
2000	450.05127	0.70099618	0.47942589	1	0	0	1
2001	472.25641	0.70099618	0.47942589	1	0	0	1
2002	479.64102	0.74316075	0.47942589	1	0	0	1
2003	528.76923	1.20697098	0.47942589	1	0	0	1

2004	735.23077	1.20697098	0.47942589	2	0	0	1
2005	1168.97437	1.37562925	0.47942589	3	1	0	1
2006	267.38461	-1.66021955	0.52425842	1	0	0	1
2007	382.25641	-1.57589041	0.52425842	3	0	0	1
2008	311.48718	-1.57589041	0.52425842	1	0	0	1
2009	229.17949	-1.49156128	0.52425842	1	0	0	1
2010	308.76923	-1.40723215	0.52425842	1	0	0	1
2011	287.64105	-1.36506758	0.52425842	1	0	0	1
2012	337.74359	-1.32290301	0.52425842	1	0	0	1
2013	294.35898	-1.28073845	0.52425842	1	0	0	1
2014	277.69232	-1.19640931	0.52425842	1	0	0	1
2015	266.46152	-1.19640931	0.52425842	1	0	0	1
2016	350.82050	-1.15424475	0.52425842	1	0	0	1
2017	455.79486	-0.98558648	0.52425842	2	0	0	1
2018	304.92307	-0.64826995	0.52425842	1	0	0	1
2019	312.41025	-0.64826995	0.52425842	1	0	0	1
2020	525.43591	-0.64826995	0.52425842	1	0	0	1
2021	292.15384	-0.60610538	0.52425842	1	0	0	1
2022	456.97433	-0.56394082	0.52425842	1	0	0	1
2023	529.23077	-0.52177625	0.52425842	2	0	0	1
2024	475.89743	-0.52177625	0.52425842	1	0	0	1
2025	534.71790	-0.39528255	0.52425842	1	0	1	1
2026	397.23074	-0.35311798	0.52425842	2	0	0	1
2027	435.02563	-0.31095342	0.52425842	1	0	0	1
2028	425.58975	-0.31095342	0.52425842	1	0	0	1
2029	482.46152	-0.18445972	0.52425842	1	0	0	1
2030	428.41028	-0.18445972	0.52425842	2	0	0	1
2031	424.71796	-0.10013058	0.52425842	1	0	0	1
2032	520.00000	-0.05796602	0.52425842	2	0	0	1
2033	444.87180	0.11069225	0.52425842	1	0	0	1
2034	376.20511	0.11069225	0.52425842	1	0	0	1
2035	621.79486	0.11069225	0.52425842	1	0	0	1
2036	541.94873	0.15285682	0.52425842	1	0	0	1
2037	589.94873	0.15285682	0.52425842	2	0	0	1
2038	233.48717	0.19502138	0.52425842	1	0	0	1
2039	327.43588	0.19502138	0.52425842	2	0	0	1
2040	364.87180	0.19502138	0.52425842	1	0	0	1
2041	333.12820	0.23718595	0.52425842	1	0	0	1
2042	392.51285	0.23718595	0.52425842	2	0	0	1
2043	378.25641	0.23718595	0.52425842	1	0	0	1
2044	475.33334	0.23718595	0.52425842	1	0	0	1
2045	345.33334	0.27935051	0.52425842	1	0	0	1
2046	521.69232	0.32151508	0.52425842	1	0	0	1

2047	582.61536	0.32151508	0.52425842	2	0	0	1
2048	408.66669	0.32151508	0.52425842	1	0	0	1
2049	430.92307	0.32151508	0.52425842	2	0	0	1
2050	474.25641	0.36367965	0.52425842	1	0	0	1
2051	259.89743	0.40584421	0.52425842	1	0	0	1
2052	487.69232	0.44800878	0.52425842	1	0	0	1
2053	342.82053	0.44800878	0.52425842	1	0	0	1
2054	326.30768	0.44800878	0.52425842	2	0	0	1
2055	455.33334	0.44800878	0.52425842	2	0	0	1
2056	481.33331	0.49017335	0.52425842	1	0	0	1
2057	664.30768	0.53233791	0.52425842	1	0	0	1
2058	665.79492	0.57450248	0.52425842	1	0	0	1
2059	554.35895	0.65883161	0.52425842	1	0	0	1
2060	530.20514	0.70099618	0.52425842	1	0	0	1
2061	445.94870	0.70099618	0.52425842	1	0	0	1
2062	595.12823	0.78532531	0.52425842	1	0	0	1
2063	486.30768	0.82748988	0.52425842	1	0	0	1
2064	479.53845	0.86965445	0.52425842	1	0	0	1
2065	478.46155	1.33346468	0.52425842	1	0	0	1
2066	480.46155	1.37562925	0.52425842	1	0	0	1
2067	865.38464	1.37562925	0.52425842	3	0	0	1
2068	584.10254	1.37562925	0.52425842	1	0	0	1
2069	788.76923	1.50212295	0.52425842	2	0	0	1
2070	1135.84607	2.00809774	0.52425842	3	1	0	1
2071	307.48715	-1.87104238	0.56909095	2	0	0	1
2072	280.30768	-1.66021955	0.56909095	2	0	0	1
2073	273.94873	-1.66021955	0.56909095	1	0	0	1
2074	223.48717	-1.61805498	0.56909095	2	0	0	1
2075	248.66666	-1.61805498	0.56909095	2	0	0	1
2076	308.35898	-1.57589041	0.56909095	1	0	0	1
2077	340.35898	-1.32290301	0.56909095	1	0	0	1
2078	330.61539	-1.02775105	0.56909095	1	0	0	1
2079	514.46155	-0.90125735	0.56909095	1	0	0	1
2080	296.82050	-0.77476365	0.56909095	1	0	0	1
2081	534.30768	-0.73259908	0.56909095	1	0	1	1
2082	506.66666	-0.73259908	0.56909095	1	0	0	1
2083	329.89743	-0.73259908	0.56909095	1	0	0	1
2084	382.51285	-0.69043452	0.56909095	1	0	0	1
2085	355.69229	-0.64826995	0.56909095	1	0	0	1
2086	511.23077	-0.56394082	0.56909095	1	0	0	1
2087	383.94873	-0.56394082	0.56909095	1	0	0	1
2088	456.20511	-0.52177625	0.56909095	1	0	0	1
2089	412.46152	-0.52177625	0.56909095	1	0	0	1

2090	402.97437	-0.52177625	0.56909095	1	0	0	1
2091	333.17950	-0.47961168	0.56909095	2	0	0	1
2092	411.38461	-0.47961168	0.56909095	1	0	0	1
2093	444.35898	-0.47961168	0.56909095	1	0	0	1
2094	238.00000	-0.39528255	0.56909095	1	0	0	1
2095	550.92310	-0.26878885	0.56909095	1	0	0	1
2096	385.23077	-0.26878885	0.56909095	1	0	0	1
2097	534.15381	0.15285682	0.56909095	2	0	0	1
2098	417.02563	0.23718595	0.56909095	1	0	0	1
2099	604.30768	0.23718595	0.56909095	1	0	0	1
2100	754.51282	0.23718595	0.56909095	3	0	0	1
2101	613.94867	0.36367965	0.56909095	3	0	0	1
2102	441.58972	0.40584421	0.56909095	1	0	0	1
2103	396.92307	0.44800878	0.56909095	1	0	0	1
2104	362.05127	0.49017335	0.56909095	1	0	0	1
2105	646.10260	0.53233791	0.56909095	2	0	1	1
2106	452.35895	0.53233791	0.56909095	1	0	0	1
2107	473.79489	0.53233791	0.56909095	1	0	0	1
2108	676.56415	0.57450248	0.56909095	2	0	0	1
2109	637.23077	0.57450248	0.56909095	1	0	0	1
2110	356.82050	0.61666705	0.56909095	2	0	0	1
2111	340.61539	0.61666705	0.56909095	2	1	0	1
2112	579.17950	0.74316075	0.56909095	1	0	0	1
2113	329.89743	0.74316075	0.56909095	3	0	0	1
2114	339.07693	0.82748988	0.56909095	1	0	0	1
2115	451.53845	1.20697098	0.56909095	2	0	0	1
2116	864.41022	1.58645208	0.56909095	3	0	0	1
2117	595.43591	1.62861664	0.56909095	1	1	0	1
2118	824.46149	1.67078121	0.56909095	2	0	0	1
2119	213.33333	-1.91320695	0.61392348	1	0	0	1
2120	218.56410	-1.87104238	0.61392348	1	0	0	1
2121	214.30769	-1.66021955	0.61392348	3	0	0	1
2122	168.97437	-1.57589041	0.61392348	1	0	0	1
2123	275.64102	-1.57589041	0.61392348	1	0	0	1
2124	242.71794	-1.53372585	0.61392348	2	0	0	1
2125	303.02567	-1.44939671	0.61392348	1	0	0	1
2126	374.10257	-1.36506758	0.61392348	1	0	0	1
2127	357.28204	-1.36506758	0.61392348	1	0	0	1
2128	415.74359	-1.36506758	0.61392348	1	0	0	1
2129	289.84616	-1.32290301	0.61392348	2	0	0	1
2130	274.97437	-1.23857388	0.61392348	1	0	0	1
2131	257.74359	-1.19640931	0.61392348	1	0	0	1
2132	264.61539	-1.15424475	0.61392348	1	0	0	1

2133	318.61539	-1.15424475	0.61392348	2	0	0	1
2134	369.79486	-1.11208018	0.61392348	1	0	0	1
2135	383.02567	-0.98558648	0.61392348	1	0	0	1
2136	296.41025	-0.90125735	0.61392348	1	0	0	1
2137	274.51282	-0.73259908	0.61392348	1	1	0	1
2138	335.58975	-0.56394082	0.61392348	1	0	0	1
2139	332.87180	-0.52177625	0.61392348	1	0	0	1
2140	454.05130	-0.47961168	0.61392348	1	0	0	1
2141	453.53848	-0.47961168	0.61392348	1	0	0	1
2142	454.00000	-0.47961168	0.61392348	1	0	0	1
2143	511.48718	-0.43744712	0.61392348	1	0	0	1
2144	466.66666	-0.39528255	0.61392348	2	0	0	1
2145	402.10254	-0.39528255	0.61392348	1	0	0	1
2146	582.92303	-0.26878885	0.61392348	1	0	0	1
2147	446.71793	-0.14229515	0.61392348	1	0	0	1
2148	366.46152	0.02636312	0.61392348	2	0	0	1
2149	557.53845	0.02636312	0.61392348	1	0	0	1
2150	601.17950	0.06852768	0.61392348	1	1	0	1
2151	521.53845	0.11069225	0.61392348	1	0	1	1
2152	349.64102	0.11069225	0.61392348	1	0	0	1
2153	471.84613	0.15285682	0.61392348	1	0	0	1
2154	418.00000	0.19502138	0.61392348	1	0	0	1
2155	645.53851	0.19502138	0.61392348	2	0	0	1
2156	347.53848	0.19502138	0.61392348	1	0	0	1
2157	308.00000	0.19502138	0.61392348	2	0	0	1
2158	404.25641	0.27935051	0.61392348	1	0	0	1
2159	398.25641	0.36367965	0.61392348	1	0	0	1
2160	417.64105	0.49017335	0.61392348	2	0	0	1
2161	732.30768	0.53233791	0.61392348	2	0	0	1
2162	518.41028	0.53233791	0.61392348	1	0	0	1
2163	359.43591	0.53233791	0.61392348	1	0	0	1
2164	638.30768	0.57450248	0.61392348	1	0	0	1
2165	417.02563	0.57450248	0.61392348	1	0	0	1
2166	424.87180	0.65883161	0.61392348	1	0	0	1
2167	799.07690	0.70099618	0.61392348	1	0	0	1
2168	533.28204	0.70099618	0.61392348	1	0	0	1
2169	703.02563	0.70099618	0.61392348	2	1	0	1
2170	672.66663	0.70099618	0.61392348	2	0	0	1
2171	698.35901	0.74316075	0.61392348	2	0	0	1
2172	422.92310	0.78532531	0.61392348	1	0	0	1
2173	556.71796	0.78532531	0.61392348	1	0	0	1
2174	649.74359	0.78532531	0.61392348	1	0	0	1
2175	574.10254	0.95398358	0.61392348	1	0	0	1

2176	570.71796	0.99614815	0.61392348	1	0	0	1
2177	751.02563	1.08047728	0.61392348	2	0	0	1
2178	883.48718	1.16480641	0.61392348	3	0	0	1
2179	763.43585	1.20697098	0.61392348	2	0	0	1
2180	740.05127	1.37562925	0.61392348	3	0	0	1
2181	529.23077	1.37562925	0.61392348	2	1	0	1
2182	591.38458	1.45995838	0.61392348	1	0	0	1
2183	605.28210	1.45995838	0.61392348	1	0	0	1
2184	578.35901	1.62861664	0.61392348	1	1	0	1
2185	356.92307	-1.82887781	0.65875601	1	0	0	1
2186	355.74359	-1.61805498	0.65875601	2	0	0	1
2187	256.00000	-1.61805498	0.65875601	2	0	0	1
2188	319.53845	-1.61805498	0.65875601	2	0	0	1
2189	293.48718	-1.57589041	0.65875601	2	0	0	1
2190	258.41025	-1.57589041	0.65875601	1	0	0	1
2191	306.76923	-1.57589041	0.65875601	2	0	0	1
2192	287.48715	-1.57589041	0.65875601	2	0	0	1
2193	327.17947	-1.53372585	0.65875601	2	0	0	1
2194	291.94870	-1.49156128	0.65875601	1	0	0	1
2195	296.41025	-1.49156128	0.65875601	2	0	0	1
2196	320.76923	-1.40723215	0.65875601	1	0	0	1
2197	398.46155	-1.36506758	0.65875601	1	0	0	1
2198	368.30771	-1.36506758	0.65875601	2	0	0	1
2199	285.64102	-1.36506758	0.65875601	1	0	0	1
2200	234.92308	-1.32290301	0.65875601	1	0	0	1
2201	394.71796	-1.28073845	0.65875601	1	0	0	1
2202	437.58975	-1.28073845	0.65875601	2	0	0	1
2203	389.58975	-1.28073845	0.65875601	1	0	0	1
2204	364.30771	-1.23857388	0.65875601	2	0	0	1
2205	319.89743	-1.23857388	0.65875601	1	0	0	1
2206	305.12820	-1.15424475	0.65875601	1	0	0	1
2207	302.92310	-1.06991561	0.65875601	1	0	0	1
2208	447.43588	-1.02775105	0.65875601	2	0	0	1
2209	288.82053	-0.94342192	0.65875601	1	0	0	1
2210	342.10254	-0.94342192	0.65875601	1	0	0	1
2211	345.38461	-0.85909278	0.65875601	1	0	0	1
2212	541.38458	-0.85909278	0.65875601	1	0	0	1
2213	606.97437	-0.77476365	0.65875601	2	0	0	1
2214	443.43591	-0.77476365	0.65875601	1	0	0	1
2215	422.25641	-0.73259908	0.65875601	1	0	0	1
2216	292.82053	-0.69043452	0.65875601	1	0	0	1
2217	526.97437	-0.69043452	0.65875601	1	0	0	1
2218	425.28204	-0.69043452	0.65875601	1	0	0	1

2219	336.41025	-0.64826995	0.65875601	1	0	0	1
2220	530.35895	-0.60610538	0.65875601	1	0	0	1
2221	316.35898	-0.52177625	0.65875601	1	0	0	1
2222	376.97433	-0.52177625	0.65875601	1	0	0	1
2223	374.61539	-0.52177625	0.65875601	1	0	0	1
2224	500.92307	-0.52177625	0.65875601	2	0	0	1
2225	407.69232	-0.47961168	0.65875601	1	0	0	1
2226	446.61539	-0.39528255	0.65875601	2	0	0	1
2227	413.17950	-0.22662428	0.65875601	1	0	0	1
2228	489.79486	-0.18445972	0.65875601	1	0	0	1
2229	709.28204	-0.18445972	0.65875601	2	0	0	1
2230	581.84613	-0.14229515	0.65875601	2	0	0	1
2231	267.89746	-0.14229515	0.65875601	1	0	0	1
2232	412.20511	-0.10013058	0.65875601	1	0	0	1
2233	520.41028	-0.05796602	0.65875601	2	0	0	1
2234	525.38464	-0.01580145	0.65875601	2	0	0	1
2235	602.87177	0.02636312	0.65875601	1	1	0	1
2236	471.79486	0.06852768	0.65875601	1	0	0	1
2237	441.69229	0.06852768	0.65875601	1	0	0	1
2238	623.43585	0.11069225	0.65875601	1	0	0	1
2239	414.35898	0.11069225	0.65875601	1	1	0	1
2240	651.84613	0.23718595	0.65875601	1	0	0	1
2241	512.61536	0.27935051	0.65875601	2	0	0	1
2242	587.58978	0.27935051	0.65875601	1	0	0	1
2243	686.41028	0.32151508	0.65875601	1	0	0	1
2244	535.07697	0.32151508	0.65875601	1	0	0	1
2245	539.58972	0.32151508	0.65875601	1	0	0	1
2246	527.23077	0.36367965	0.65875601	1	0	0	1
2247	549.64105	0.44800878	0.65875601	1	0	0	1
2248	537.69232	0.44800878	0.65875601	1	0	0	1
2249	270.92307	0.49017335	0.65875601	1	0	0	1
2250	480.15384	0.49017335	0.65875601	1	0	0	1
2251	375.69229	0.53233791	0.65875601	1	0	0	1
2252	331.64102	0.53233791	0.65875601	1	0	0	1
2253	521.28204	0.57450248	0.65875601	2	0	0	1
2254	510.66666	0.61666705	0.65875601	1	0	0	1
2255	505.69229	0.82748988	0.65875601	1	0	0	1
2256	528.92310	0.82748988	0.65875601	1	0	0	1
2257	512.71796	0.82748988	0.65875601	1	0	0	1
2258	522.05127	0.82748988	0.65875601	1	0	0	1
2259	626.97437	0.86965445	0.65875601	2	0	0	1
2260	468.71796	0.91181901	0.65875601	1	0	0	1
2261	751.94873	0.95398358	0.65875601	1	0	0	1

2262	448.10257	1.08047728	0.65875601	2	0	0	1
2263	649.38464	1.12264185	0.65875601	1	0	0	1
2264	529.79486	1.16480641	0.65875601	1	0	0	1
2265	637.48718	1.24913555	0.65875601	1	0	0	1
2266	259.84616	1.24913555	0.65875601	2	0	0	1
2267	390.76923	1.24913555	0.65875601	2	0	0	1
2268	636.41028	1.37562925	0.65875601	2	0	0	1
2269	610.10254	1.41779381	0.65875601	1	0	0	1
2270	556.76923	1.71294578	0.65875601	1	0	0	1
2271	1138.76929	2.21892058	0.65875601	3	0	0	1
2272	330.82050	-1.99753608	0.70358854	2	0	0	1
2273	211.64104	-1.95537151	0.70358854	2	0	0	1
2274	238.82053	-1.87104238	0.70358854	2	0	0	1
2275	284.35898	-1.74454868	0.70358854	1	0	0	1
2276	291.79486	-1.74454868	0.70358854	2	0	0	0
2277	326.35898	-1.70238411	0.70358854	2	0	0	1
2278	387.07690	-1.70238411	0.70358854	2	0	0	1
2279	169.89743	-1.70238411	0.70358854	2	0	0	1
2280	334.51282	-1.66021955	0.70358854	2	0	0	1
2281	414.15384	-1.61805498	0.70358854	1	0	0	1
2282	336.97433	-1.57589041	0.70358854	1	0	0	1
2283	298.56412	-1.57589041	0.70358854	1	0	0	1
2284	306.20511	-1.57589041	0.70358854	2	0	0	1
2285	353.07693	-1.57589041	0.70358854	2	0	0	1
2286	269.33334	-1.57589041	0.70358854	2	0	0	1
2287	200.82051	-1.57589041	0.70358854	1	0	0	1
2288	283.33334	-1.57589041	0.70358854	2	0	0	1
2289	378.05130	-1.53372585	0.70358854	1	0	0	1
2290	301.69229	-1.53372585	0.70358854	1	0	0	1
2291	333.53848	-1.49156128	0.70358854	1	0	0	1
2292	356.25641	-1.49156128	0.70358854	1	0	0	1
2293	136.66667	-1.49156128	0.70358854	1	0	0	1
2294	379.58975	-1.49156128	0.70358854	1	0	0	1
2295	346.97433	-1.49156128	0.70358854	2	0	0	1
2296	264.41025	-1.49156128	0.70358854	1	0	0	1
2297	235.38461	-1.49156128	0.70358854	2	0	0	1
2298	256.97437	-1.49156128	0.70358854	1	0	0	1
2299	233.79488	-1.49156128	0.70358854	1	0	0	1
2300	343.23077	-1.44939671	0.70358854	1	0	0	1
2301	255.58974	-1.44939671	0.70358854	1	0	0	1
2302	349.94873	-1.44939671	0.70358854	2	0	0	1
2303	316.30768	-1.44939671	0.70358854	1	0	0	1
2304	396.00000	-1.44939671	0.70358854	2	0	0	1

2305	256.66666	-1.44939671	0.70358854	1	0	0	1
2306	329.69232	-1.40723215	0.70358854	2	0	0	1
2307	341.33331	-1.40723215	0.70358854	1	0	0	1
2308	320.10257	-1.36506758	0.70358854	1	0	0	1
2309	311.12820	-1.36506758	0.70358854	1	0	0	1
2310	344.56412	-1.36506758	0.70358854	2	0	0	1
2311	270.10257	-1.36506758	0.70358854	1	0	0	1
2312	344.46155	-1.36506758	0.70358854	2	0	0	1
2313	330.35898	-1.36506758	0.70358854	1	0	0	1
2314	202.25641	-1.32290301	0.70358854	1	0	0	1
2315	367.33331	-1.32290301	0.70358854	2	0	0	1
2316	240.87180	-1.32290301	0.70358854	1	0	0	1
2317	404.25641	-1.32290301	0.70358854	2	0	0	1
2318	326.25641	-1.32290301	0.70358854	1	0	0	1
2319	331.02563	-1.28073845	0.70358854	2	0	0	1
2320	353.94873	-1.28073845	0.70358854	1	0	0	1
2321	334.41025	-1.23857388	0.70358854	1	0	1	1
2322	423.28207	-1.23857388	0.70358854	2	0	0	1
2323	182.30769	-1.23857388	0.70358854	1	0	0	1
2324	429.28204	-1.23857388	0.70358854	2	0	0	1
2325	282.92310	-1.19640931	0.70358854	2	0	0	1
2326	434.66666	-1.19640931	0.70358854	2	0	0	1
2327	349.38461	-1.15424475	0.70358854	1	0	0	1
2328	274.30771	-1.15424475	0.70358854	1	0	0	1
2329	336.05127	-1.11208018	0.70358854	1	0	0	1
2330	444.25641	-1.11208018	0.70358854	1	0	0	1
2331	557.33337	-1.11208018	0.70358854	2	0	0	1
2332	435.58975	-1.02775105	0.70358854	1	0	0	1
2333	502.00000	-1.02775105	0.70358854	1	0	0	1
2334	359.33334	-0.98558648	0.70358854	1	0	0	1
2335	372.97437	-0.98558648	0.70358854	2	0	0	1
2336	304.00000	-0.94342192	0.70358854	2	0	0	1
2337	270.87180	-0.94342192	0.70358854	1	0	0	1
2338	266.87180	-0.85909278	0.70358854	1	0	0	1
2339	309.07693	-0.85909278	0.70358854	1	0	0	1
2340	388.00000	-0.85909278	0.70358854	1	0	0	1
2341	254.05128	-0.85909278	0.70358854	1	0	0	1
2342	390.51282	-0.81692822	0.70358854	3	0	0	1
2343	388.25641	-0.81692822	0.70358854	1	0	0	1
2344	426.51282	-0.81692822	0.70358854	1	0	0	1
2345	278.41028	-0.81692822	0.70358854	1	0	0	1
2346	250.35898	-0.81692822	0.70358854	1	0	0	1
2347	314.20514	-0.77476365	0.70358854	1	0	0	1

2348	322.51285	-0.77476365	0.70358854	1	0	0	1
2349	325.23077	-0.77476365	0.70358854	1	0	0	1
2350	453.84616	-0.73259908	0.70358854	1	0	0	1
2351	475.79486	-0.73259908	0.70358854	2	0	0	1
2352	615.07697	-0.73259908	0.70358854	1	1	0	1
2353	505.89743	-0.73259908	0.70358854	1	0	0	1
2354	394.61539	-0.73259908	0.70358854	1	0	0	1
2355	501.02563	-0.69043452	0.70358854	1	0	0	1
2356	280.61539	-0.69043452	0.70358854	1	0	0	1
2357	474.35898	-0.64826995	0.70358854	1	0	0	1
2358	246.15384	-0.64826995	0.70358854	1	0	0	1
2359	495.17947	-0.64826995	0.70358854	1	1	0	1
2360	367.02563	-0.60610538	0.70358854	1	0	0	1
2361	337.38461	-0.60610538	0.70358854	2	0	0	1
2362	415.33334	-0.56394082	0.70358854	1	0	0	1
2363	368.56412	-0.56394082	0.70358854	2	0	0	1
2364	356.30768	-0.56394082	0.70358854	1	0	0	1
2365	485.33334	-0.52177625	0.70358854	2	0	0	1
2366	237.02565	-0.52177625	0.70358854	1	0	0	1
2367	512.97437	-0.47961168	0.70358854	1	0	0	1
2368	385.17947	-0.47961168	0.70358854	1	0	0	1
2369	479.12820	-0.47961168	0.70358854	1	0	0	1
2370	369.33334	-0.43744712	0.70358854	1	0	0	1
2371	650.66669	-0.43744712	0.70358854	2	0	0	1
2372	595.53851	-0.43744712	0.70358854	2	0	0	1
2373	414.97437	-0.43744712	0.70358854	1	1	0	1
2374	576.10260	-0.43744712	0.70358854	2	0	0	1
2375	390.71796	-0.43744712	0.70358854	2	0	0	1
2376	339.48718	-0.39528255	0.70358854	1	0	0	1
2377	566.76923	-0.39528255	0.70358854	1	0	0	1
2378	376.61539	-0.39528255	0.70358854	1	0	0	1
2379	463.64102	-0.39528255	0.70358854	1	0	0	1
2380	366.46152	-0.39528255	0.70358854	1	0	0	1
2381	434.46155	-0.35311798	0.70358854	1	0	0	1
2382	368.46155	-0.31095342	0.70358854	1	0	0	1
2383	599.89746	-0.31095342	0.70358854	2	0	0	1
2384	536.61542	-0.31095342	0.70358854	2	0	0	1
2385	610.82050	-0.31095342	0.70358854	1	0	0	1
2386	372.87180	-0.31095342	0.70358854	1	0	0	1
2387	574.05127	-0.26878885	0.70358854	1	0	0	1
2388	307.79489	-0.22662428	0.70358854	2	0	0	1
2389	312.46152	-0.22662428	0.70358854	2	0	0	1
2390	644.97430	-0.22662428	0.70358854	1	0	0	1

2391	231.94872	-0.22662428	0.70358854	1	0	0	1
2392	537.12823	-0.14229515	0.70358854	1	0	0	1
2393	510.25641	-0.14229515	0.70358854	1	0	0	1
2394	432.51285	-0.14229515	0.70358854	1	0	0	1
2395	388.87180	-0.14229515	0.70358854	1	0	0	1
2396	391.84613	-0.14229515	0.70358854	1	0	0	1
2397	543.23077	-0.14229515	0.70358854	1	0	0	1
2398	620.66669	-0.10013058	0.70358854	1	0	0	1
2399	568.61542	-0.10013058	0.70358854	1	0	1	1
2400	579.28204	-0.10013058	0.70358854	1	0	0	1
2401	643.74359	-0.10013058	0.70358854	2	0	0	1
2402	314.10257	-0.10013058	0.70358854	1	0	0	1
2403	448.71796	-0.10013058	0.70358854	1	0	0	1
2404	551.28204	-0.05796602	0.70358854	1	0	0	1
2405	409.38461	-0.05796602	0.70358854	1	0	0	1
2406	590.15387	-0.05796602	0.70358854	1	1	0	1
2407	291.38461	-0.01580145	0.70358854	1	0	0	1
2408	513.89740	-0.01580145	0.70358854	1	1	0	1
2409	402.25641	-0.01580145	0.70358854	1	0	0	1
2410	573.07690	-0.01580145	0.70358854	1	1	0	1
2411	495.07693	0.02636312	0.70358854	1	0	0	1
2412	435.74359	0.02636312	0.70358854	2	0	0	1
2413	372.00000	0.02636312	0.70358854	1	0	0	1
2414	505.17947	0.02636312	0.70358854	1	0	0	1
2415	602.82050	0.02636312	0.70358854	1	0	0	1
2416	220.87180	0.02636312	0.70358854	1	0	0	1
2417	649.84613	0.06852768	0.70358854	1	0	0	1
2418	467.48715	0.06852768	0.70358854	1	0	0	1
2419	534.56409	0.06852768	0.70358854	1	0	0	1
2420	522.56409	0.11069225	0.70358854	1	0	0	1
2421	600.97437	0.11069225	0.70358854	1	0	0	1
2422	366.51282	0.11069225	0.70358854	2	0	0	1
2423	655.58978	0.11069225	0.70358854	1	0	0	1
2424	331.17947	0.15285682	0.70358854	1	0	0	1
2425	478.05130	0.19502138	0.70358854	1	0	0	1
2426	571.58972	0.19502138	0.70358854	1	0	0	1
2427	314.46155	0.19502138	0.70358854	1	0	0	1
2428	703.89740	0.19502138	0.70358854	1	0	0	1
2429	522.46155	0.19502138	0.70358854	2	1	0	1
2430	605.12823	0.19502138	0.70358854	1	0	0	1
2431	661.17950	0.19502138	0.70358854	1	0	0	1
2432	521.69232	0.19502138	0.70358854	2	0	0	1
2433	871.33331	0.23718595	0.70358854	3	0	0	1

2434	784.76923	0.23718595	0.70358854	2	0	0	1
2435	227.79488	0.23718595	0.70358854	1	0	0	0
2436	400.15384	0.27935051	0.70358854	1	0	0	1
2437	596.35901	0.32151508	0.70358854	1	0	0	1
2438	524.66663	0.32151508	0.70358854	1	0	0	1
2439	548.51282	0.32151508	0.70358854	1	0	0	1
2440	392.92310	0.32151508	0.70358854	2	0	0	1
2441	624.25641	0.40584421	0.70358854	1	1	1	1
2442	451.23077	0.40584421	0.70358854	2	0	0	1
2443	795.58978	0.44800878	0.70358854	2	0	1	1
2444	284.41025	0.49017335	0.70358854	1	0	0	1
2445	349.33334	0.49017335	0.70358854	1	0	0	1
2446	530.61536	0.49017335	0.70358854	2	0	0	1
2447	658.66669	0.49017335	0.70358854	1	1	0	1
2448	426.87180	0.49017335	0.70358854	1	0	0	1
2449	368.76923	0.53233791	0.70358854	1	0	0	1
2450	513.17950	0.53233791	0.70358854	2	0	0	1
2451	440.66666	0.53233791	0.70358854	1	0	0	1
2452	597.48718	0.53233791	0.70358854	2	0	0	1
2453	852.66663	0.53233791	0.70358854	3	0	0	1
2454	738.00000	0.53233791	0.70358854	2	0	0	1
2455	156.00000	0.57450248	0.70358854	2	0	0	0
2456	416.82050	0.61666705	0.70358854	1	0	0	1
2457	500.97437	0.65883161	0.70358854	1	0	0	1
2458	448.30771	0.65883161	0.70358854	1	0	0	1
2459	438.41028	0.65883161	0.70358854	2	0	0	1
2460	684.66663	0.65883161	0.70358854	1	0	0	1
2461	488.30771	0.65883161	0.70358854	1	0	0	1
2462	453.53848	0.65883161	0.70358854	1	0	0	1
2463	383.28207	0.70099618	0.70358854	2	0	0	1
2464	451.69229	0.74316075	0.70358854	1	1	0	1
2465	435.53845	0.74316075	0.70358854	1	0	0	1
2466	597.28204	0.78532531	0.70358854	2	0	0	1
2467	648.00000	0.78532531	0.70358854	1	0	0	1
2468	373.02567	0.82748988	0.70358854	1	0	0	1
2469	601.28204	0.82748988	0.70358854	1	0	0	1
2470	476.15384	0.82748988	0.70358854	2	1	0	1
2471	664.92303	0.82748988	0.70358854	2	0	0	1
2472	445.02563	0.82748988	0.70358854	1	0	0	1
2473	494.87180	0.86965445	0.70358854	1	0	0	1
2474	527.38464	1.03831271	0.70358854	1	0	0	1
2475	437.17947	1.03831271	0.70358854	1	0	0	1
2476	878.30768	1.08047728	0.70358854	1	1	0	1

2477	897.74359	1.12264185	0.70358854	1	0	0	1
2478	1029.17944	1.12264185	0.70358854	2	0	1	1
2479	794.97430	1.16480641	0.70358854	2	1	0	1
2480	472.61539	1.24913555	0.70358854	1	0	0	1
2481	530.41028	1.29130011	0.70358854	2	0	0	1
2482	736.15387	1.37562925	0.70358854	1	0	0	1
2483	568.15387	1.37562925	0.70358854	1	0	0	1
2484	929.89746	1.37562925	0.70358854	1	1	0	1
2485	720.46155	1.62861664	0.70358854	2	1	0	1
2486	420.56409	1.67078121	0.70358854	2	0	0	1
2487	992.87177	1.67078121	0.70358854	1	1	1	1
2488	815.58978	1.67078121	0.70358854	3	0	0	1
2489	962.05127	1.92376861	0.70358854	1	1	0	1
2490	329.79486	-1.87104238	0.74842108	1	0	0	1
2491	237.74359	-1.74454868	0.74842108	2	0	0	1
2492	353.74359	-1.57589041	0.74842108	1	0	0	1
2493	278.61539	-1.57589041	0.74842108	1	0	0	1
2494	201.53847	-1.49156128	0.74842108	1	0	0	1
2495	336.10257	-1.49156128	0.74842108	2	0	0	1
2496	238.56410	-1.44939671	0.74842108	1	0	0	1
2497	407.84616	-1.44939671	0.74842108	1	0	0	1
2498	482.61539	-1.36506758	0.74842108	1	0	0	1
2499	296.00000	-1.15424475	0.74842108	1	0	0	1
2500	303.74359	-1.11208018	0.74842108	1	0	0	1
2501	329.58975	-1.06991561	0.74842108	1	0	0	1
2502	261.12820	-1.06991561	0.74842108	1	0	0	1
2503	394.46155	-0.98558648	0.74842108	1	1	0	1
2504	446.15384	-0.98558648	0.74842108	1	0	0	1
2505	311.69229	-0.98558648	0.74842108	1	0	0	1
2506	430.92307	-0.94342192	0.74842108	2	0	0	1
2507	395.58975	-0.73259908	0.74842108	1	0	0	1
2508	424.05130	-0.69043452	0.74842108	2	0	0	1
2509	359.84616	-0.60610538	0.74842108	1	0	0	1
2510	472.56412	-0.60610538	0.74842108	1	0	1	1
2511	475.07693	-0.47961168	0.74842108	1	0	1	1
2512	622.10254	-0.47961168	0.74842108	2	0	0	1
2513	552.46155	-0.47961168	0.74842108	2	0	0	1
2514	367.64105	-0.35311798	0.74842108	2	0	0	1
2515	498.00000	-0.31095342	0.74842108	2	0	0	1
2516	632.41022	-0.22662428	0.74842108	2	0	0	1
2517	384.51282	-0.22662428	0.74842108	1	0	0	1
2518	400.10257	-0.14229515	0.74842108	2	0	0	1
2519	448.20514	-0.14229515	0.74842108	2	0	0	1

2520	490.00000	-0.01580145	0.74842108	2	1	0	1
2521	466.51282	0.02636312	0.74842108	1	0	0	1
2522	479.43591	0.02636312	0.74842108	1	0	0	1
2523	395.53845	0.06852768	0.74842108	2	1	0	1
2524	514.00000	0.11069225	0.74842108	1	0	0	1
2525	429.43591	0.15285682	0.74842108	1	0	0	1
2526	484.97437	0.23718595	0.74842108	1	0	0	1
2527	571.64099	0.27935051	0.74842108	1	0	0	1
2528	634.46149	0.32151508	0.74842108	2	0	0	1
2529	555.48718	0.44800878	0.74842108	2	0	0	1
2530	696.15387	0.49017335	0.74842108	2	0	0	1
2531	434.00000	0.53233791	0.74842108	1	0	0	1
2532	444.97437	0.53233791	0.74842108	1	0	0	1
2533	435.48718	0.53233791	0.74842108	1	0	0	1
2534	384.15384	0.57450248	0.74842108	1	0	0	1
2535	568.10260	0.61666705	0.74842108	2	1	0	1
2536	571.43591	0.65883161	0.74842108	1	0	0	1
2537	565.02570	0.70099618	0.74842108	1	0	0	1
2538	443.17950	0.70099618	0.74842108	1	0	0	1
2539	530.46155	0.70099618	0.74842108	1	0	0	1
2540	420.76923	0.70099618	0.74842108	1	1	0	1
2541	473.89743	0.74316075	0.74842108	2	0	0	1
2542	465.12820	0.74316075	0.74842108	1	0	0	1
2543	485.58975	0.78532531	0.74842108	1	0	0	1
2544	731.94873	0.91181901	0.74842108	2	0	0	1
2545	604.97430	0.91181901	0.74842108	1	0	0	1
2546	749.23077	1.03831271	0.74842108	1	0	0	1
2547	572.10254	1.12264185	0.74842108	1	0	0	1
2548	595.38464	1.24913555	0.74842108	1	0	0	1
2549	945.84619	1.29130011	0.74842108	3	0	0	1
2550	543.74359	1.58645208	0.74842108	1	0	0	1
2551	1170.30774	1.79727491	0.74842108	3	0	0	1
2552	304.05130	-1.66021955	0.79325361	1	0	0	1
2553	318.51282	-1.61805498	0.79325361	1	0	0	1
2554	391.38461	-1.57589041	0.79325361	2	0	0	1
2555	300.25641	-1.49156128	0.79325361	1	0	0	1
2556	286.30768	-1.40723215	0.79325361	1	0	0	1
2557	278.82053	-1.36506758	0.79325361	1	0	0	1
2558	364.71796	-1.28073845	0.79325361	1	0	0	1
2559	354.30771	-1.19640931	0.79325361	1	0	0	1
2560	254.46155	-1.06991561	0.79325361	2	0	0	1
2561	298.15387	-0.94342192	0.79325361	3	0	0	1
2562	346.87180	-0.64826995	0.79325361	1	0	0	1

2563	388.15387	-0.60610538	0.79325361	2	0	0	1
2564	510.00000	-0.22662428	0.79325361	2	0	0	1
2565	386.41025	-0.10013058	0.79325361	1	0	0	1
2566	248.56410	0.19502138	0.79325361	1	0	0	1
2567	515.58978	0.19502138	0.79325361	1	0	0	1
2568	522.56409	0.32151508	0.79325361	2	0	0	1
2569	586.82056	0.40584421	0.79325361	2	1	0	1
2570	338.05130	0.53233791	0.79325361	1	0	0	1
2571	368.05130	0.70099618	0.79325361	1	0	0	1
2572	666.66669	1.24913555	0.79325361	1	1	0	1
2573	482.71796	1.29130011	0.79325361	1	0	0	1
2574	237.17949	-1.36506758	0.83808614	1	0	1	1
2575	525.89746	-1.02775105	0.83808614	2	1	0	1
2576	310.51282	-0.56394082	0.83808614	1	0	0	1
2577	577.17950	0.06852768	0.83808614	1	0	0	1
2578	523.53845	0.32151508	0.83808614	2	0	0	1
2579	483.38461	0.40584421	0.83808614	1	1	0	1
2580	533.53845	0.49017335	0.83808614	2	1	0	1
2581	851.84613	1.58645208	0.83808614	1	0	0	1
2582	410.05127	-1.36506758	0.88291867	1	0	0	1
2583	439.64102	-0.56394082	0.88291867	2	0	0	1
2584	431.07690	-0.31095342	0.88291867	1	0	0	1
2585	805.79492	0.19502138	0.88291867	2	0	0	1
2586	578.41028	0.53233791	0.88291867	2	0	0	1
2587	653.38458	0.91181901	0.88291867	1	0	0	1
2588	929.33331	1.16480641	0.88291867	3	0	0	1
2589	377.94873	-1.44939671	0.92775120	2	0	0	1
2590	314.30771	-1.40723215	0.92775120	2	0	0	1
2591	335.33334	-0.81692822	0.92775120	2	0	0	1
2592	797.58978	-0.43744712	0.92775120	3	0	0	0
2593	573.33331	-0.14229515	0.92775120	2	0	0	1
2594	600.05127	0.23718595	0.92775120	1	0	0	1
2595	523.02563	0.36367965	0.92775120	1	0	0	1
2596	744.35895	0.61666705	0.92775120	2	0	0	1
2597	543.48718	0.86965445	0.92775120	1	0	0	1
2598	1133.12830	1.62861664	0.92775120	2	0	0	1
2599	373.69232	0.19502138	0.97258373	1	0	0	1
2600	167.23077	0.32151508	0.97258373	1	0	0	1
2601	528.10260	0.49017335	0.97258373	1	0	0	1
2602	446.97433	-1.36506758	1.01741627	1	0	0	1
2603	333.43591	-1.32290301	1.01741627	1	0	0	1
2604	424.56412	-0.73259908	1.01741627	1	0	0	1
2605	363.84616	-0.56394082	1.01741627	1	0	1	1

2606	609.48718	-0.47961168	1.01741627	1	0	0	1
2607	498.00000	-0.31095342	1.01741627	1	0	0	1
2608	490.66666	-0.18445972	1.01741627	1	0	1	1
2609	578.10260	-0.05796602	1.01741627	2	0	0	1
2610	502.15384	0.02636312	1.01741627	1	0	0	1
2611	641.48718	0.02636312	1.01741627	1	0	0	1
2612	685.53851	0.82748988	1.01741627	2	1	1	1
2613	572.66663	0.91181901	1.01741627	1	0	1	1
2614	817.43591	0.95398358	1.01741627	1	0	1	1
2615	592.46155	0.99614815	1.01741627	1	0	0	1
2616	680.30768	1.12264185	1.01741627	1	0	0	1
2617	629.53845	1.20697098	1.01741627	1	0	0	1
2618	683.53845	1.41779381	1.01741627	1	0	0	1
2619	737.02563	1.45995838	1.01741627	1	0	0	1
2620	823.94867	1.79727491	1.01741627	1	0	0	1
2621	1067.12817	2.38757884	1.01741627	1	0	0	1
2622	688.97437	2.42974341	1.01741627	1	0	0	1
2623	941.69232	3.31519931	1.01741627	1	0	0	1
2624	426.15384	-0.60610538	1.06224880	1	0	0	1
2625	519.79486	-0.39528255	1.06224880	2	0	0	1
2626	571.84613	-0.39528255	1.06224880	1	0	0	1
2627	424.97437	0.23718595	1.06224880	1	0	0	1
2628	515.43591	0.44800878	1.06224880	2	0	0	1
2629	682.20514	0.53233791	1.06224880	2	1	0	1
2630	792.56409	0.74316075	1.06224880	2	0	0	1
2631	632.46155	0.78532531	1.06224880	2	0	0	1
2632	675.33337	0.82748988	1.06224880	1	0	0	1
2633	356.25641	0.91181901	1.06224880	2	1	0	1
2634	765.38464	1.16480641	1.06224880	1	1	0	1
2635	1208.05127	1.41779381	1.06224880	2	0	0	1
2636	793.12817	1.58645208	1.06224880	1	1	0	1
2637	937.79486	1.58645208	1.06224880	1	0	0	1
2638	839.12823	1.58645208	1.06224880	1	1	0	1
2639	847.53845	2.09242688	1.06224880	1	1	0	1
2640	392.56412	-1.02775105	1.10708133	1	0	0	1
2641	469.89743	-0.85909278	1.10708133	2	0	0	1
2642	517.53845	-0.52177625	1.10708133	1	0	0	1
2643	631.38458	-0.52177625	1.10708133	1	0	0	1
2644	443.23077	-0.43744712	1.10708133	1	0	0	1
2645	445.02563	-0.35311798	1.10708133	1	0	0	1
2646	524.05127	-0.31095342	1.10708133	2	0	0	1
2647	651.12817	-0.14229515	1.10708133	2	0	0	1
2648	646.97437	-0.05796602	1.10708133	1	0	0	1

2649	540.82050	0.02636312	1.10708133	2	0	0	1
2650	560.20514	0.02636312	1.10708133	2	0	0	1
2651	746.30774	0.23718595	1.10708133	2	0	1	1
2652	657.43591	0.40584421	1.10708133	2	1	0	1
2653	775.43591	0.44800878	1.10708133	1	0	0	1
2654	704.71790	0.44800878	1.10708133	1	1	0	1
2655	711.12817	0.53233791	1.10708133	3	1	0	1
2656	706.20514	0.70099618	1.10708133	1	0	0	1
2657	695.12823	0.70099618	1.10708133	1	1	1	1
2658	713.53845	0.82748988	1.10708133	1	0	0	1
2659	712.00000	0.82748988	1.10708133	1	1	0	1
2660	746.05133	0.91181901	1.10708133	2	0	0	1
2661	684.66663	1.12264185	1.10708133	1	0	0	1
2662	734.51282	1.37562925	1.10708133	2	1	0	1
2663	319.43591	-1.11208018	1.15191386	2	0	0	1
2664	467.58975	-0.73259908	1.15191386	2	0	0	1
2665	373.53848	-0.73259908	1.15191386	1	0	0	1
2666	541.38458	-0.69043452	1.15191386	1	0	0	1
2667	266.61539	-0.60610538	1.15191386	2	0	0	1
2668	259.69229	-0.47961168	1.15191386	1	0	0	1
2669	485.89743	-0.18445972	1.15191386	1	0	0	1
2670	558.20514	-0.18445972	1.15191386	1	0	0	1
2671	531.69232	-0.10013058	1.15191386	2	0	1	1
2672	261.28204	0.19502138	1.15191386	1	0	0	1
2673	865.94873	0.32151508	1.15191386	2	0	0	1
2674	653.69226	0.44800878	1.15191386	1	0	0	1
2675	808.76923	0.53233791	1.15191386	2	0	0	1
2676	341.79486	0.61666705	1.15191386	1	0	0	1
2677	792.46155	0.86965445	1.15191386	1	1	0	1
2678	860.30768	1.12264185	1.15191386	1	0	0	1
2679	727.17950	1.50212295	1.15191386	1	1	1	1
2680	286.10257	1.58645208	1.15191386	2	1	1	1
2681	322.82053	-1.82887781	1.19674639	2	0	0	1
2682	350.66666	-1.70238411	1.19674639	1	0	0	1
2683	282.61539	-1.61805498	1.19674639	1	0	0	1
2684	246.76924	-1.57589041	1.19674639	1	0	0	1
2685	313.28207	-1.57589041	1.19674639	2	0	0	1
2686	282.56412	-1.53372585	1.19674639	1	0	0	1
2687	328.82053	-1.49156128	1.19674639	1	0	0	1
2688	272.20511	-1.44939671	1.19674639	1	0	0	1
2689	291.17947	-1.44939671	1.19674639	1	0	1	1
2690	307.48715	-1.23857388	1.19674639	1	0	0	1
2691	567.12823	-1.19640931	1.19674639	2	0	0	1

2692	143.74358	-1.15424475	1.19674639	2	0	0	1
2693	529.84613	-1.06991561	1.19674639	1	0	0	1
2694	372.92310	-1.06991561	1.19674639	1	0	0	1
2695	352.87180	-1.02775105	1.19674639	1	0	0	1
2696	469.43591	-1.02775105	1.19674639	1	0	0	1
2697	396.20511	-1.02775105	1.19674639	1	0	0	1
2698	484.92307	-0.94342192	1.19674639	2	0	0	1
2699	581.43591	-0.85909278	1.19674639	1	0	0	1
2700	431.89743	-0.85909278	1.19674639	1	0	0	1
2701	409.79486	-0.81692822	1.19674639	1	0	0	1
2702	454.35898	-0.81692822	1.19674639	1	0	0	1
2703	522.25641	-0.81692822	1.19674639	1	0	0	1
2704	420.20514	-0.77476365	1.19674639	2	0	0	1
2705	375.12820	-0.77476365	1.19674639	1	0	0	1
2706	616.10260	-0.64826995	1.19674639	2	0	0	1
2707	517.07690	-0.60610538	1.19674639	3	0	0	1
2708	501.89743	-0.60610538	1.19674639	1	0	0	1
2709	359.33334	-0.56394082	1.19674639	1	0	0	1
2710	498.97437	-0.56394082	1.19674639	2	0	0	1
2711	478.92310	-0.56394082	1.19674639	1	0	0	1
2712	348.82053	-0.56394082	1.19674639	2	0	0	1
2713	612.76923	-0.52177625	1.19674639	1	0	0	1
2714	269.23077	-0.52177625	1.19674639	1	0	0	1
2715	507.33331	-0.47961168	1.19674639	1	0	1	1
2716	430.71796	-0.47961168	1.19674639	2	0	0	1
2717	565.12823	-0.39528255	1.19674639	2	0	0	1
2718	344.97437	-0.31095342	1.19674639	1	0	0	1
2719	556.76923	-0.31095342	1.19674639	1	0	0	1
2720	515.33337	-0.31095342	1.19674639	1	0	0	1
2721	435.43588	-0.26878885	1.19674639	1	0	0	1
2722	566.41028	-0.26878885	1.19674639	1	0	0	1
2723	498.51282	-0.26878885	1.19674639	1	0	0	1
2724	618.00000	-0.22662428	1.19674639	1	0	1	1
2725	609.38464	-0.22662428	1.19674639	2	0	0	1
2726	670.71796	-0.10013058	1.19674639	1	0	0	1
2727	503.17950	-0.10013058	1.19674639	1	0	0	1
2728	512.71796	-0.05796602	1.19674639	1	0	1	1
2729	575.07697	0.02636312	1.19674639	1	0	0	1
2730	652.61536	0.02636312	1.19674639	2	0	0	1
2731	402.82053	0.06852768	1.19674639	2	1	0	1
2732	768.71796	0.11069225	1.19674639	1	0	0	1
2733	548.51282	0.15285682	1.19674639	1	0	0	1
2734	574.20508	0.15285682	1.19674639	2	0	0	1

2735	300.66666	0.15285682	1.19674639	1	0	0	1
2736	466.15384	0.19502138	1.19674639	1	1	0	1
2737	541.74359	0.27935051	1.19674639	1	0	0	1
2738	814.56409	0.36367965	1.19674639	1	0	1	1
2739	498.82053	0.36367965	1.19674639	2	0	0	1
2740	770.71796	0.53233791	1.19674639	3	0	0	1
2741	753.17944	0.53233791	1.19674639	1	0	0	1
2742	624.87177	0.65883161	1.19674639	1	0	0	1
2743	660.92310	0.70099618	1.19674639	2	0	0	1
2744	631.12817	0.78532531	1.19674639	1	0	0	1
2745	748.71796	0.86965445	1.19674639	1	1	0	1
2746	475.17947	0.86965445	1.19674639	1	0	0	1
2747	754.97430	0.86965445	1.19674639	1	0	0	1
2748	839.58972	0.91181901	1.19674639	2	0	0	1
2749	1175.33325	0.91181901	1.19674639	1	0	0	1
2750	846.10260	0.95398358	1.19674639	2	0	1	1
2751	821.48718	1.16480641	1.19674639	2	0	0	1
2752	531.28204	1.20697098	1.19674639	1	1	0	1
2753	719.94873	1.37562925	1.19674639	2	0	0	1
2754	613.79486	1.37562925	1.19674639	1	0	0	1
2755	1188.00000	1.54428751	1.19674639	3	1	1	1
2756	780.51282	1.79727491	1.19674639	2	0	0	1
2757	683.53845	1.79727491	1.19674639	1	0	0	1
2758	818.61542	2.21892058	1.19674639	2	1	0	1
2759	345.94870	-1.40723215	1.24157892	2	0	0	1
2760	380.46155	-1.36506758	1.24157892	1	0	0	1
2761	372.05127	-1.02775105	1.24157892	1	0	0	1
2762	430.61539	-0.98558648	1.24157892	2	0	0	1
2763	251.84616	-0.94342192	1.24157892	1	0	1	1
2764	471.74359	-0.94342192	1.24157892	2	0	0	1
2765	448.41028	-0.85909278	1.24157892	1	0	0	1
2766	431.89743	-0.85909278	1.24157892	1	0	1	1
2767	486.87180	-0.73259908	1.24157892	2	0	0	1
2768	562.41022	-0.73259908	1.24157892	2	0	0	0
2769	411.89743	-0.69043452	1.24157892	1	0	0	1
2770	397.84616	-0.60610538	1.24157892	1	0	0	1
2771	556.82056	-0.56394082	1.24157892	1	0	0	1
2772	523.79486	-0.56394082	1.24157892	1	0	0	1
2773	600.97437	-0.56394082	1.24157892	2	0	0	1
2774	491.74359	-0.52177625	1.24157892	1	0	0	1
2775	505.64102	-0.52177625	1.24157892	2	0	0	1
2776	594.71790	-0.47961168	1.24157892	2	1	1	1
2777	505.02563	-0.47961168	1.24157892	1	0	0	1

2778	511.28204	-0.47961168	1.24157892	2	0	0	1
2779	569.23077	-0.39528255	1.24157892	1	0	0	1
2780	532.92303	-0.39528255	1.24157892	1	0	0	1
2781	476.71793	-0.35311798	1.24157892	1	0	0	1
2782	449.28204	-0.35311798	1.24157892	1	0	0	1
2783	520.35901	-0.31095342	1.24157892	2	0	0	1
2784	517.07690	-0.31095342	1.24157892	1	0	0	1
2785	601.17950	-0.22662428	1.24157892	1	0	0	1
2786	1022.41022	-0.22662428	1.24157892	2	1	0	1
2787	575.89746	-0.22662428	1.24157892	1	0	0	1
2788	566.61542	-0.14229515	1.24157892	2	0	0	1
2789	914.05127	-0.10013058	1.24157892	2	0	0	1
2790	682.20514	-0.10013058	1.24157892	1	0	0	1
2791	459.74359	-0.01580145	1.24157892	1	0	0	1
2792	658.46155	-0.01580145	1.24157892	2	0	0	1
2793	544.00000	0.06852768	1.24157892	2	0	0	1
2794	660.15387	0.11069225	1.24157892	1	0	0	1
2795	679.33331	0.11069225	1.24157892	2	0	0	1
2796	634.46149	0.27935051	1.24157892	1	0	0	1
2797	704.10254	0.32151508	1.24157892	1	0	0	1
2798	457.94873	0.32151508	1.24157892	1	1	0	1
2799	647.58978	0.44800878	1.24157892	2	0	0	1
2800	642.10254	0.53233791	1.24157892	2	0	0	1
2801	610.66669	0.61666705	1.24157892	2	0	0	1
2802	984.51282	0.61666705	1.24157892	2	0	0	1
2803	599.94873	0.61666705	1.24157892	1	0	0	1
2804	936.87183	0.61666705	1.24157892	3	0	0	1
2805	880.61536	0.82748988	1.24157892	2	1	1	1
2806	802.35895	0.91181901	1.24157892	1	0	1	1
2807	700.46155	0.99614815	1.24157892	1	0	0	1
2808	932.00000	1.16480641	1.24157892	3	1	0	1
2809	842.20514	1.54428751	1.24157892	2	0	0	1
2810	419.64102	-1.32290301	1.28641146	3	0	0	1
2811	451.79486	-1.19640931	1.28641146	2	0	0	1
2812	347.07690	-0.98558648	1.28641146	1	0	0	1
2813	509.17950	-0.90125735	1.28641146	1	0	0	1
2814	517.69232	-0.81692822	1.28641146	1	0	0	1
2815	649.48718	-0.47961168	1.28641146	2	0	0	1
2816	529.43591	-0.39528255	1.28641146	1	0	1	1
2817	505.23077	-0.22662428	1.28641146	1	0	0	1
2818	625.53851	-0.18445972	1.28641146	1	0	0	1
2819	741.48718	-0.10013058	1.28641146	1	0	1	1
2820	396.05127	-0.01580145	1.28641146	2	0	0	1

2821	740.76923	0.11069225	1.28641146	3	0	0	1
2822	549.58972	0.27935051	1.28641146	1	0	0	1
2823	595.84619	0.57450248	1.28641146	1	0	1	1
2824	538.46155	0.61666705	1.28641146	1	0	0	1
2825	670.82050	0.65883161	1.28641146	1	0	0	1
2826	665.64105	0.74316075	1.28641146	2	0	0	1
2827	589.02563	0.78532531	1.28641146	1	1	0	1
2828	689.84613	0.95398358	1.28641146	1	0	0	1
2829	953.84613	0.99614815	1.28641146	2	0	0	1
2830	807.23077	1.24913555	1.28641146	1	0	0	1
2831	745.07697	1.37562925	1.28641146	1	0	0	1
2832	465.89743	-1.36506758	1.33124399	1	0	0	1
2833	358.76923	-1.23857388	1.33124399	1	0	0	1
2834	133.33333	-1.19640931	1.33124399	1	0	1	1
2835	403.64102	-1.11208018	1.33124399	1	0	0	1
2836	414.61539	-0.85909278	1.33124399	1	0	0	1
2837	422.56412	-0.73259908	1.33124399	2	0	0	1
2838	463.48718	-0.73259908	1.33124399	1	0	0	1
2839	247.23077	-0.69043452	1.33124399	1	0	0	1
2840	158.56410	-0.64826995	1.33124399	2	0	0	1
2841	487.53848	-0.64826995	1.33124399	2	0	0	1
2842	542.71796	-0.56394082	1.33124399	2	0	0	1
2843	447.33331	-0.43744712	1.33124399	1	0	0	1
2844	482.10254	-0.43744712	1.33124399	1	0	0	1
2845	552.46155	-0.39528255	1.33124399	1	0	0	1
2846	621.53845	-0.39528255	1.33124399	1	0	0	1
2847	471.79486	-0.35311798	1.33124399	1	1	0	1
2848	512.20514	-0.35311798	1.33124399	1	0	0	1
2849	573.28204	-0.31095342	1.33124399	2	1	0	1
2850	652.20514	0.15285682	1.33124399	2	0	1	1
2851	768.46155	0.27935051	1.33124399	2	0	0	1
2852	794.25641	0.44800878	1.33124399	1	0	0	1
2853	661.74359	0.49017335	1.33124399	1	0	0	1
2854	581.79486	0.57450248	1.33124399	1	1	1	1
2855	817.23077	0.65883161	1.33124399	2	0	1	1
2856	641.58972	0.70099618	1.33124399	1	0	0	1
2857	1168.00000	1.20697098	1.33124399	2	0	0	1
2858	501.38461	-0.35311798	1.37607652	1	0	0	1
2859	625.79492	-0.31095342	1.37607652	2	0	1	0
2860	830.15387	0.11069225	1.37607652	3	0	0	1
2861	583.23077	0.11069225	1.37607652	1	0	0	1
2862	345.23077	0.36367965	1.37607652	1	0	0	1
2863	365.53845	-1.23857388	1.42090905	1	0	0	1

2864	254.30769	-1.23857388	1.42090905	1	0	0	1
2865	493.02567	-0.81692822	1.42090905	1	0	0	1
2866	868.35901	-0.77476365	1.42090905	2	0	0	1
2867	518.56409	-0.60610538	1.42090905	1	0	0	1
2868	538.30768	-0.56394082	1.42090905	2	0	0	1
2869	425.53845	-0.47961168	1.42090905	1	0	0	1
2870	512.25641	-0.43744712	1.42090905	1	0	0	1
2871	785.33337	0.70099618	1.42090905	2	0	0	1
2872	267.02563	0.74316075	1.42090905	1	1	0	1
2873	668.10260	1.29130011	1.42090905	2	0	0	1
2874	512.56409	-1.23857388	1.46574158	2	0	0	1
2875	616.05133	-0.73259908	1.46574158	1	0	0	1
2876	524.41022	-0.69043452	1.46574158	1	0	0	1
2877	504.15384	-0.69043452	1.46574158	1	0	0	1
2878	544.30768	-0.22662428	1.46574158	1	0	0	1
2879	541.17950	0.11069225	1.46574158	2	0	0	1
2880	792.25641	0.11069225	1.46574158	2	0	0	1
2881	795.07697	0.11069225	1.46574158	1	0	0	1
2882	700.97437	0.36367965	1.46574158	1	0	0	1
2883	410.30768	0.49017335	1.46574158	1	0	0	1
2884	635.02570	0.65883161	1.46574158	2	0	0	1
2885	312.76926	0.82748988	1.46574158	3	0	1	1
2886	450.71796	0.95398358	1.46574158	1	0	0	1
2887	915.64105	1.03831271	1.46574158	1	0	0	1
2888	385.89743	1.08047728	1.46574158	2	0	0	1
2889	1158.25647	1.37562925	1.46574158	2	1	0	1
2890	911.53845	1.71294578	1.46574158	1	0	0	1
2891	414.35898	-1.15424475	1.51057411	2	0	0	1
2892	519.28204	-0.85909278	1.51057411	2	0	0	1
2893	284.61539	-0.73259908	1.51057411	2	0	0	1
2894	591.23077	-0.73259908	1.51057411	2	0	0	1
2895	473.17950	-0.56394082	1.51057411	1	0	0	1
2896	520.92310	-0.56394082	1.51057411	2	0	0	1
2897	558.82050	-0.47961168	1.51057411	3	1	0	1
2898	196.15384	-0.35311798	1.51057411	3	0	0	1
2899	517.02563	-0.31095342	1.51057411	1	0	1	1
2900	1169.58972	1.54428751	1.51057411	2	1	0	1
2901	1236.82056	1.79727491	1.51057411	1	0	0	1
2902	323.58975	-1.66021955	1.55540665	1	0	0	1
2903	495.07693	-1.49156128	1.55540665	2	1	0	1
2904	372.51285	-1.44939671	1.55540665	2	0	0	1
2905	383.79489	-1.40723215	1.55540665	1	0	0	1
2906	407.48715	-1.28073845	1.55540665	1	0	0	1

2907	460.10257	-1.19640931	1.55540665	1	0	0	1
2908	595.64105	-1.15424475	1.55540665	2	0	0	1
2909	502.66669	-1.15424475	1.55540665	1	0	0	1
2910	505.89743	-1.06991561	1.55540665	1	0	0	1
2911	366.76923	-1.02775105	1.55540665	2	0	0	1
2912	438.30771	-1.02775105	1.55540665	2	0	0	1
2913	397.74359	-0.98558648	1.55540665	1	0	0	1
2914	463.43591	-0.98558648	1.55540665	1	0	0	1
2915	513.28204	-0.94342192	1.55540665	1	0	0	1
2916	375.58975	-0.90125735	1.55540665	1	0	0	1
2917	532.61536	-0.90125735	1.55540665	2	0	0	1
2918	436.30768	-0.90125735	1.55540665	1	0	0	1
2919	566.92310	-0.77476365	1.55540665	1	0	0	1
2920	526.51282	-0.73259908	1.55540665	1	0	0	1
2921	379.48718	-0.73259908	1.55540665	1	0	1	1
2922	550.00000	-0.73259908	1.55540665	2	0	0	1
2923	701.64099	-0.56394082	1.55540665	1	0	0	1
2924	495.12820	-0.56394082	1.55540665	1	0	0	1
2925	619.53845	-0.39528255	1.55540665	2	0	0	1
2926	657.07697	-0.35311798	1.55540665	3	0	0	1
2927	644.82050	-0.31095342	1.55540665	2	0	0	1
2928	440.10257	-0.31095342	1.55540665	2	1	0	1
2929	686.25641	-0.31095342	1.55540665	1	1	0	1
2930	674.76923	-0.31095342	1.55540665	2	0	0	1
2931	467.89746	-0.26878885	1.55540665	1	0	0	1
2932	553.69226	-0.18445972	1.55540665	1	0	0	1
2933	584.30768	-0.18445972	1.55540665	1	0	0	1
2934	815.74359	-0.18445972	1.55540665	1	0	0	1
2935	672.71796	-0.14229515	1.55540665	2	0	0	1
2936	670.61536	-0.14229515	1.55540665	2	0	0	1
2937	572.76923	-0.05796602	1.55540665	1	0	1	1
2938	663.43585	-0.01580145	1.55540665	2	0	0	1
2939	658.20514	-0.01580145	1.55540665	1	0	1	1
2940	598.20514	0.02636312	1.55540665	2	1	0	1
2941	654.76923	0.06852768	1.55540665	1	0	1	1
2942	779.79486	0.06852768	1.55540665	2	1	0	1
2943	507.89746	0.11069225	1.55540665	1	1	0	1
2944	745.43591	0.11069225	1.55540665	2	0	1	1
2945	813.79486	0.11069225	1.55540665	3	1	0	1
2946	678.30768	0.11069225	1.55540665	1	0	0	1
2947	511.74359	0.19502138	1.55540665	2	0	0	1
2948	964.41022	0.23718595	1.55540665	2	1	1	1
2949	754.61536	0.36367965	1.55540665	2	1	0	1

2950	537.33337	0.44800878	1.55540665	1	0	0	1
2951	505.02563	0.49017335	1.55540665	1	0	0	1
2952	422.00000	0.49017335	1.55540665	2	0	0	1
2953	386.30768	0.49017335	1.55540665	1	0	0	1
2954	732.30768	0.53233791	1.55540665	2	0	0	1
2955	727.33337	0.65883161	1.55540665	1	0	0	1
2956	464.87180	0.70099618	1.55540665	1	0	0	1
2957	944.00000	0.78532531	1.55540665	1	1	1	1
2958	918.41028	0.78532531	1.55540665	2	0	0	1
2959	945.33337	0.82748988	1.55540665	1	0	0	1
2960	857.79486	0.86965445	1.55540665	1	0	1	1
2961	838.46155	0.95398358	1.55540665	1	0	1	1
2962	860.10254	1.16480641	1.55540665	1	0	0	1
2963	522.61536	1.20697098	1.55540665	2	0	0	1
2964	848.76923	1.20697098	1.55540665	1	1	0	1
2965	988.97437	1.33346468	1.55540665	1	1	0	1
2966	1035.43591	1.58645208	1.55540665	2	1	0	1
2967	619.94873	2.00809774	1.55540665	1	0	1	1
2968	678.10260	2.42974341	1.55540665	2	0	0	1
2969	290.25641	-1.66021955	1.60023918	2	0	0	1
2970	382.15384	-0.94342192	1.60023918	1	0	0	1
2971	357.07690	-0.94342192	1.60023918	1	0	1	1
2972	496.20511	-0.94342192	1.60023918	2	0	0	1
2973	429.84616	-0.94342192	1.60023918	2	0	0	1
2974	543.84613	-0.85909278	1.60023918	2	0	0	1
2975	535.74359	-0.64826995	1.60023918	2	0	0	1
2976	565.07697	-0.47961168	1.60023918	1	0	1	1
2977	628.15387	-0.31095342	1.60023918	1	0	1	1
2978	616.30774	-0.18445972	1.60023918	1	0	1	1
2979	754.05127	0.23718595	1.60023918	1	0	0	1
2980	738.25641	0.36367965	1.60023918	1	0	0	1
2981	851.17950	0.74316075	1.60023918	1	0	0	1
2982	1216.71802	1.79727491	1.60023918	1	1	0	1
2983	341.84613	-1.40723215	1.64507171	1	0	0	1
2984	540.76923	-0.73259908	1.64507171	1	0	0	1
2985	510.41025	-0.60610538	1.64507171	1	0	1	1
2986	670.20514	-0.56394082	1.64507171	1	0	0	1
2987	584.41022	-0.39528255	1.64507171	2	0	0	1
2988	768.76923	-0.22662428	1.64507171	2	0	0	1
2989	678.05127	-0.10013058	1.64507171	1	0	1	1
2990	588.41028	-0.10013058	1.64507171	1	1	1	1
2991	774.25641	-0.01580145	1.64507171	1	0	1	1
2992	380.46155	0.19502138	1.64507171	2	1	0	1

2993	615.33337	0.23718595	1.64507171	1	0	0	1
2994	859.02563	0.32151508	1.64507171	2	0	1	1
2995	778.76923	0.57450248	1.64507171	1	0	0	1
2996	269.33334	-1.44939671	1.68990424	1	0	0	1
2997	239.02565	-1.36506758	1.68990424	2	0	0	1
2998	297.23074	-1.11208018	1.68990424	1	0	0	1
2999	497.33331	-0.98558648	1.68990424	1	0	0	1
3000	364.05130	-0.77476365	1.68990424	2	0	0	1
3001	495.28204	-0.77476365	1.68990424	1	0	1	1
3002	739.84613	-0.77476365	1.68990424	2	1	1	1
3003	568.71796	-0.73259908	1.68990424	2	0	0	1
3004	628.61542	-0.64826995	1.68990424	2	0	0	1
3005	606.10260	-0.39528255	1.68990424	2	0	0	1
3006	671.89740	-0.39528255	1.68990424	1	0	0	1
3007	577.64105	-0.35311798	1.68990424	1	0	0	1
3008	606.97437	-0.10013058	1.68990424	1	0	1	1
3009	395.38461	-0.10013058	1.68990424	2	0	0	1
3010	845.48718	0.15285682	1.68990424	2	0	0	1
3011	768.97437	0.78532531	1.68990424	1	0	0	1
3012	960.15387	0.78532531	1.68990424	2	0	0	1
3013	610.10254	1.24913555	1.68990424	1	0	0	1
3014	904.71790	1.37562925	1.68990424	1	0	0	1
3015	1459.23071	3.65251584	1.68990424	1	0	1	1
3016	412.30768	-1.49156128	1.71232051	1	0	0	1
3017	338.46155	-1.36506758	1.71232051	1	0	0	1
3018	422.87180	-1.36506758	1.71232051	2	0	0	1
3019	380.05127	-1.23857388	1.71232051	1	0	0	1
3020	237.79488	-1.15424475	1.71232051	1	0	0	1
3021	262.30768	-1.02775105	1.71232051	2	0	0	1
3022	512.41028	-0.85909278	1.71232051	1	0	0	1
3023	587.38464	-0.43744712	1.71232051	1	1	0	1
3024	705.38464	-0.39528255	1.71232051	2	0	0	1
3025	319.74359	-0.39528255	1.71232051	1	0	0	1
3026	683.89740	-0.31095342	1.71232051	1	0	0	1
3027	548.51282	-0.22662428	1.71232051	2	1	0	1
3028	376.97433	-0.22662428	1.71232051	1	0	0	1
3029	399.74359	-0.18445972	1.71232051	1	0	0	1
3030	521.43591	0.11069225	1.71232051	1	0	0	1
3031	375.38461	0.23718595	1.71232051	1	0	0	1
3032	601.53845	0.32151508	1.71232051	1	1	0	1
3033	408.05130	0.32151508	1.71232051	1	1	0	1
3034	865.28210	0.40584421	1.71232051	2	0	1	1
3035	726.76923	0.44800878	1.71232051	1	0	0	1

3036	618.51282	0.44800878	1.71232051	1	0	0	1
3037	576.20514	0.49017335	1.71232051	1	0	0	1
3038	327.43588	0.65883161	1.71232051	1	0	0	1
3039	786.82056	0.74316075	1.71232051	1	0	0	1
3040	521.94873	0.95398358	1.71232051	2	0	0	1
3041	977.17950	1.03831271	1.71232051	2	1	0	1
3042	805.84619	1.24913555	1.71232051	1	0	0	1
3043	912.20514	1.67078121	1.71232051	1	0	0	1
3044	848.92310	1.71294578	1.71232051	1	1	1	1
3045	345.64102	-1.36506758	1.73473677	1	0	0	1
3046	249.12820	-1.23857388	1.73473677	1	0	0	1
3047	403.64102	-1.15424475	1.73473677	1	0	1	1
3048	387.38461	-0.98558648	1.73473677	2	0	0	1
3049	383.23077	-0.56394082	1.73473677	2	0	0	1
3050	451.17947	-0.43744712	1.73473677	2	0	0	1
3051	646.76923	-0.14229515	1.73473677	2	0	1	1
3052	687.17950	0.02636312	1.73473677	2	0	0	1
3053	612.92303	0.11069225	1.73473677	1	0	0	1
3054	667.38464	0.11069225	1.73473677	1	0	0	1
3055	795.02570	0.44800878	1.73473677	2	0	0	1
3056	623.89740	0.61666705	1.73473677	2	0	0	1
3057	871.02563	0.74316075	1.73473677	1	1	0	1
3058	972.05127	0.78532531	1.73473677	2	1	0	1
3059	1334.97437	2.72489538	1.73473677	2	0	0	1
3060	378.30771	-1.61805498	1.77956930	1	0	0	1
3061	413.23077	-1.19640931	1.77956930	1	0	0	1
3062	417.02563	-0.94342192	1.77956930	2	0	0	1
3063	425.23077	-0.94342192	1.77956930	1	0	0	1
3064	523.38458	-0.81692822	1.77956930	1	0	0	1
3065	551.94873	-0.60610538	1.77956930	1	0	1	1
3066	413.48718	-0.60610538	1.77956930	1	0	1	1
3067	445.33334	-0.56394082	1.77956930	1	0	0	1
3068	506.30768	-0.47961168	1.77956930	1	0	0	1
3069	383.07693	-0.43744712	1.77956930	1	0	0	1
3070	384.71796	-0.43744712	1.77956930	1	0	0	1
3071	660.71796	-0.22662428	1.77956930	2	0	1	1
3072	590.71796	0.19502138	1.77956930	1	1	0	1
3073	322.25641	0.57450248	1.77956930	1	0	1	1
3074	667.33337	0.61666705	1.77956930	1	0	1	1
3075	665.79492	0.74316075	1.77956930	1	0	1	1
3076	857.84619	0.95398358	1.77956930	1	0	0	1
3077	585.07697	1.29130011	1.77956930	1	0	0	1
3078	1517.94873	2.47190798	1.77956930	2	1	0	1

```
3079  358.76923 -0.52177625  1.82440184         1     0         0          1
3080  746.82056  0.06852768  1.82440184         1     0         0          1
3081  593.94867  0.74316075  1.82440184         1     0         0          1
3082 1232.20520  2.38757884  1.82440184         2     0         0          1
```

As we mention before scaling to normality is equivalent of fitting a Normal variable first. The problem with scaling to normality is that if the variables are highly skew or kurtotic scaling them to normality does correct for skewness or kurtosis. Using a parametric distribution with four parameters some of which are skewness and kurtosis parameters it may correct that. Next we standardised using the `SHASHo` distribution;

```
#head(data_scale(da, response=rent,  family="SHASHo"))
```

The second form od standardisation is to transform all continuous x-variables to a range zero to one. Here is how this is done;

```
#head(data_scale(da, response=rent,  scale.to="0to1"))
```

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Transformations

In the context of outliers, the focus on how far a single observation deviates from the rest of the data. However, when considering **transformations**, the goal shifts toward a better capturing of the relationship between a continuous predictor and the response variable. Specifically, we aim to model **peaks and troughs** in the relationship between a single x-variable and the response better. To achieve this, it is sometimes sufficient to stretch the x-axis so that the predictor variable is more evenly distributed within its its range. A common approach for this is to use a power transformation of the form $T = X^{\lambda}$. This transoformation can reduce the curvature in the response and make the curve fitting easier. Importantly, the power transformation smoothly transitions into a logarithmic transformation as $\lambda \to 0$, since: $\frac{X^{\lambda}-1}{\lambda} \to \log(X) \quad \text{as } \lambda \to 0$. Thus, in the limit as $\lambda \to 0$ is the power transformation is the log transformation $T = \log(X)$.

> **Note**
>
> The power transformation is a subclass of the shifted power transformation $T = (\alpha + X)^{\lambda}$ which is not consider here

The aim of the power transformation is to make the model fitting process more robust and reliable and maximizing the information extracted from the data. There are three different functions in `gamlss.prepdata` dealling with power transformations, `xy_Ptrans()`, `data_Ptrans()` and `data_Ptrans_plot()`

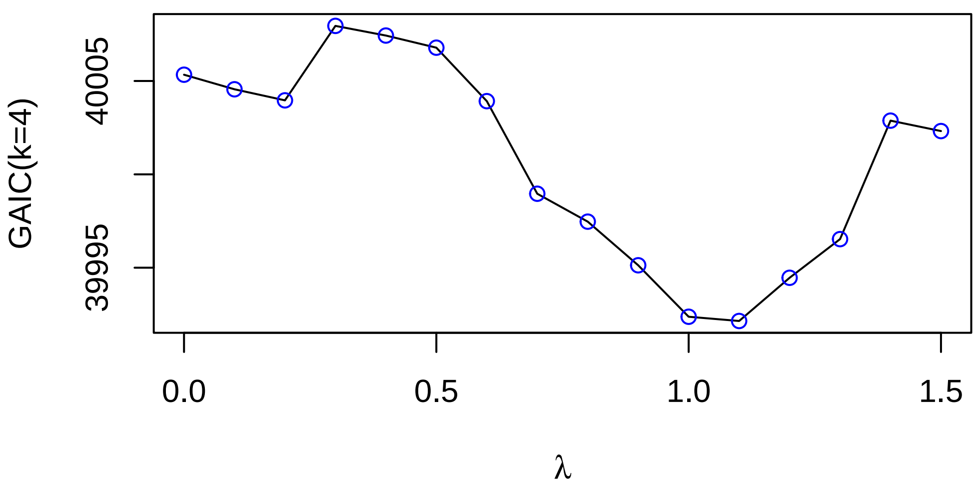
> **Note**
>
> It's important to distinguish between the concepts of *transformations* (the subject of this section) and of *feature extraction*, a term often used in machine learning. We refer to transformations as functions applied to a individual explanatory variables while we use feature extraction when multiple explanatory variables are involved. In both cases, the goal is to enhance the model capabilities.

`xy_Ptrans()`

```
pp<-gamlss.prepdata::xy_Ptrans(da$area, da$rent)
pp
```

```
[1] 1.06212
```

```
gamlss.prepdata::xy_Ptrans(da$area, da$rent, prof=TRUE, k=4)
```



`data_Ptrans()`

```r
gamlss.prepdata:::data_Ptrans(da, response=rent)
```

```
 100 % of data are saved,
that is, 3082 observations.

$area
[1] 1.06212

$yearc
[1] 1.499942
```

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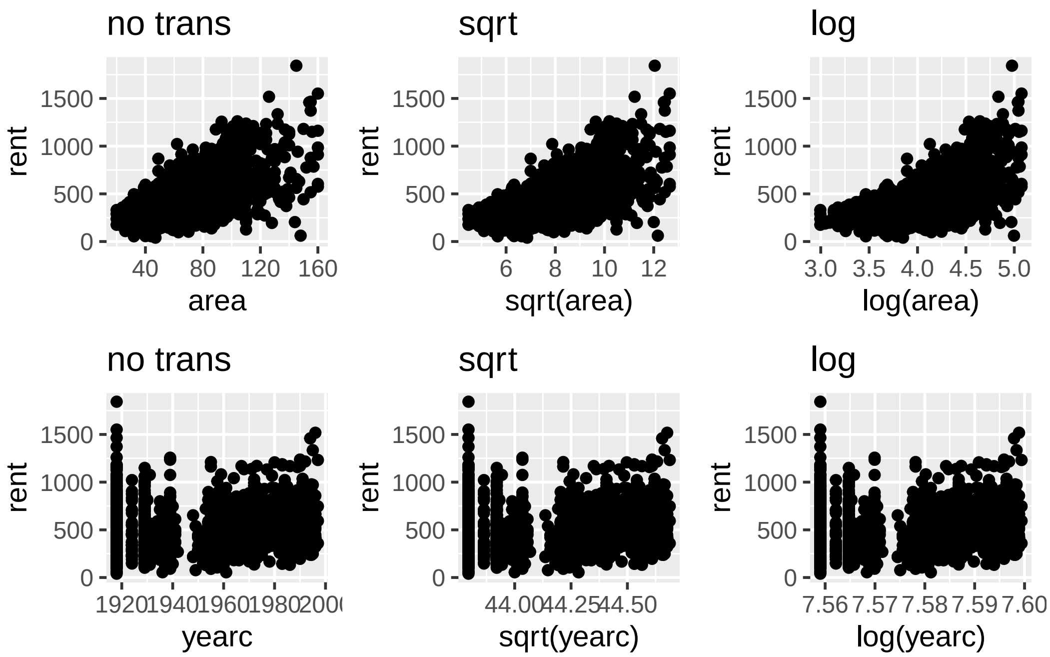
`data_Ptrans_plot()` (repeat)

```r
gamlss.prepdata:::data_Ptrans_plot(da, rent)
```

```
 100 % of data are saved,
that is, 3082 observations.
```



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Dates and Time

Time series data sets consist of observations collected at consistent time intervals—e.g., daily, weekly, or monthly. These data require special treatment because observations that are closer together in time tend to be more similar, violating the usual assumption of independence between observations. A similar issue arises in spatial data sets, where observations that are geographically closer often exhibit spatial correlation, again violating independence.

Many data sets include spatial or temporal features —such as longitude and latitude or timestamps (e.g., 12/10/2012)— those variables are can be used for for modeling directly, but also for interpreting or visualization of the results. If temporal or spatial information exists in the data, it is essential to ensure these features are correctly read and interpreted in `R`. For instance, dates in `R` can be handled using the `as.Date()` function. To understand its functionality and options associated with the function please consult the help documentation via `help("as.Date")`. Below, we provide two simple functions of how to work with date-time variables.

`time_dt2dh()`

Suppose the variable `dt` contains both a date and a time. We may want to extract and separate this into two components: one containing the date, and another capturing the hour of the day.

```
dt <- c("01/01/2011 00:00", "01/01/2011 01:00", "01/01/2011 02:00", "01/01/2011 03:00", "01/0
```

The function to do this separation can lok like;

```
time_dt2dh <- function(datetime, format=NULL)
  {
          X <- t(as.data.frame(strsplit(datetime,' ')))
rownames(X) <- NULL
colnames(X) <- c("date", "time")
       hour <- as.numeric(sub(":",".",X[,2]))
      date <- as.Date(X[,1],format=format)
      data.frame(date, hour)
  }
```

```
P <- time_dt2dh(dt, "%d/%m/%Y")
P
```

```
        date hour
1 2011-01-01    0
```

```
2 2011-01-01    1
3 2011-01-01    2
4 2011-01-01    3
5 2011-01-01    4
6 2011-01-01    5
```

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time_hour2num()

The second example contains time given in its common format, `c("12:35", "10:50")` and we would like t change it to numeric which we can use in the model.

```
t <- c("12:35", "10:50")
```

The function to change this to numeric;

```
time_hour2num <- function(time, pattern=":")
  {
  t <- gsub(pattern, ".", time)
   as.numeric(t)
  }
time_hour2num(t)
```

```
[1] 12.35 10.50
```

For user who deal with a lot of time series try the function `POSXct()` and also the function `select()` from the `dplur` package.

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Data Partition

Introduction to data partition

Partitioning the data is essential for comparing different models and checking for overfitting. Overfitting occurs when a model fails to generalize well to new data, typically because it is too closely fitted to the data. In contrast, underfitting refers to a poor model that fails to capture the essential patterns in both fitted and the new data, resulting to a model that is far the population intended to represent. It common to refer to data used for fitting the model as **training** data and the new data as **test** data.

Figure 1 shows different ways the data can be partitioned. A dataset can be split once, in **holdout** samples, or split multiple times using cross validation, **CV**, or **bootstrap**. For large datasets, it is common to split the data into a *training* set and two holdout samples, the *test* set, and, if needed the *validation* set. The training set is used for fitting the models, the test set is used to evaluate the prediction power and also to compare models, and the validation set is used for tuning the `hyperparameters`. [Note that for additive smoothing regression models, the hyper parameters are the `smoothing` parameters.] Observations in the training set are often referred to as the `in-bag` data, while observations in the test or validation set are termed the `out-of-bag` data.

Partitioning the data into training, test, and validation sets should be done randomly. This bring us to the two possible problems which could occurred when splitting the data; The **zero variance variable** and the **extreme values** problems. The zero variance variable (or nearly zero variance problem) occurs when a variable in the data (could be a factor) has only few distinct values and the frequency one of them is very low. For example, if we have the factor `gender` with mostly `female` values and only few `male` and we randomly portioning the data, there is a good change that all males could end up in only one of the partitions. That is, one of the partition would have factor gender properly with two levels, while the other partition with only one level. This would make the fitting and interpretation of the model difficult. One should check if this is the case. The extreme values problem is similar in nature but will effect overfitting. Extreme values could occurred in both the explanatory and the response variables. for the first case we have deal with `outliers` in Section we concentrate here on extreme values in the response. Extreme values in the response (if they genuine) are of great importance for risk analysis. There is a danger thet such evens when splitting the data can go to one partition and not to the other. Any evaluation of risks in this case could be potentially wrong. One possible solution could be **repeated** partition, i.e bootstrap.

Bootstrap, middle of 1, is a repeated partition method. In the classical, non-parametric bootstrap, the index of the data i.e. $1, 2, \dots, n$ is sampled with replacement B times and the model is refitted the on the re-sample data. In the Bayesian bootstrap the data are weighted using weighted samples from a multinomial distribution. B re-weighted models are refitted. While both classical and Bayesian bootstraps are computationally expensive, they provide additional insights into the model and in particular the variability of all of its parameters. A sample of length B is obtained for all the parameters of interest.

In K-fold cross-validation, (right part of Figure 1), the data are split into K sub-samples. The model is then fitted K times, each time using $K-1$ sub-samples for training and the remaining one sub-sample as test data set. By the end of the process, all n observations will have been used as `test` data exactly once. This is the main advantage of K-fold cross-validation is that provides reliable test data for model evaluation, (with the cost of fitting K models). Leave-One-Out Cross-Validation (LOO) is a special case of K-fold cross-validation, where the model is trained on $n-1$ observations and tested on the remaining single observation, iterating this process n times. Consequently, n different models are refitted.

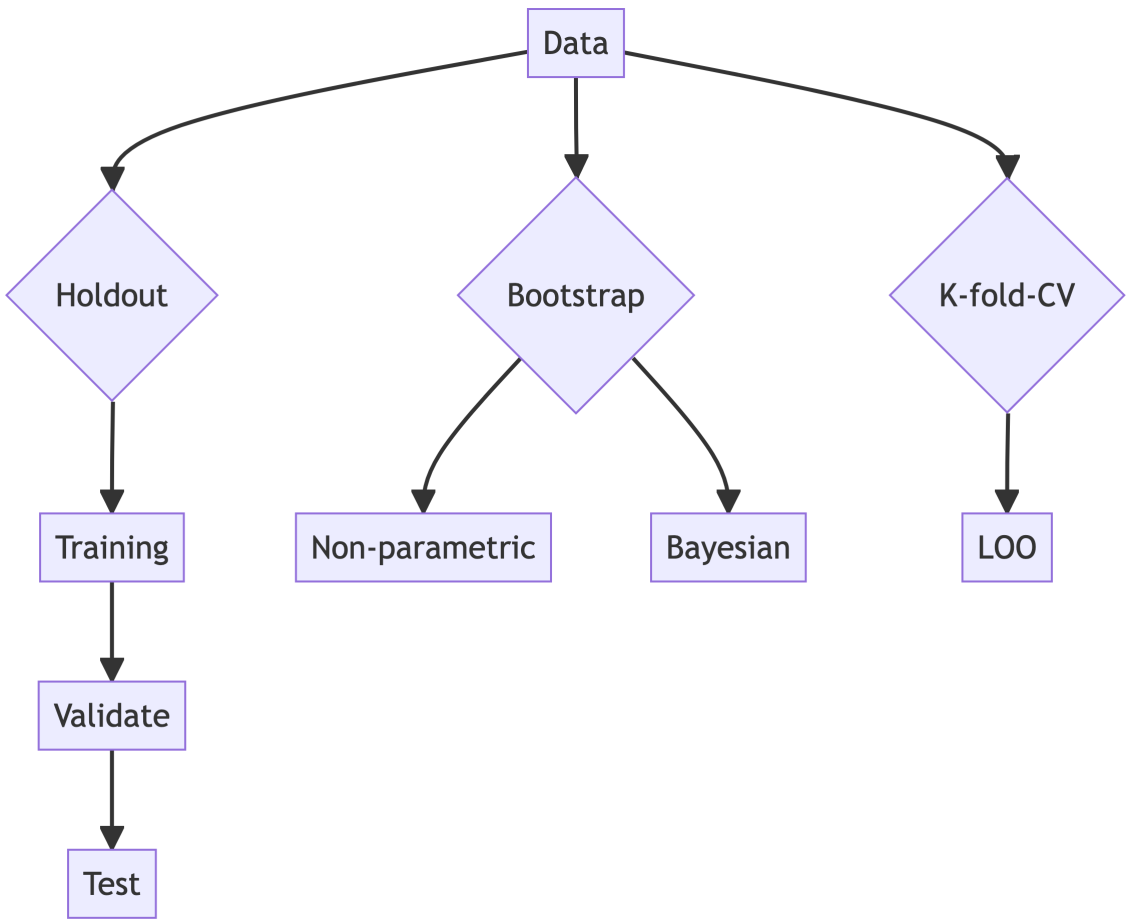

Figure 1: Different ways of splitting data to get in order to get more information”

Note

Repeated partitions (i.e. bootstrapping and k-fold cross validation) could mitigating some of the problems associated from a single partitions.

Data partitioning help in selection of models, model diagnostics and detection of overfitting. Here are some rules of thump applied to distributional regression models;

- if the residuals from the training data and the test data sets behave well, one is confidence, that the distributional assumption is adequate.
- if the fitted model performs reasonably well (using a measure of goodness-of-fit) on both the training and the test data sets, one can be confident that overfitting has been avoided.

All methods in Figure 1, need performance metrics. Those metrics can help can: i) facilitate model comparison, ii) detect overfitting, and iii) select hyper-parameters.

Various goodness-of-fit measures can be used to check and compare different distributional regression models. One of the most popular methods which do not required partition of the data is the Generalized Akaike Information Criterion (GAIC), see Akaike (1983). To calculate the GAIC, only the in-bag data are required. However, calculating the GAIC requires an accurate measure of the degrees of freedom used to fit the model. For mathematical (stochastic) models, the degrees of freedom are straight forward to define as they correspond to the number of estimated parameters. In contrast, for over-parametrised algorithmic models like neural networks, estimating the degrees of freedom is challenging. Note that the degrees of freedom serve as a measure of the model's complexity. We expect more complex models to be closer to the training data, which may reduce their ability to generalize well to new data (overfitting).

Note

Note that the most common of goodness-of-fit measures used in machine learning models requaring data partition are **not** appropriate for distributional regression model like GAMLSS. This because those measures concentrate in the behaviour at the centre of the distribution of the response which may or not be the focus of the study.

The most common measures of goodness-of-fit used in machine learning models with partitioning are;

- $RMSE$: root mean squared error;
- MAE: mean absolute error;
- R^2: and
- $MAPE$: mean absolute percentage error.

All those measures of goodness-of-fit are OK is someone is interested how the middle part of the distribution of the response is behaving but are NOT god if the interest is in the tail or if the distribution is very skew. For distributional regression model the following are more appropriate measures of goodness-of-fits.

- the **log-likelihood**: i.e. $\ell = \log L$, or equivalent the **deviance** $d = -2\ell$
- *CRPS*; the continuous rank probability scores.

Note that in general goodness-of-fit measure could be can applied to both the `in bag` and `out of bag` data sets;

Computationally partition of the data for any distributional regression model, can be achieved in different ways;

1. by *splitting* the original data frame in different subsets;
2. by using a *factor* which classifies the partitions of data sets;
3. by *indexing* the original data, i.e. `data[index,]`,
4. by using *prior weights* in the fitting.

So while we would like to think data partitioning as cutting of the original data into different subset (case 1 above), there is no inherent need for physically partition into the subgroups when fitting the models since the same results can be achieved with different methods. For example using prior weights in a K-fold cross-validation, we need K dummy vectors i_k. The dummy vectors take the value of 1 for the k-th set of observations and 0 for the rest.

The function `data_part()` creates a factor for holdout or k-fold CV partitions. The `data_part_list()` creates a list with different `data.frames` for holdout or k-fold CV partitions. The function `data_boot_index()` and `data_Kfold_index()` creates appropriate vector for indexing, while `data_boot_weights()` and `data_Kfold_weights()` generates the appropriate vectors for prior weights.

Table 4: The data partition functions

Functions	Usage
`data_part()`	Creates a single or multiple (CV) partitions by introducing a factor with different levels
`data_part_list()`	Creates a single or multiple (CV) partitions with output a list of `data.frames`
`data_boot_index()`	Creates two lists. The in-bag,`IB`, indices for fitting and the out-of-bag, `OOB`, indices for prediction

Functions	Usage
data_boot_weights()	Create a $n \times B$ matrix with columns weights for bootstrap fits
data_Kfold_index()	Creates a $n \times K$ matrix of variables to be used for cross validation data indexing
data_Kfold_weights()	Creates a $n \times K$ matrix of dummy variables to be used for cross validation weighted fits
data_cut()	This is not a partition funtion but randomly select a specified porpotion of the data

Here are the data partition functions.

data_part()

The function `data_part()` it does not partitioning the data as such but adds a factor called `partition` to the data indicating the partition. By default the data are partitioned into two sets the training ,`train`, and the test data, `test`.

```
dap <- gamlss.prepdata::data_part(da)
```

```
data partition into two sets
```

```
head(dap)
```

```
      rent area yearc location bath kitchen cheating partition
1 109.9487   26  1918        2    0       0        0     train
2 243.2820   28  1918        2    0       0        1      test
3 261.6410   30  1918        1    0       0        1     train
4 106.4103   30  1918        2    0       0        0      test
5 133.3846   30  1918        2    0       0        1      test
6 339.0256   30  1918        2    0       0        1     train
```

If the user would like to split the data in two separate `data.frame`'s she/he can use;

```
daTrain <- subset(dap, partition=="train")
daTest <- subset(dap, partition=="test")
dim(daTrain)
```

```
[1] 1846    8
```

```
dim(daTest)
```

```
[1] 1236    8
```

Note, that use of the option `partition` has the following behaviour;

i) `partition=2L` (the default) the factor has two levels `train`, and `test`.
ii) `partition=3L` the factor has three levels `train`, `val` (for validation) and `test`.
iii) `partition > 4L` say `K` then the levels are `1`, `2`...`K`. The factor then can be used to identify K-fold cross validation, see also the function `data_Kfols_CV()`.

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`data_part_list()`

The function `data_part_list()` creates a list of `dataframes` which can be used either for single partition or for cross validation. The argument `partition` allows the splitting of the data to up to 20 subsets. The default is a list of 2 with elements `training` and `test` (single partition).

Here is a single partition in `train` and `test` data sets.

```
allda <-  data_part_list(rent)
length(allda)
```

```
[1] 2
```

```
dim(allda[["train"]]) # training data
```

```
[1] 1200    9
```

```
dim(allda[["test"]]) # test data
```

```
[1] 769   9
```

Here is a multiple partition for cross validation.

```r
allda <-  data_part_list(rent, partition=10)
```

```
10-fold data partition
```

```r
length(allda)
```

```
[1] 10
```

```r
names(allda)
```

```
 [1] "1"  "2"  "3"  "4"  "5"  "6"  "7"  "8"  "9"  "10"
```

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`data_boot_index()`

The function `data_boot_index()` create two lists. The first called `IB` (for in-bag) and the second `OOB`, (out-of-bag). Each ellement of the list `IB` can be use to select a bootstrap sample from the original `data.frame` while each element of `OOB` can be use to select the out of sample data points. Note that each bootstap sample approximately contains 2/3 of the original data so we expact on avarage to 1/3 to be in `OOB` sample.

```r
DD <- data_boot_index(rent, B=10)
```

The class, length and names of the created lists;

```r
class(DD)
```

```
[1] "list"
```

```r
length(DD)
```

```
[1] 2
```

```r
names(DD)
```

```
[1] "IB"  "OOB"
```

the first obsevations of the

```r
head(DD$IB[[1]]) # in bag
```

```
[1] 1 2 4 6 8 8
```

```r
head(DD$OOB[[1]]) # out-of-bag
```

```
[1]  3  5  7 15 21 22
```

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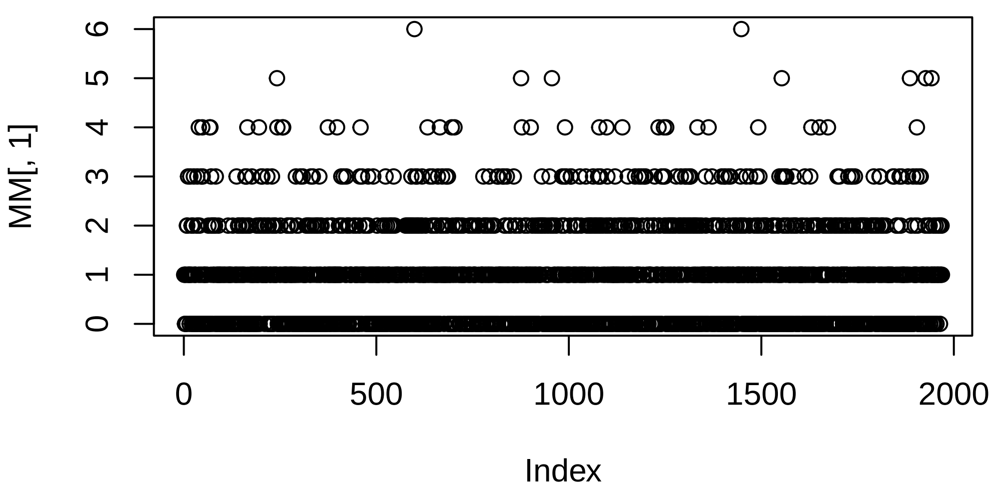
data_boot_weights()

The function `data_boot_weights()` create a $n \times B.$ matrix with columns possible weights for bootstap fits. Note that each bootstap sample approximately contains .666 of the original obsbation so we in general we expect 0.333 of the data to have zero weights.

```r
MM <- data_boot_weights(rent, B=10)
```

The `M` is a matrix which columns can be used as weights in bootstrap fits; Here is a plot of the first column

```r
plot(MM[,1])
```



and here is the first 6 rows os the matrix;

```r
head(MM)
```

```
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
[1,]    1    2    2    1    0    1    3    1    3     1
[2,]    1    1    0    2    2    1    2    1    1     2
[3,]    0    0    2    1    2    1    2    1    1     0
[4,]    1    0    1    0    3    0    1    1    0     2
[5,]    0    2    3    1    1    1    2    1    2     1
[6,]    1    0    0    1    0    1    4    0    1     0
```

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data_Kfold_index()

The function `data_Kfold()` creates a matrix which columns can be use for cross validation either as indices to select data for fitting or as prior weights.

```r
PP <- data_Kfold_index(rent, K=10)
head(PP)
```

```
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
[1,]    1    1    1    1    0    1    1    1    1     1
[2,]    2    2    0    2    2    2    2    2    2     2
[3,]    3    3    3    3    3    3    3    3    0     3
[4,]    4    4    4    4    4    0    4    4    4     4
[5,]    5    5    5    5    0    5    5    5    5     5
[6,]    6    6    6    6    6    6    6    0    6     6
```

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data_Kfold_weights()

The function `data_Kfold()` creates a matrix which columns can be use for cross validation either as indices to select data for fitting or as prior weights.

```r
PP <- data_Kfold_weights(rent, K=10)
head(PP)
```

```
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
[1,]    1    1    1    1    0    1    1    1    1     1
[2,]    1    1    0    1    1    1    1    1    1     1
[3,]    1    1    1    1    1    1    1    1    0     1
[4,]    1    1    1    1    1    0    1    1    1     1
[5,]    1    1    1    1    0    1    1    1    1     1
[6,]    1    1    1    1    1    1    1    0    1     1
```

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data_cut()

The function `data_cut()` is not included in the partition Section for its data partitioning properties. It is designed to select a random subset of the data specifically for plotting purposes. This is especially useful when working with large datasets, where plotting routines—such as those from the `ggplot2` package—can become very slow.

The `data_cut()` function can either: • Automatically reduce the data size based on the total number of observations, or • Subset the data according to a user-specified percentage.

Here is an example of the function assuming that that the user requires 50 of the data;

```
da1 <- data_cut(rent99, percentage=.5)
```

```
 50 % of data are saved, 
that is, 1541 observations.
```

```
dim(rent99)
```

```
[1] 3082    9
```

```
dim(da1)
```

```
[1] 1541    9
```

The function `data_cut()` is used extensively in many plotting routines within the `gamlss.ggplots` package. When the percentage option is not specified, a default rule is applied to determine the proportion of the data used for plotting. This approach balances performance and visualization quality when working with large datasets.

Let n denote the total number of observations. Then:

- if $n \leq 50,000$: All data are used for plotting.
- if $50,000 < n \leq 100,000$: 50% of the data are used.
- If $100,000 < n \leq 1,000,000$: 20% of the data are used.
- If $n > 1,000,000$: 10% of the data are used.

This default behaviour ensures that plotting remains efficient even with very large datasets.

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Distribution Families

> **Note**
>
> All functions listed below belong to the `gamlss.ggplots` package, not the `gamlss.prepdata` package. However, since they can be useful during the pre-modeling stage, they are presented here. The later version of the `gamlss.ggplots` can be found in https://github.com/gamlss-dev/gamlss.ggplots

In the following function, no fitted model is required—only values for the parameters need to be specified.

`family_pdf()`

The function `family_pdf()` plots individual probability density functions (PDFs) from distributions in the `gamlss.family` package. Although it typically requires the family argument, it defaults to the Normal distribution (NO) if none is provided.

```
family_pdf(from=-5,to=5, mu=0, sigma=c(.5,1,2))
```

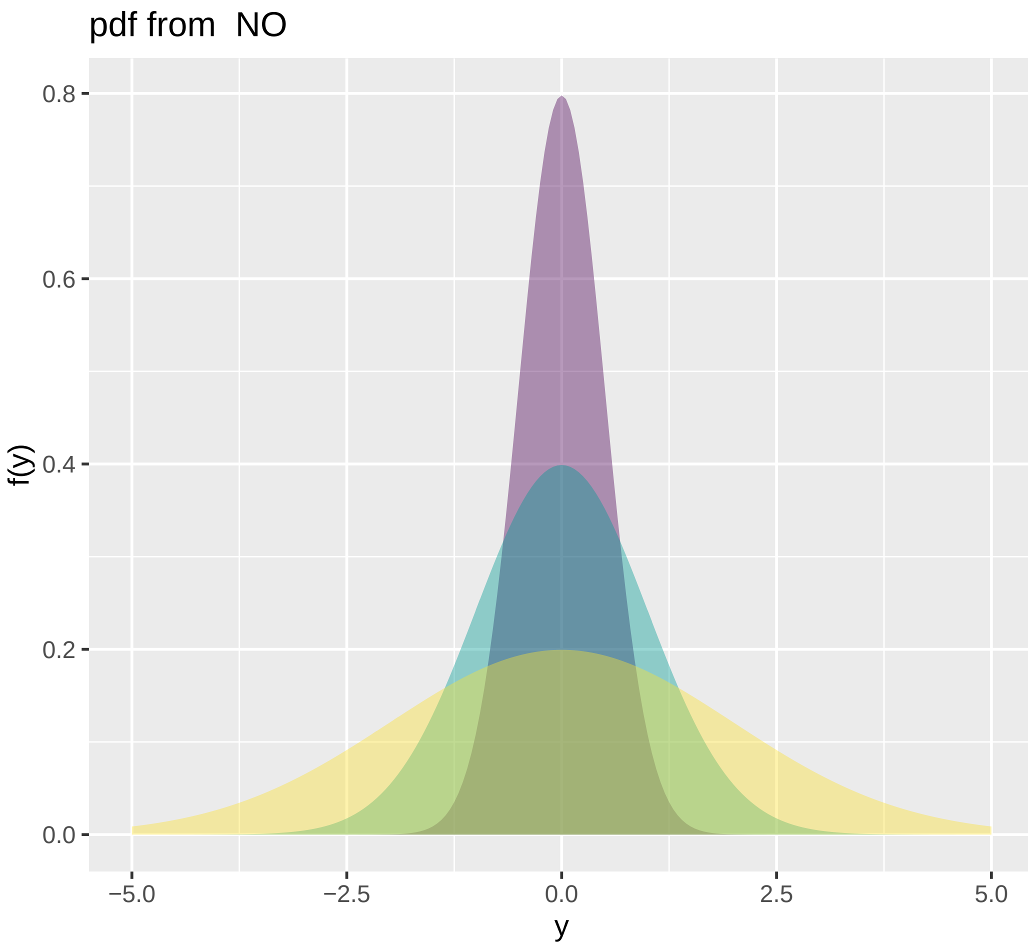

Figure 2: Continuous response example: plotting the pdf of a normal random variable.

The following demonstrates how discrete distributions are displayed. We begin with a distribution that can take infinitely many count values—the Negative Binomial distribution;

```
family_pdf(NBI, to=15, mu=1, sigma=c(.5,1,2), alpha=.9, size.seqment = 3)
```

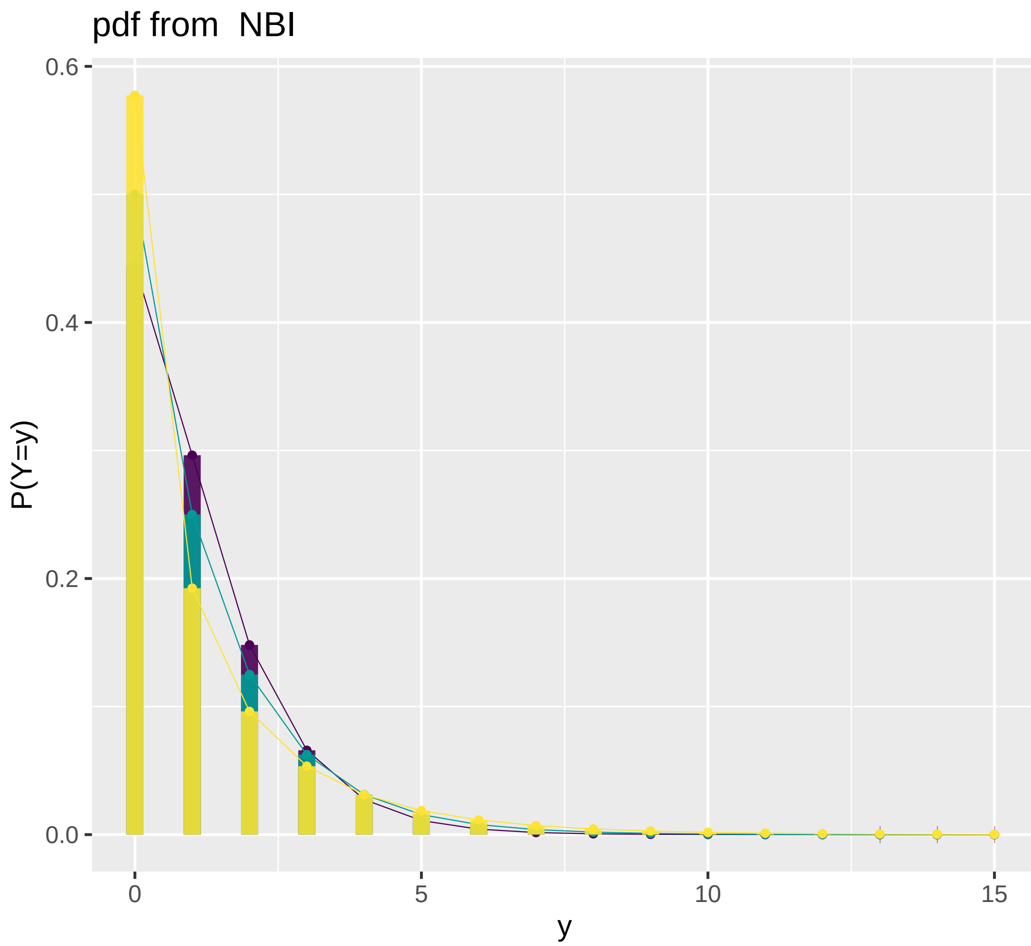

Figure 3: Count response example: plotting the pdf of a beta binomial.

Here we use a discrete distribution with a finite range of possible values: the Beta-Binomial distribution.

```
family_pdf(BB, to=15, mu=.5, sigma=c(.5,1,2),  alpha=.9, , size.seqment = 3)
```

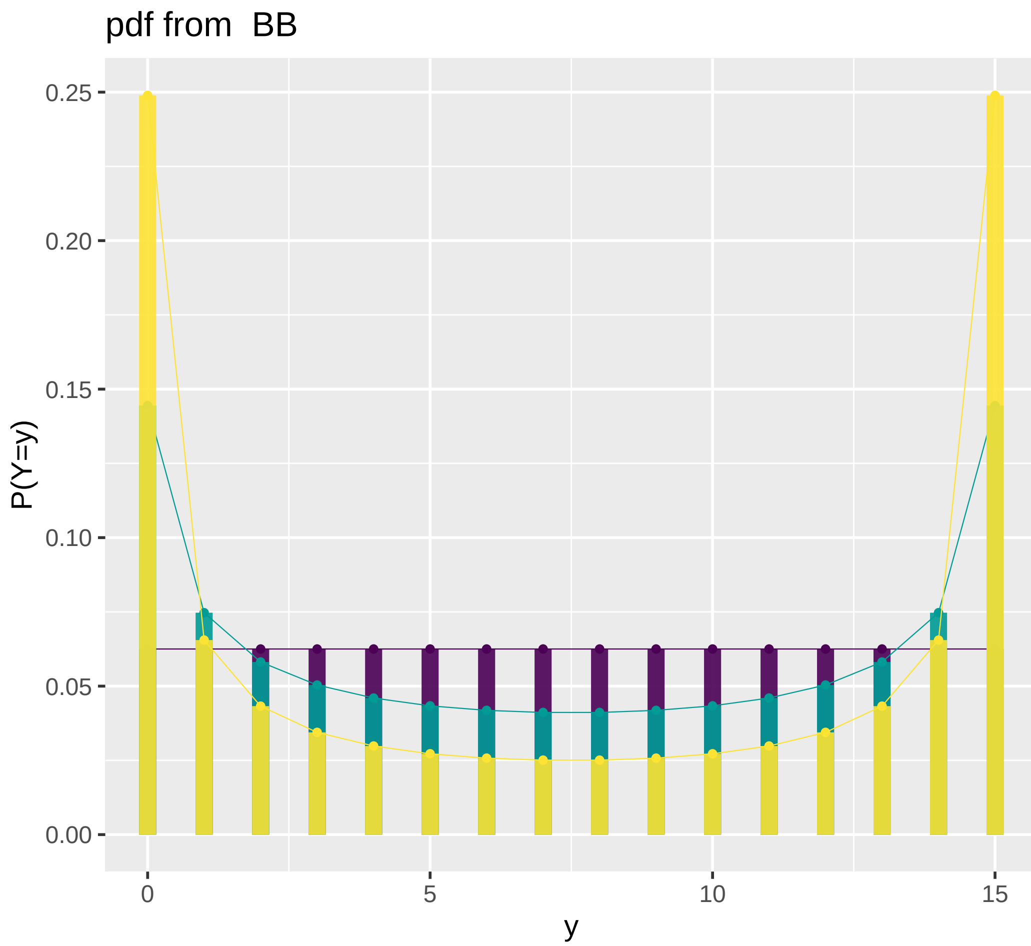

Figure 4: Beta binomial response example: plotting the pdf of a beta binomial.

`family_cdf`

The function `family_cdf()`

The function `family_cdf()` plots cumulative distribution functions (CDFs) from the **gamlss.family** distributions. The primary argument required is the family specifying the distribution to be used.

```
family_cdf(NBI, to=15, mu=1, sigma=c(.5,1,2), alpha=.9, size.seqment = 3)
```

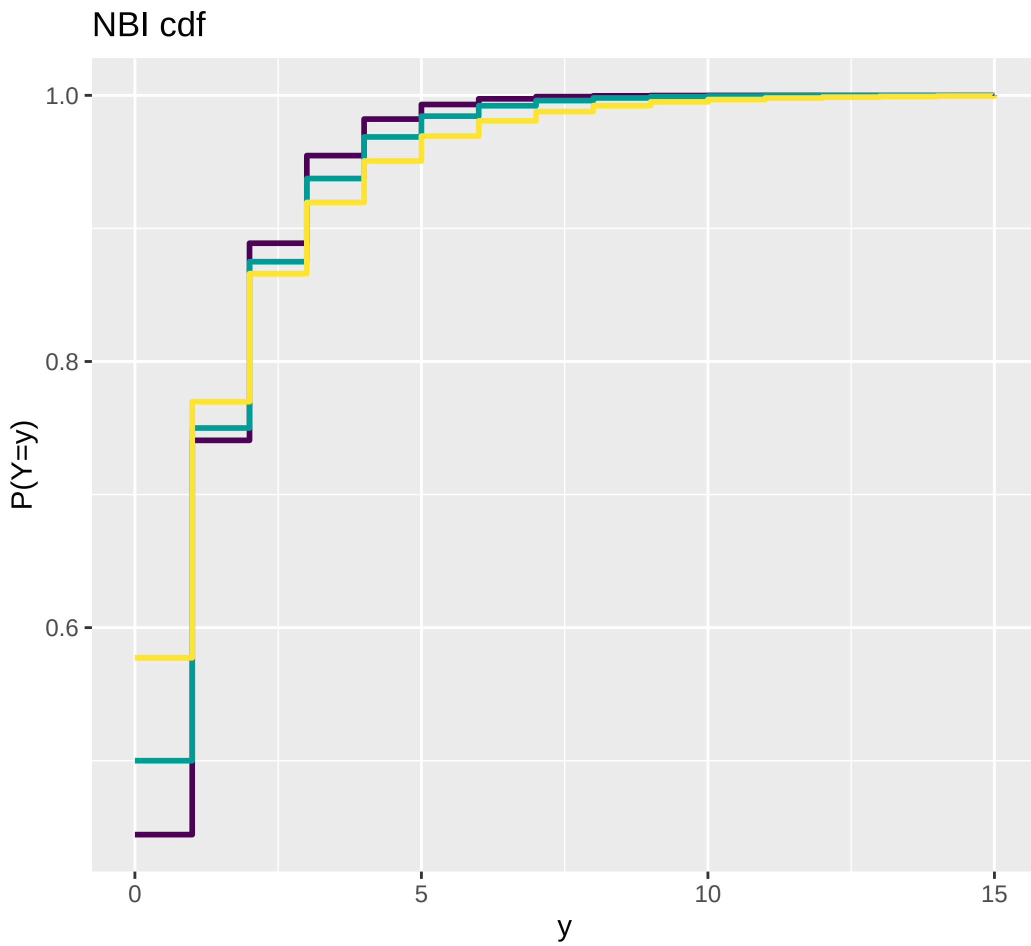

Figure 5: Count response example: plotting the cdf of a negative binomial.

The CDf of the negative binomial;

```
family_cdf(BB, to=15, mu=.5, sigma=c(.5,1,2),  alpha=.9, , size.seqment = 3)
```

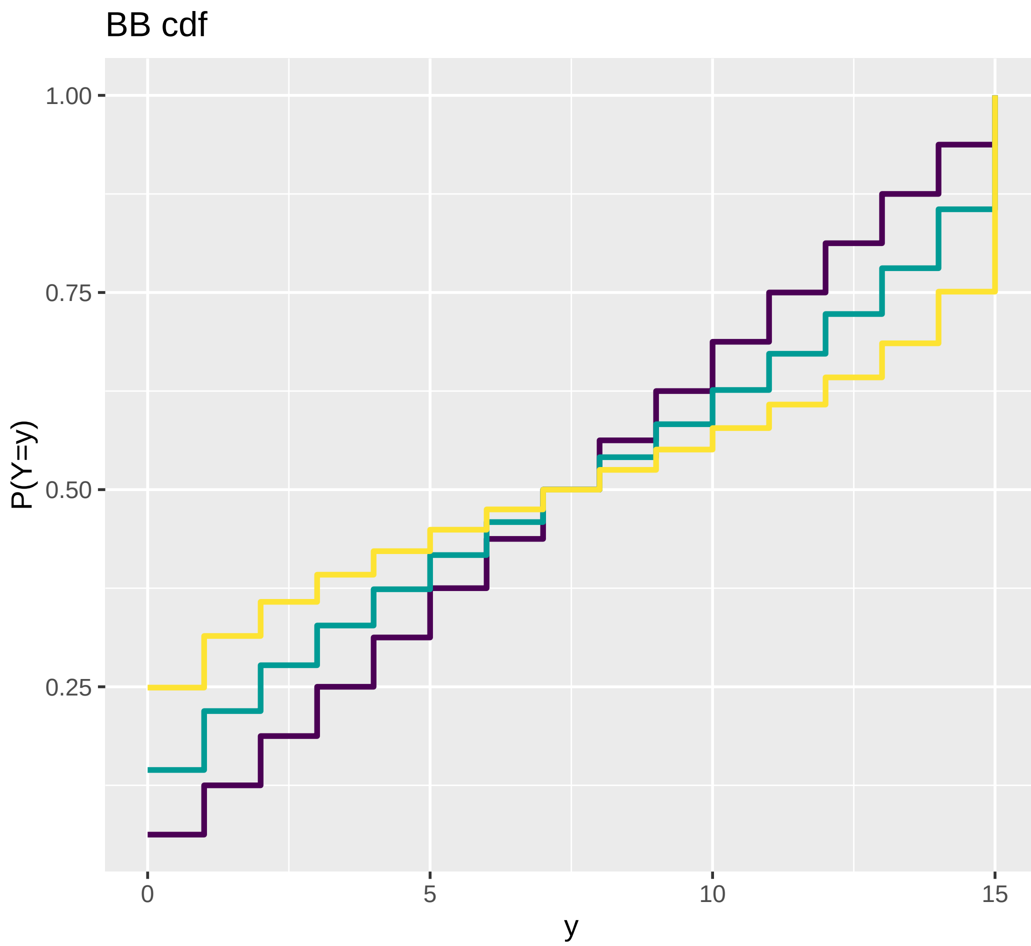

Figure 6: Count response example: plotting the cdf of a beta binomial.

`family_cor()`

The function `family_cor()` offers a basic method for examining the inter-correlation among the parameters of any distribution from the **gamlss.family**. It performs the following steps:

- Generates 10,000 random values from the specified distribution.
- Fits the same distribution to the generated data.
- Extracts and plots the correlation coefficients of the distributional parameters.

These correlation coefficients are derived from the variance-covariance matrix of the fitted model.

> ⚠ Warning
>
> This method provides only a rough indication of how the parameters are correlated at specific values of the distribution's parameters. The correlation structure may vary significantly at different points in the parameter space, as the distribution can behave quite differently depending on those values.

```
#source("~/Dropbox/github/gamlss-ggplots/R/family_cor.R")
gamlss.ggplots:::family_cor("BCTo", mu=1, sigma=0.11, nu=1, tau=5, no.sim=10000)
```

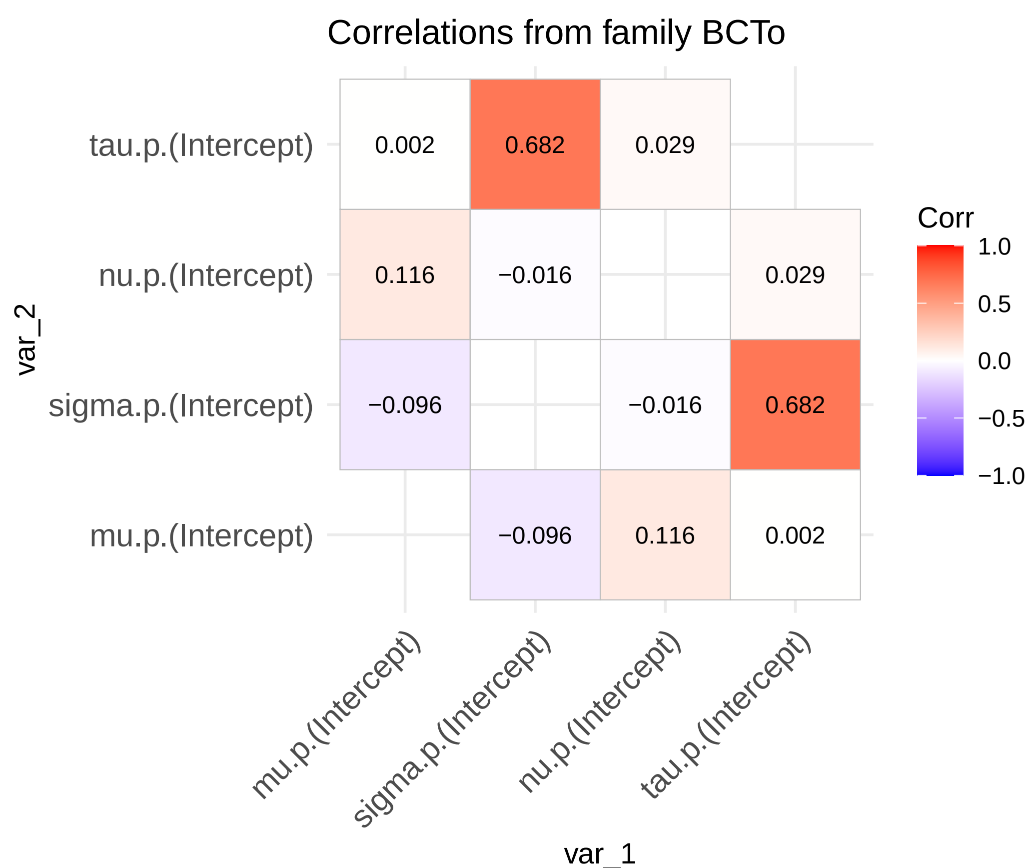

Figure 7: Family correlation of a BCTo distribution at specified values of the parameters.

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